GLENCOE MATHEMATICS

Algebra 2

Chapter 14
Resource Masters

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Consumable Workbooks

Many of the worksheets contained in the Chapter Resource Masters booklets are available as consumable workbooks.

Study Guide and Intervention Workbook	0-07-828029-X
Skills Practice Workbook	0-07-828023-0
Practice Workbook	0-07-828024-9

ANSWERS FOR WORKBOOKS The answers for Chapter 14 of these workbooks can be found in the back of this Chapter Resource Masters booklet.

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Algebra 2
Chapter 14 Resource Masters

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Teacher's Guide to Using the Chapter 14 Resource Masters

The ***Fast File*** Chapter Resource system allows you to conveniently file the resources you use most often. The *Chapter 14 Resource Masters* includes the core materials needed for Chapter 14. These materials include worksheets, extensions, and assessment options. The answers for these pages appear at the back of this booklet.

All of the materials found in this booklet are included for viewing and printing in the *Algebra 2 TeacherWorks* CD-ROM.

Vocabulary Builder Pages vii–viii include a student study tool that presents up to twenty of the key vocabulary terms from the chapter. Students are to record definitions and/or examples for each term. You may suggest that students highlight or star the terms with which they are not familiar.

WHEN TO USE Give these pages to students before beginning Lesson 14-1. Encourage them to add these pages to their Algebra 2 Study Notebook. Remind them to add definitions and examples as they complete each lesson.

Study Guide and Intervention Each lesson in *Algebra 2* addresses two objectives. There is one Study Guide and Intervention master for each objective.

WHEN TO USE Use these masters as reteaching activities for students who need additional reinforcement. These pages can also be used in conjunction with the Student Edition as an instructional tool for students who have been absent.

Skills Practice There is one master for each lesson. These provide computational practice at a basic level.

WHEN TO USE These masters can be used with students who have weaker mathematics backgrounds or need additional reinforcement.

Practice There is one master for each lesson. These problems more closely follow the structure of the Practice and Apply section of the Student Edition exercises. These exercises are of average difficulty.

WHEN TO USE These provide additional practice options or may be used as homework for second day teaching of the lesson.

Reading to Learn Mathematics One master is included for each lesson. The first section of each master asks questions about the opening paragraph of the lesson in the Student Edition. Additional questions ask students to interpret the context of and relationships among terms in the lesson. Finally, students are asked to summarize what they have learned using various representation techniques.

WHEN TO USE This master can be used as a study tool when presenting the lesson or as an informal reading assessment after presenting the lesson. It is also a helpful tool for ELL (English Language Learner) students.

Enrichment There is one extension master for each lesson. These activities may extend the concepts in the lesson, offer an historical or multicultural look at the concepts, or widen students' perspectives on the mathematics they are learning. These are not written exclusively for honors students, but are accessible for use with all levels of students.

WHEN TO USE These may be used as extra credit, short-term projects, or as activities for days when class periods are shortened.

Assessment Options

The assessment masters in the *Chapter 14 Resource Masters* offer a wide range of assessment tools for intermediate and final assessment. The following lists describe each assessment master and its intended use.

Chapter Assessment

CHAPTER TESTS

- *Form 1* contains multiple-choice questions and is intended for use with basic level students.
- *Forms 2A and 2B* contain multiple-choice questions aimed at the average level student. These tests are similar in format to offer comparable testing situations.
- *Forms 2C and 2D* are composed of free-response questions aimed at the average level student. These tests are similar in format to offer comparable testing situations. Grids with axes are provided for questions assessing graphing skills.
- *Form 3* is an advanced level test with free-response questions. Grids without axes are provided for questions assessing graphing skills.

  All of the above tests include a free-response Bonus question.
- The **Open-Ended Assessment** includes performance assessment tasks that are suitable for all students. A scoring rubric is included for evaluation guidelines. Sample answers are provided for assessment.
- A **Vocabulary Test**, suitable for all students, includes a list of the vocabulary words in the chapter and ten questions assessing students' knowledge of those terms. This can also be used in conjunction with one of the chapter tests or as a review worksheet.

Intermediate Assessment

- Four free-response **quizzes** are included to offer assessment at appropriate intervals in the chapter.
- A **Mid-Chapter Test** provides an option to assess the first half of the chapter. It is composed of both multiple-choice and free-response questions.

Continuing Assessment

- The **Cumulative Review** provides students an opportunity to reinforce and retain skills as they proceed through their study of Algebra 2. It can also be used as a test. This master includes free-response questions.
- The **Standardized Test Practice** offers continuing review of algebra concepts in various formats, which may appear on the standardized tests that they may encounter. This practice includes multiple-choice, grid-in, and quantitative-comparison questions. Bubble-in and grid-in answer sections are provided on the master.

Answers

- Page A1 is an answer sheet for the Standardized Test Practice questions that appear in the Student Edition on pages 810–811. This improves students' familiarity with the answer formats they may encounter in test taking.
- The answers for the lesson-by-lesson masters are provided as reduced pages with answers appearing in red.
- Full-size answer keys are provided for the assessment masters in this booklet.

NAME _____ DATE _____ PERIOD _____

14 Reading to Learn Mathematics

Vocabulary Builder

This is an alphabetical list of the key vocabulary terms you will learn in Chapter 14. As you study the chapter, complete each term's definition or description. Remember to add the page number where you found the term. Add these pages to your Algebra Study Notebook to review vocabulary at the end of the chapter.

Vocabulary Term	Found on Page	Definition/Description/Example
amplitude AM·pluh·TOOD		
double-angle formula		
half-angle formula		
midline		
phase shift FAYZ		

(continued on the next page)

14 Reading to Learn Mathematics

Vocabulary Builder *(continued)*

Vocabulary Term	Found on Page	Definition/Description/Example
trigonometric equation		
trigonometric identity		
vertical shift		

NAME _____ DATE _____ PERIOD _____

14-1 Study Guide and Intervention

Graphing Trigonometric Functions

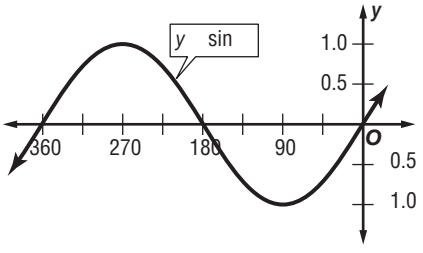
Graph Trigonometric Functions To graph a trigonometric function, make a table of values for known degree measures (0°, 30°, 45°, 60°, 90°, and so on). Round function values to the nearest tenth, and plot the points. Then connect the points with a smooth, continuous curve. The *period* of the sine, cosine, secant, and cosecant functions is 360° or 2π radians.

Amplitude of a Function	The **amplitude** of the graph of a periodic function is the absolute value of half the difference between its maximum and minimum values.

Example **Graph $y = \sin\theta$ for $-360° \le \theta \le 0°$.**

First make a table of values.

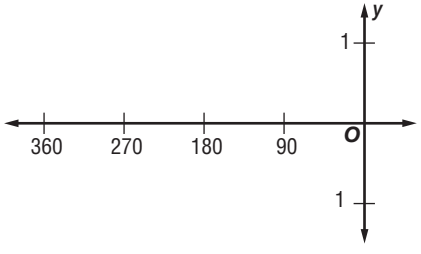
θ	$-360°$	$-330°$	$-315°$	$-300°$	$-270°$	$-240°$	$-225°$	$-210°$	$-180°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
θ	$-150°$	$-135°$	$-120°$	$-90°$	$-60°$	$-45°$	$-30°$	0°	
$\sin\theta$	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{1}{2}$	0	



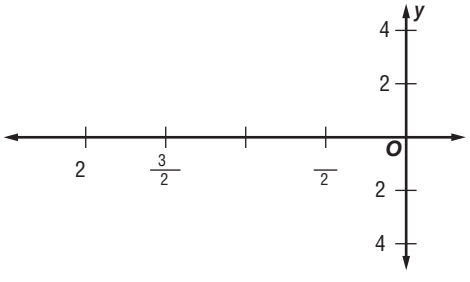
Exercises

Graph the following functions for the given domain.

1. $\cos\theta$, $-360° \le \theta \le 0°$

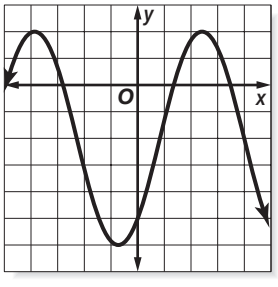


2. $\tan\theta$, $-2\pi \le \theta \le 0$

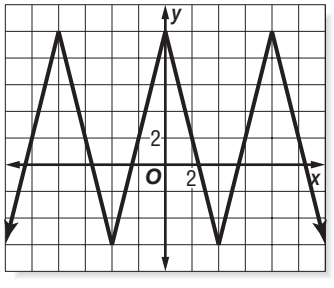


What is the amplitude of each function?

3.



4.



Lesson 14-1

14-1 Study Guide and Intervention *(continued)*

Graphing Trigonometric Functions

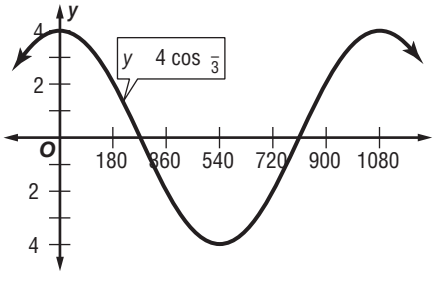
Variations of Trigonometric Functions

Amplitudes and Periods	For functions of the form $y = a \sin b\theta$ and $y = a \cos b\theta$, the amplitude is $\lvert a \rvert$, and the period is $\frac{360°}{\lvert b \rvert}$ or $\frac{2\pi}{\lvert b \rvert}$. For functions of the form $y = a \tan b\theta$, the amplitude is not defined, and the period is $\frac{180°}{\lvert b \rvert}$ or $\frac{\pi}{\lvert b \rvert}$.

Example **Find the amplitude and period of each function. Then graph the function.**

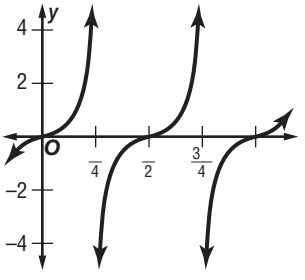
a. $y = 4 \cos \frac{\theta}{3}$

First, find the amplitude.
$\lvert a \rvert = \lvert 4 \rvert$, so the amplitude is 4.
Next find the period.
$\frac{360°}{\left\lvert \frac{1}{3} \right\rvert} = 1080°$
Use the amplitude and period to help graph the function.



b. $y = -\frac{1}{2} \tan 2\theta$

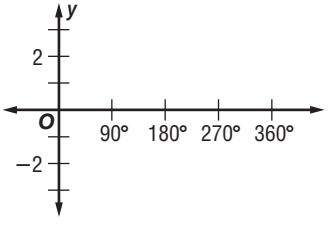
The amplitude is not defined, and the period is $\frac{\pi}{2}$.



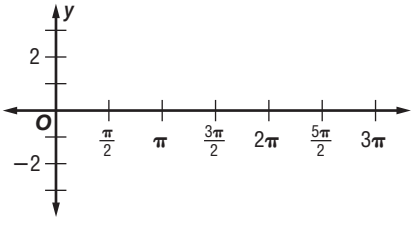
Exercises

Find the amplitude, if it exists, and period of each function. Then graph each function.

1. $y = -3 \sin \theta$



2. $y = 2 \tan \frac{\theta}{2}$



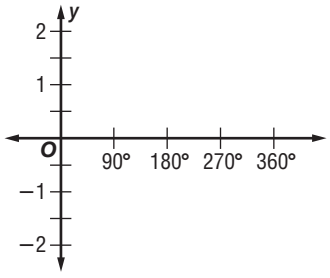
NAME _____ DATE _____ PERIOD _____

14-1 Skills Practice

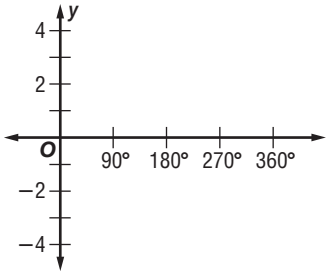
Graphing Trigonometric Functions

Find the amplitude, if it exists, and period of each function. Then graph each function.

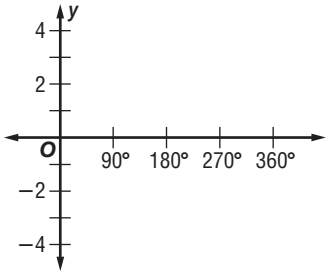
1. $y = 2 \cos \theta$



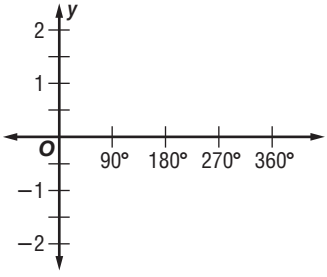
2. $y = 4 \sin \theta$



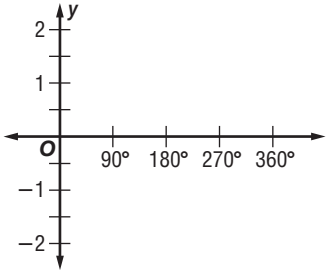
3. $y = 2 \sec \theta$



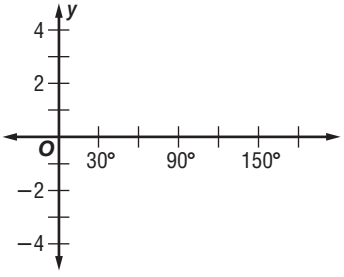
4. $y = \frac{1}{2} \tan \theta$



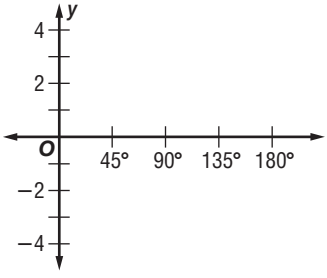
5. $y = \sin 3\theta$



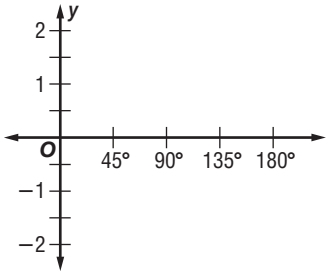
6. $y = \csc 3\theta$



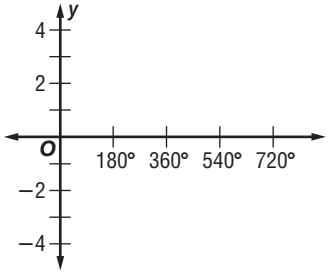
7. $y = \tan 2\theta$



8. $y = \cos 2\theta$



9. $y = 4 \sin \frac{1}{2}\theta$



Lesson 14-1

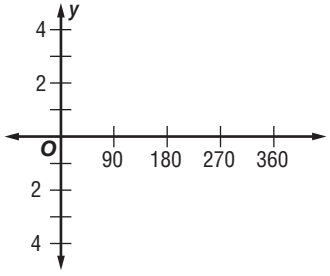
NAME _____ DATE _____ PERIOD _____

14-1 Practice

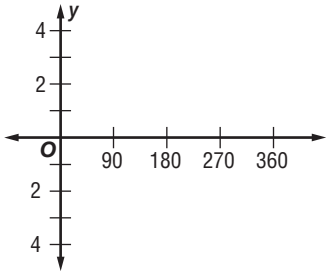
Graphing Trigonometric Functions

Find the amplitude, if it exists, and period of each function. Then graph each function.

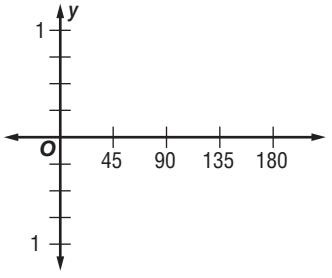
1. $y = -4 \sin \theta$



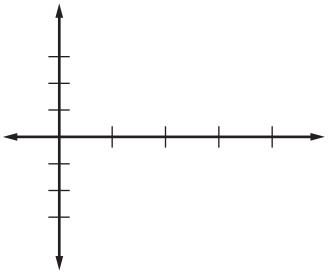
2. $y = \cot \frac{1}{2}\theta$



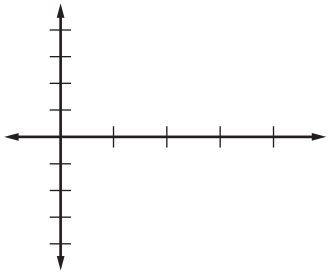
3. $y = \cos 5\theta$



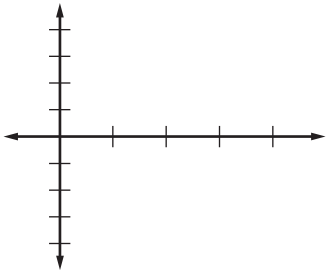
4. $y = \csc \frac{3}{4}\theta$



5. $y = 2 \tan \frac{1}{2}\theta$

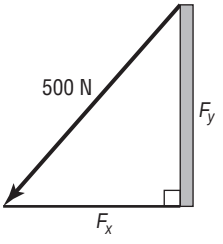


6. $2y = \sin \theta$



FORCE For Exercises 7 and 8, use the following information.

An anchoring cable exerts a force of 500 Newtons on a pole. The force has the horizontal and vertical components F_x and F_y. (A force of one Newton (N), is the force that gives an acceleration of 1 m/sec^2 to a mass of 1 kg.)



7. The function $F_x = 500 \cos \theta$ describes the relationship between the angle θ and the horizontal force. What are the amplitude and period of this function?

8. The function $F_y = 500 \sin \theta$ describes the relationship between the angle θ and the vertical force. What are the amplitude and period of this function?

WEATHER For Exercises 9 and 10, use the following information.

The function $y = 60 + 25 \sin \frac{\pi}{6}t$, where t is in months and $t = 0$ corresponds to April 15, models the average high temperature in degrees Fahrenheit in Centerville.

9. Determine the period of this function. What does this period represent?

10. What is the maximum high temperature and when does this occur?

NAME _____ DATE _____ PERIOD _____

14-1 Reading to Learn Mathematics

Graphing Trigonometric Functions

Pre-Activity **Why can you predict the behavior of tides?**

Read the introduction to Lesson 14-1 at the top of page 762 in your textbook.

Consider the tides of the Atlantic Ocean as a function of time. Approximately what is the period of this function?

Reading the Lesson

1. Determine whether each statement is *true* or *false*.

a. The period of a function is the distance between the maximum and minimum points.

b. The amplitude of a function is the difference between its maximum and minimum values.

c. The amplitude of the function $y = \sin \theta$ is 2π.

d. The function $y = \cot \theta$ has no amplitude.

e. The period of the function $y = \sec \theta$ is π.

f. The amplitude of the function $y = 2 \cos \theta$ is 4.

g. The function $y = \sin 2\theta$ has a period of π.

h. The period of the function $y = \cot 3\theta$ is $\frac{\pi}{3}$.

i. The amplitude of the function $y = -5 \sin \theta$ is -5.

j. The period of the function $y = \csc \frac{1}{4}\theta$ is 4π.

k. The graph of the function $y = \sin \theta$ has no asymptotes.

l. The graph of the function $y = \tan \theta$ has an asymptote at $\theta = 180°$.

m. When $\theta = 360°$, the values of $\cos \theta$ and $\sec \theta$ are equal.

n. When $\theta = 270°$, $\cot \theta$ is undefined.

o. When $\theta = 180°$, $\csc \theta$ is undefined.

Helping You Remember

2. What is an easy way to remember the periods of $y = a \sin b\theta$ and $y = a \cos b\theta$?

Lesson 14-1

NAME _____ DATE _____ PERIOD _____

14-1 Enrichment

Blueprints

Interpreting blueprints requires the ability to select and use trigonometric functions and geometric properties. The figure below represents a plan for an improvement to a roof. The metal fitting shown makes a 30° angle with the horizontal. The vertices of the geometric shapes are *not* labeled in these plans. Relevant information must be selected and the appropriate function used to find the unknown measures.

Example **Find the unknown measures in the figure at the right.**

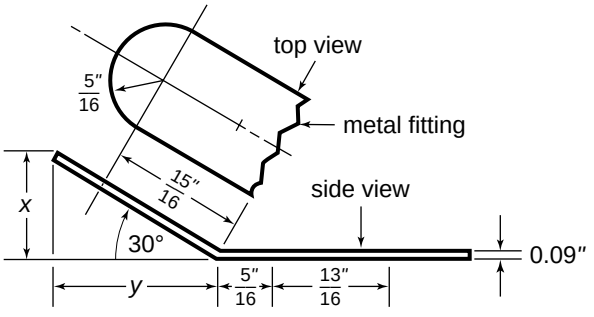
The measures x and y are the legs of a right triangle.

The measure of the hypotenuse is $\frac{15}{16}$ in. + $\frac{5}{16}$ in. or $\frac{20}{16}$ in.

$\frac{y}{\frac{20}{16}} = \cos 30°$ $\quad$ $\frac{x}{\frac{20}{16}} = \sin 30°$

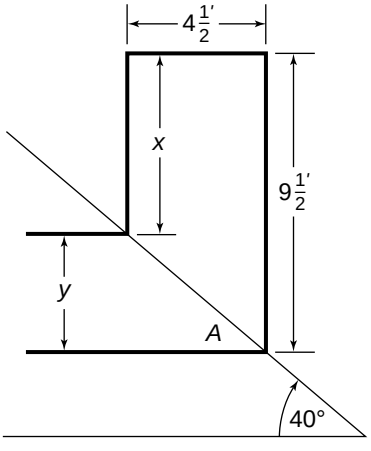
$y = 1.08$ in. $\quad$ $x = 0.63$ in.

Roofing Improvement

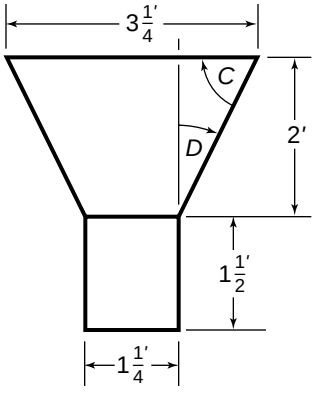


Find the unknown measures of each of the following.

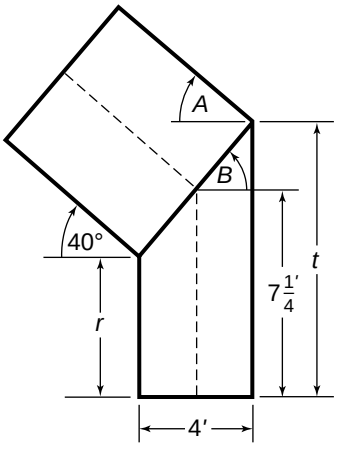
1. Chimney on roof



2. Air vent



3. Elbow joint



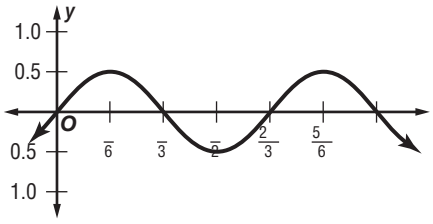
14-2 Study Guide and Intervention

Translations of Trigonometric Graphs

Horizontal Translations When a constant is subtracted from the angle measure in a trigonometric function, a **phase shift** of the graph results.

Phase Shift	The horizontal phase shift of the graphs of the functions $y = a \sin b(\theta - h)$, $y = a \cos b(\theta - h)$, and $y = a \tan b(\theta - h)$ is h, where $b > 0$. If $h > 0$, the shift is to the right. If $h < 0$, the shift is to the left.

Example **State the amplitude, period, and phase shift for $y = \frac{1}{2} \cos 3\left(\theta - \frac{\pi}{2}\right)$. Then graph the function.**



Amplitude: $a = \left|\frac{1}{2}\right|$ or $\frac{1}{2}$

Period: $\frac{2\pi}{|b|} = \frac{2\pi}{|3|}$ or $\frac{2\pi}{3}$

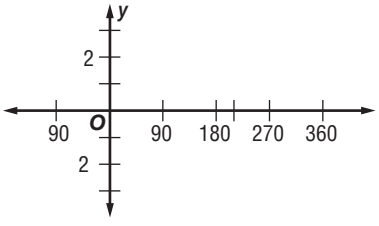
Phase Shift: $h = \frac{\pi}{2}$

The phase shift is to the right since $\frac{\pi}{2} > 0$.

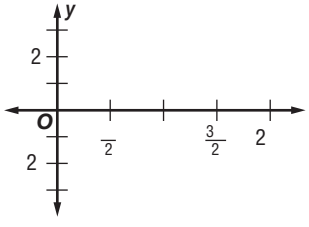
Exercises

State the amplitude, period, and phase shift for each function. Then graph the function.

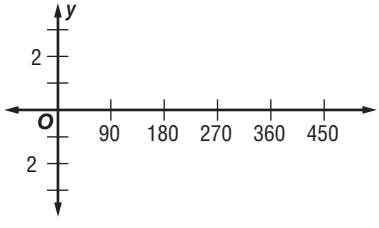
1. $y = 2 \sin (\theta + 60°)$



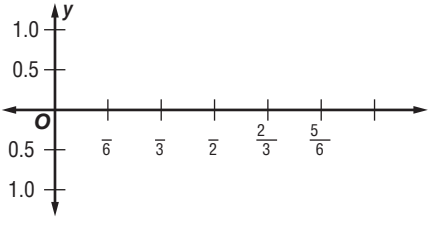
2. $y = \tan \left(\theta - \frac{\pi}{2}\right)$



3. $y = 3 \cos (\theta - 45°)$



4. $y = \frac{1}{2} \sin 3\left(\theta - \frac{\pi}{3}\right)$



Lesson 14-2

14-2 Study Guide and Intervention *(continued)*

Translations of Trigonometric Graphs

Vertical Translations When a constant is added to a trigonometric function, the graph is shifted vertically.

Vertical Shift	The vertical shift of the graphs of the functions $y = a \sin b(\theta - h) + k$, $y = a \cos b(\theta - h) + k$, and $y = a \tan b(\theta - h) + k$ is k. If $k > 0$, the shift is up. If $k < 0$, the shift is down.

The **midline** of a vertical shift is $y = k$.

Graphing Trigonometric Functions	**Step 1** Determine the vertical shift, and graph the midline. **Step 2** Determine the amplitude, if it exists. Use dashed lines to indicate the maximum and minimum values of the function. **Step 3** Determine the period of the function and graph the appropriate function. **Step 4** Determine the phase shift and translate the graph accordingly.

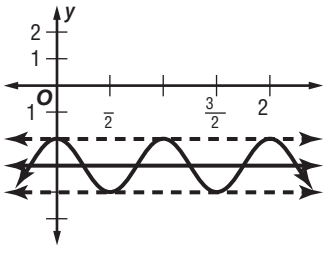
Example **State the vertical shift, equation of the midline, amplitude, and period for $y = \cos 2\theta - 3$. Then graph the function.**

Vertical Shift: $k = -3$, so the vertical shift is 3 units down.

The equation of the midline is $y = -3$.

Amplitude: $|a| = |1|$ or 1

Period: $\frac{2\pi}{|b|} = \frac{2\pi}{|2|}$ or π

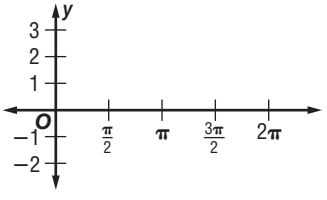


Since the amplitude of the function is 1, draw dashed lines parallel to the midline that are 1 unit above and below the midline. Then draw the cosine curve, adjusted to have a period of π.

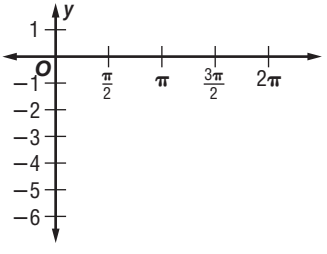
Exercises

State the vertical shift, equation of the midline, amplitude, and period for each function. Then graph the function.

1. $y = \frac{1}{2} \cos \theta + 2$



2. $y = 3 \sin \theta - 2$

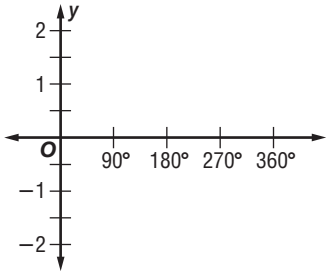


14-2 Skills Practice

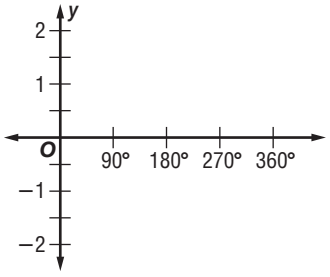
Translations of Trigonometric Graphs

State the amplitude, period, and phase shift for each function. Then graph the function.

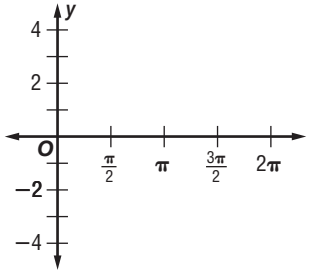
1. $y = \sin(\theta + 90°)$



2. $y = \cos(\theta - 45°)$

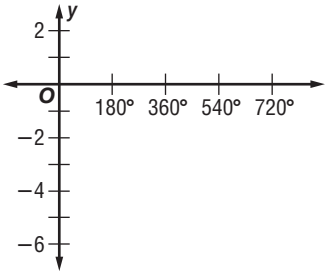


3. $y = \tan\left(\theta - \frac{\pi}{2}\right)$

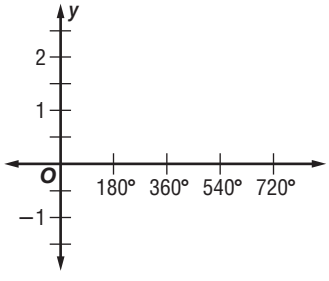


State the vertical shift, equation of the midline, amplitude, and period for each function. Then graph the function.

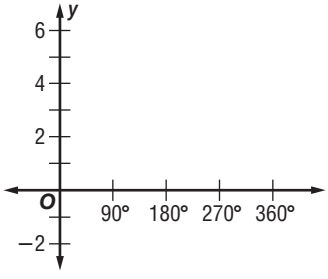
4. $y = \csc\theta - 2$



5. $y = \cos\theta + 1$

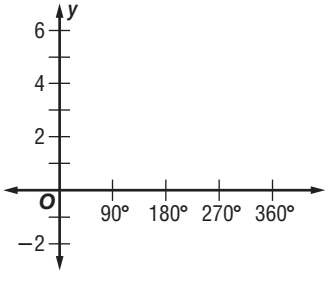


6. $y = \sec\theta + 3$

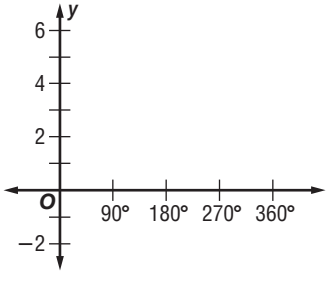


State the vertical shift, amplitude, period, and phase shift of each function. Then graph the function.

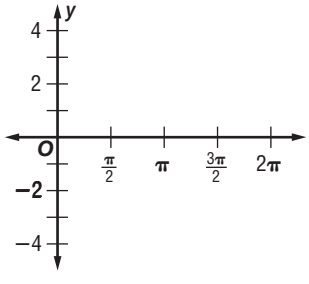
7. $y = 2\cos[3(\theta + 45°)] + 2$



8. $y = 3\sin[2(\theta - 90°)] + 2$



9. $y = 4\cot\left[\frac{4}{3}\left(\theta + \frac{\pi}{4}\right)\right] - 2$



Lesson 14-2

NAME _____ DATE _____ PERIOD _____

14-2 Practice

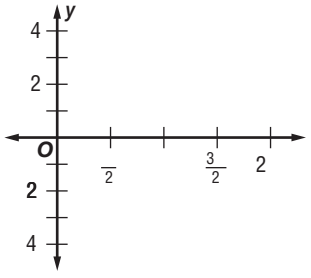
Translations of Trigonometric Graphs

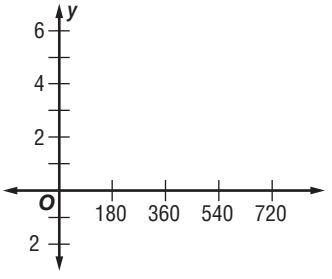
State the vertical shift, amplitude, period, and phase shift for each function. Then graph the function.

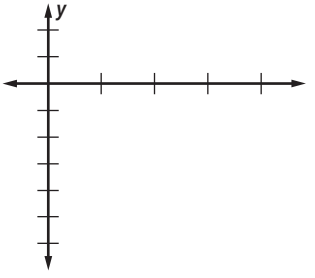
1. $y = \frac{1}{2}\tan\left(\theta - \frac{\pi}{2}\right)$

2. $y = 2\cos(\theta + 30°) + 3$

3. $y = 3\csc(2\theta + 60°) - 2.5$







ECOLOGY For Exercises 4–6, use the following information.

The population of an insect species in a stand of trees follows the growth cycle of a particular tree species. The insect population can be modeled by the function $y = 40 + 30\sin 6t$, where t is the number of years since the stand was first cut in November, 1920.

4. How often does the insect population reach its maximum level?

5. When did the population last reach its maximum?

6. What condition in the stand do you think corresponds with a minimum insect population?

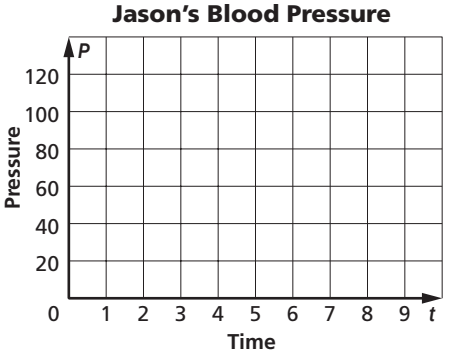
BLOOD PRESSURE For Exercises 7–9, use the following information.

Jason's blood pressure is 110 over 70, meaning that the pressure oscillates between a maximum of 110 and a minimum of 70. Jason's heart rate is 45 beats per minute. The function that represents Jason's blood pressure P can be modeled using a sine function with no phase shift.

7. Find the amplitude, midline, and period in seconds of the function.

8. Write a function that represents Jason's blood pressure P after t seconds.

9. Graph the function.



NAME _____ DATE _____ PERIOD _____

14-2 Reading to Learn Mathematics

Translations of Trigonometric Graphs

Pre-Activity **How can translations of trigonometric graphs be used to show animal populations?**

Read the introduction to Lesson 14-2 at the top of page 769 in your textbook.

According to the model given in your textbook, what would be the estimated rabbit population for January 1, 2005?

Reading the Lesson

1. Determine whether the graph of each function represents a shift of the parent function *to the left*, *to the right*, *upward*, or *downward*. (Do not actually graph the functions.)

a. $y = \sin(\theta + 90°)$

b. $y = \sin\theta + 3$

c. $y = \cos\left(\theta - \frac{\pi}{3}\right)$

d. $y = \tan\theta - 4$

2. Determine whether the graph of each function has an *amplitude change*, *period change*, *phase shift*, or *vertical shift* compared to the graph of the parent function. (More than one of these may apply to each function. Do not actually graph the functions.)

a. $y = 3\sin\left(\theta + \frac{5\pi}{6}\right)$

b. $y = \cos(2\theta + 70°)$

c. $y = -4\cos 3\theta$

d. $y = \sec\frac{1}{2}\theta + 3$

e. $y = \tan\left(\theta - \frac{\pi}{4}\right) - 1$

f. $y = 2\sin\left(\frac{1}{3}\theta + \frac{\pi}{6}\right) - 4$

Helping You Remember

3. Many students have trouble remembering which of the functions $y = \sin(\theta + \alpha)$ and $y = \sin(\theta - \alpha)$ represents a shift to the left and which represents a shift to the right. Using $\alpha = 45°$, explain a good way to remember which is which.

Lesson 14-2

NAME _____ DATE _____ PERIOD _____

14-2 Enrichment

Translating Graphs of Trigonometric Functions

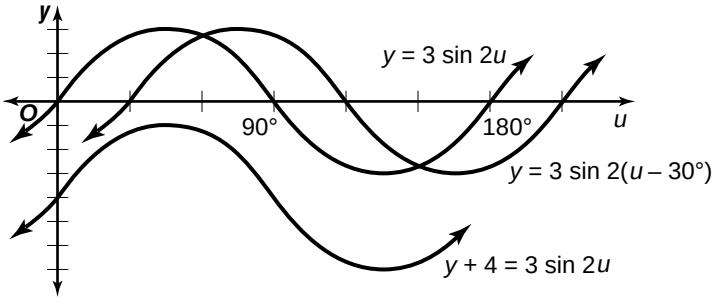
Three graphs are shown at the right:

$y = 3 \sin 2\theta$

$y = 3 \sin 2(\theta - 30°)$

$y + 4 = 3 \sin 2\theta$

Replacing θ with $(\theta - 30°)$ translates the graph to the right. Replacing y with $y + 4$ translates the graph 4 units down.



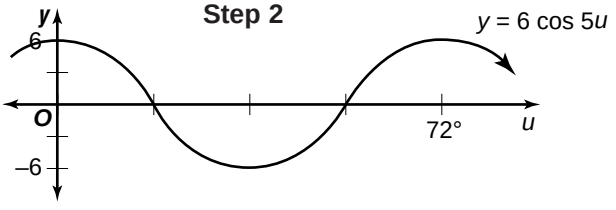
Example **Graph one cycle of $y = 6 \cos (5\theta + 80°) + 2$.**

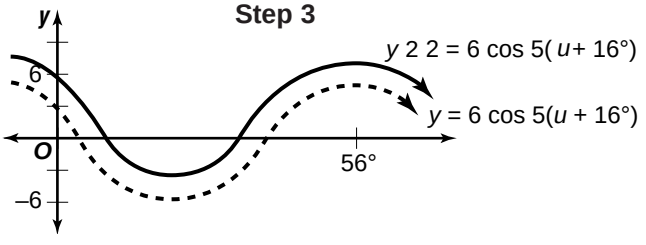
Step 1 Transform the equation into the form $y - k = a \cos b(\theta - h)$.

$y - 2 = 6 \cos 5(\theta + 16°)$

Step 2 Sketch $y = 6 \cos 5\theta$.

Step 3 Translate $y = 6 \cos 5\theta$ to obtain the desired graph.





Sketch these graphs on the same coordinate system.

1. $y = 3 \sin 2(\theta + 45°)$ **2.** $y - 1 = 3 \sin 2\theta$ **3.** $y + 5 = 3 \sin 2(\theta + 90°)$

On another piece of paper, graph one cycle of each curve.

4. $y = 2 \sin 4(\theta - 50°)$ **5.** $y = 5 \sin (3\theta + 90°)$

6. $y = 6 \cos (4\theta + 360°) + 3$ **7.** $y = 6 \cos 4\theta + 3$

8. The graphs for problems 6 and 7 should be the same. Use the sum formula for cosine of a sum to show that the equations are equivalent.

14-3 Study Guide and Intervention

Trigonometric Identities

Find Trigonometric Values A **trigonometric identity** is an equation involving trigonometric functions that is true for all values for which every expression in the equation is defined.

Basic Trigonometric Identities	**Quotient Identities**	$\tan\theta = \frac{\sin\theta}{\cos\theta}$	$\cot\theta = \frac{\cos\theta}{\sin\theta}$	
	Reciprocal Identities	$\csc\theta = \frac{1}{\sin\theta}$	$\sec\theta = \frac{1}{\cos\theta}$	$\cot\theta = \frac{1}{\tan\theta}$
	Pythagorean Identities	$\cos^2\theta + \sin^2\theta = 1$	$\tan^2\theta + 1 = \sec^2\theta$	$\cot^2\theta + 1 = \csc^2\theta$

Example **Find the value of $\cot\theta$ if $\csc\theta = -\frac{11}{5}$; $180° < \theta < 270°$.**

$\cot^2\theta + 1 = \csc^2\theta$ Trigonometric identity

$\cot^2\theta + 1 = \left(-\frac{11}{5}\right)^2$ Substitute $-\frac{11}{5}$ for $\csc\theta$.

$\cot^2\theta + 1 = \frac{121}{25}$ Square $-\frac{11}{5}$.

$\cot^2\theta = \frac{96}{25}$ Subtract 1 from each side.

$\cot\theta = \pm\frac{4\sqrt{6}}{5}$ Take the square root of each side.

Since θ is in the third quadrant, $\cot\theta$ is positive, Thus $\cot\theta = \frac{4\sqrt{6}}{5}$.

Exercises

Find the value of each expression.

1. $\tan\theta$, if $\cot\theta = 4$; $180° < \theta < 270°$

2. $\csc\theta$, if $\cos\theta = \frac{\sqrt{3}}{2}$; $0° \le \theta < 90°$

3. $\cos\theta$, if $\sin\theta = \frac{3}{5}$; $0° \le \theta < 90°$

4. $\sec\theta$, if $\sin\theta = \frac{1}{3}$; $0° \le \theta < 90°$

5. $\cos\theta$, if $\tan\theta = -\frac{4}{3}$; $90° < \theta < 180°$

6. $\tan\theta$, if $\sin\theta = \frac{3}{7}$; $0° \le \theta < 90°$

7. $\sec\theta$, if $\cos\theta = -\frac{7}{8}$; $90° < \theta < 180°$

8. $\sin\theta$, if $\cos\theta = \frac{6}{7}$; $270° \le \theta < 360°$

9. $\cot\theta$, if $\csc\theta = \frac{12}{5}$; $90° < \theta < 180°$

10. $\sin\theta$, if $\csc\theta = -\frac{9}{4}$; $270° < \theta < 360°$

Lesson 14-3

14-3 Study Guide and Intervention *(continued)*

Trigonometric Identities

Simplify Expressions The simplified form of a trigonometric expression is written as a numerical value or in terms of a single trigonometric function, if possible. Any of the trigonometric identities on page 849 can be used to simplify expressions containing trigonometric functions.

Example 1 **Simplify $(1 - \cos^2 \theta) \sec \theta \cot \theta + \tan \theta \sec \theta \cos^2 \theta$.**

$$\begin{aligned}(1 - \cos^2 \theta) \sec \theta \cot \theta + \tan \theta \sec \theta \cos^2 \theta &= \sin^2 \theta \cdot \frac{1}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\cos \theta} \cdot \cos^2 \theta \\ &= \sin \theta + \sin \theta \\ &= 2 \sin \theta\end{aligned}$$

Example 2 **Simplify $\dfrac{\sec \theta \cdot \cot \theta}{1 - \sin \theta} - \dfrac{\csc \theta}{1 + \sin \theta}$.**

$$\begin{aligned}\frac{\sec \theta \cdot \cot \theta}{1 - \sin \theta} - \frac{\csc \theta}{1 + \sin \theta} &= \frac{\frac{1}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta}}{1 - \sin \theta} - \frac{\frac{1}{\sin \theta}}{1 + \sin \theta} \\ &= \frac{\frac{1}{\sin \theta}(1 + \sin \theta) - \frac{1}{\sin \theta}(1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)} \\ &= \frac{\frac{1}{\sin \theta} + 1 - \frac{1}{\sin \theta} + 1}{1 - \sin^2 \theta} \\ &= \frac{2}{\cos^2 \theta}\end{aligned}$$

Exercises

Simplify each expression.

1. $\dfrac{\tan \theta \cdot \csc \theta}{\sec \theta}$

2. $\dfrac{\sin \theta \cdot \cot \theta}{\sec^2 \theta - \tan^2 \theta}$

3. $\dfrac{\sin^2 \theta - \cot \theta \cdot \tan \theta}{\cot \theta \cdot \sin \theta}$

4. $\dfrac{\cos \theta}{\sec \theta - \tan \theta}$

5. $\dfrac{\tan \theta \cdot \cos \theta}{\sin \theta} + \cot \theta \cdot \sin \theta \cdot \tan \theta \cdot \csc \theta$

6. $\dfrac{\csc^2 \theta - \cot^2 \theta}{\tan \theta \cdot \cos \theta}$

7. $3 \tan \theta \cdot \cot \theta + 4 \sin \theta \cdot \csc \theta + 2 \cos \theta \cdot \sec \theta$

8. $\dfrac{1 - \cos^2 \theta}{\tan \theta \cdot \sin \theta}$

NAME _____ DATE _____ PERIOD _____

14-3 Skills Practice

Trigonometric Identities

Find the value of each expression.

1. $\sin\theta$, if $\cos\theta = -\frac{4}{5}$ and $90° < \theta < 180°$

2. $\cos\theta$, if $\tan\theta = 1$ and $180° < \theta < 270°$

3. $\sec\theta$, if $\tan\theta = 1$ and $0° \le \theta < 90°$

4. $\cos\theta$, if $\tan\theta = \frac{1}{2}$ and $0° \le \theta < 90°$

5. $\tan\theta$, if $\sin\theta = -\frac{\sqrt{2}}{2}$ and $180° < \theta < 270°$

6. $\cos\theta$, if $\sec\theta = 2$ and $270° < \theta < 360°$

7. $\cos\theta$, if $\csc\theta = -2$ and $180° < \theta < 270°$

8. $\tan\theta$, if $\cos\theta = -\frac{2\sqrt{5}}{5}$ and $180° < \theta < 270°$

9. $\cos\theta$, if $\cot\theta = -\frac{3}{2}$ and $90° < \theta < 180°$

10. $\csc\theta$, if $\cos\theta = \frac{8}{17}$ and $0° < \theta < 90°$

11. $\cot\theta$, if $\csc\theta = -2$ and $180° < \theta < 270°$

12. $\tan\theta$, if $\sin\theta = -\frac{5}{13}$ and $180° < \theta < 270°$

Simplify each expression.

13. $\sin\theta\sec\theta$

14. $\csc\theta\sin\theta$

15. $\cot\theta\sec\theta$

16. $\frac{\cos\theta}{\sec\theta}$

17. $\tan\theta + \cot\theta$

18. $\csc\theta\tan\theta - \tan\theta\sin\theta$

19. $\frac{1 - \sin^2\theta}{\sin\theta + 1}$

20. $\csc\theta + \cot\theta$

21. $\frac{\sin^2\theta + \cos^2\theta}{1 - \cos^2\theta}$

22. $1 + \frac{\tan^2\theta}{1 + \sec\theta}$

Lesson 14-3

NAME _____ DATE _____ PERIOD _____

14-3 Practice

Trigonometric Identities

Find the value of each expression.

1. $\sin\theta$, if $\cos\theta = \frac{5}{13}$ and $0° \le \theta < 90°$

2. $\sec\theta$, if $\sin\theta = -\frac{15}{17}$ and $180° < \theta < 270°$

3. $\cot\theta$, if $\cos\theta = \frac{3}{10}$ and $270° < \theta < 360°$

4. $\sin\theta$, if $\cot\theta = \frac{1}{2}$ and $0° \le \theta < 90°$

5. $\cot\theta$, if $\csc\theta = -\frac{3}{2}$ and $180° < \theta < 270°$

6. $\sec\theta$, if $\csc\theta = -8$ and $270° < \theta < 360°$

7. $\sec\theta$, if $\tan\theta = 4$ and $180° < \theta < 270°$

8. $\sin\theta$, if $\tan\theta = -\frac{1}{2}$ and $270° < \theta < 360°$

9. $\cot\theta$, if $\tan\theta = \frac{2}{5}$ and $0° \le \theta < 90°$

10. $\cot\theta$, if $\cos\theta = \frac{1}{3}$ and $270° < \theta < 360°$

Simplify each expression.

11. $\csc\theta\tan\theta$

12. $\frac{\sin^2\theta}{\tan^2\theta}$

13. $\sin^2\theta\cot^2\theta$

14. $\cot^2\theta + 1$

15. $\frac{\csc^2\theta - \cot^2\theta}{1 - \cos^2\theta}$

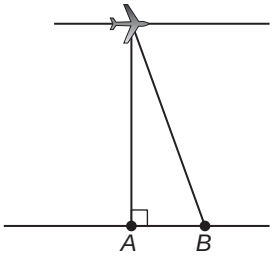
16. $\frac{\csc\theta - \sin\theta}{\cos\theta}$

17. $\sin\theta + \cos\theta\cot\theta$

18. $\frac{\cos\theta}{1 - \sin\theta} - \frac{\cos\theta}{1 + \sin\theta}$

19. $\sec^2\theta\cos^2\theta - \tan^2\theta$

20. AERIAL PHOTOGRAPHY The illustration shows a plane taking an aerial photograph of point A. Because the point is directly below the plane, there is no distortion in the image. For any point B not directly below the plane, however, the increase in distance creates distortion in the photograph. This is because as the distance from the camera to the point being photographed increases, the exposure of the film reduces by $(\sin\theta)(\csc\theta - \sin\theta)$. Express $(\sin\theta)(\csc\theta - \sin\theta)$ in terms of $\cos\theta$ only.



21. TSUNAMIS The equation $y = a\sin\theta t$ represents the height of the waves passing a buoy at a time t in seconds. Express a in terms of $\csc\theta t$.

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14-3 Reading to Learn Mathematics

Trigonometric Identities

Pre-Activity **How can trigonometry be used to model the path of a baseball?**

Read the introduction to Lesson 14-3 at the top of page 777 in your textbook.

Suppose that a baseball is hit from home plate with an initial velocity of 58 feet per second at an angle of 36° with the horizontal from an initial height of 5 feet. Show the equation that you would use to find the height of the ball 10 seconds after the ball is hit. (Show the formula with the appropriate numbers substituted, but do not do any calculations.)

Reading the Lesson

1. Match each expression from the list on the left with an expression from the list on the right that is equal to it for all values for which each expression is defined. (Some of the expressions from the list on the right may be used more than once or not at all.)

a. $\sec^2 \theta - \tan^2 \theta$	**i.** $\frac{1}{\sin \theta}$
b. $\cot^2 \theta + 1$	**ii.** $\tan \theta$
c. $\frac{\sin \theta}{\cos \theta}$	**iii.** 1
d. $\sin^2 \theta + \cos^2 \theta$	**iv.** $\sec \theta$
e. $\csc \theta$	**v.** $\csc^2 \theta$
f. $\frac{1}{\cos \theta}$	**vi.** $\cot \theta$
g. $\frac{\cos \theta}{\sin \theta}$	

2. Write an identity that you could use to find each of the indicated trigonometric values and tell whether that value is positive or negative. (Do not actually find the values.)

a. $\tan \theta$, if $\sin \theta = -\frac{4}{5}$ and $180° < \theta < 270°$

b. $\sec \theta$, if $\tan \theta = -3$ and $90° < \theta < 180°$

Helping You Remember

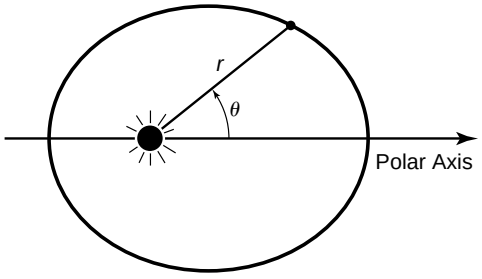
3. A good way to remember something new is to relate it to something you already know. How can you use the unit circle definitions of the sine and cosine that you learned in Chapter 13 to help you remember the Pythagorean identity $\cos^2 \theta + \sin^2 \theta = 1$?

Lesson 14-3

14-3 Enrichment

Planetary Orbits

The orbit of a planet around the sun is an ellipse with the sun at one focus. Let the pole of a polar coordinate system be that focus and the polar axis be toward the other focus. The polar equation of an ellipse is



$r = \dfrac{2ep}{1 - e\cos\theta}$. Since $2p = \dfrac{b^2}{c}$ and $b^2 = a^2 - c^2$,

$2p = \dfrac{a^2 - c^2}{c} = \dfrac{a^2}{c}\left(1 - \dfrac{c^2}{a^2}\right)$. Because $e = \dfrac{c}{a}$,

$2p = a\left(\dfrac{a}{c}\right)\left(1 - \left(\dfrac{c}{a}\right)^2\right) = a\left(\dfrac{1}{e}\right)(1 - e^2)$.

Therefore $2ep = a(1 - e^2)$. Substituting into the polar equation of an ellipse yields an equation that is useful for finding distances from the planet to the sun.

$$r = \frac{a(1 - e^2)}{1 - e\cos\theta}$$

Note that e is the eccentricity of the orbit and a is the length of the semi-major axis of the ellipse. Also, a is the mean distance of the planet from the sun.

Example **The mean distance of Venus from the sun is 67.24×10^6 miles and the eccentricity of its orbit is .006788. Find the minimum and maximum distances of Venus from the sun.**

The minimum distance occurs when $\theta = \pi$.

$$r = \frac{67.24 \times 10^6(1 - 0.006788^2)}{1 - 0.006788\cos\pi} = 66.78 \times 10^6 \text{ miles}$$

The maximum distance occurs when $\theta = 0$.

$$r = \frac{67.24 \times 10^6(1 - 0.006788^2)}{1 - 0.006788\cos 0} = 67.70 \times 10^6 \text{ miles}$$

Complete each of the following.

1. The mean distance of Mars from the sun is 141.64×10^6 miles and the eccentricity of its orbit is 0.093382. Find the minimum and maximum distances of Mars from the sun.

2. The minimum distance of Earth from the sun is 91.445×10^6 miles and the eccentricity of its orbit is 0.016734. Find the mean and maximum distances of Earth from the sun.

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14-4 Study Guide and Intervention

Verifying Trigonometric Identities

Transform One Side of an Equation Use the basic trigonometric identities along with the definitions of the trigonometric functions to verify trigonometric identities. Often it is easier to begin with the more complicated side of the equation and transform that expression into the form of the simpler side.

Example **Verify that each of the following is an identity.**

a. $\dfrac{\sin\theta}{\cot\theta} - \sec\theta = -\cos\theta$

Transform the left side.

$$\frac{\sin\theta}{\cot\theta} - \sec\theta \stackrel{?}{=} -\cos\theta$$

$$\frac{\sin\theta}{\frac{\cos\theta}{\sin\theta}} - \frac{1}{\cos\theta} \stackrel{?}{=} -\cos\theta$$

$$\frac{\sin^2\theta}{\cos\theta} - \frac{1}{\cos\theta} \stackrel{?}{=} -\cos\theta$$

$$\frac{\sin^2 - 1}{\cos\theta} \stackrel{?}{=} -\cos\theta$$

$$\frac{-\cos^2\theta}{\cos\theta} \stackrel{?}{=} -\cos\theta$$

$$-\cos\theta = -\cos\theta$$

b. $\dfrac{\tan\theta}{\csc\theta} + \cos\theta = \sec\theta$

Transform the left side.

$$\frac{\tan\theta}{\csc\theta} + \cos\theta \stackrel{?}{=} \sec\theta$$

$$\frac{\frac{\sin\theta}{\cos\theta}}{\frac{1}{\sin\theta}} + \cos\theta \stackrel{?}{=} \sec\theta$$

$$\frac{\sin^2\theta}{\cos\theta} + \cos\theta \stackrel{?}{=} \sec\theta$$

$$\frac{\sin^2\theta + \cos^2\theta}{\cos\theta} \stackrel{?}{=} \sec\theta$$

$$\frac{1}{\cos\theta} \stackrel{?}{=} \sec\theta$$

$$\sec\theta = \sec\theta$$

Exercises

Verify that each of the following is an identity.

1. $1 + \csc^2\theta \cdot \cos^2\theta = \csc^2\theta$

2. $\dfrac{\sin\theta}{1 - \cos\theta} - \dfrac{\cot\theta}{1 + \cos\theta} = \dfrac{1 - \cos^3\theta}{\sin^3\theta}$

Lesson 14-4

14-4 Study Guide and Intervention *(continued)*

Verifying Trigonometric Identities

Transform Both Sides of an Equation The following techniques can be helpful in verifying trigonometric identities.

- Substitute one or more basic identities to simplify an expression.
- Factor or multiply to simplify an expression.
- Multiply both numerator and denominator by the same trigonometric expression.
- Write each side of the identity in terms of sine and cosine only. Then simplify each side.

Example **Verify that $\dfrac{\tan^2\theta + 1}{\sin\theta \cdot \tan\theta \cdot \sec\theta + 1} = \sec^2\theta - \tan^2\theta$ is an identity.**

$$\frac{\tan^2\theta + 1}{\sin\theta \cdot \tan\theta \cdot \sec\theta + 1} \stackrel{?}{=} \sec^2\theta - \tan^2\theta$$

$$\frac{\sec^2\theta}{\sin\theta \cdot \frac{\sin\theta}{\cos\theta} \cdot \frac{1}{\cos\theta} + 1} \stackrel{?}{=} \frac{1}{\cos^2\theta} - \frac{\sin^2\theta}{\cos^2\theta}$$

$$\frac{\frac{1}{\cos^2\theta}}{\frac{\sin^2\theta}{\cos^2\theta} + 1} \stackrel{?}{=} \frac{1 - \sin^2\theta}{\cos^2\theta}$$

$$\frac{\frac{1}{\cos^2\theta}}{\frac{\sin^2\theta + \cos^2\theta}{\cos^2\theta}} \stackrel{?}{=} \frac{\cos^2\theta}{\cos^2\theta}$$

$$\frac{1}{\sin^2\theta + \cos^2\theta} \stackrel{?}{=} 1$$

$$1 = 1$$

Exercises

Verify that each of the following is an identity.

1. $\csc\theta \cdot \sec\theta = \cot\theta + \tan\theta$

2. $\dfrac{\tan^2\theta}{1 - \cos^2\theta} = \dfrac{\sec\theta}{\cos\theta}$

3. $\dfrac{\cos\theta \cdot \cot\theta}{\sin\theta} = \dfrac{\csc\theta}{\sin\theta \cdot \sec^2\theta}$

4. $\dfrac{\csc^2\theta - \cot^2\theta}{\sec^2\theta} = \cot^2\theta(1 - \cos^2\theta)$

NAME _____ DATE _____ PERIOD _____

14-4 Skills Practice

Verifying Trigonometric Identities

Verify that each of the following is an identity.

1. $\tan \theta \cos \theta = \sin \theta$

2. $\cot \theta \tan \theta = 1$

3. $\csc \theta \cos \theta = \cot \theta$

4. $\dfrac{1 - \sin^2 \theta}{\cos \theta} = \cos \theta$

5. $(\tan \theta)(1 - \sin^2 \theta) = \sin \theta \cos \theta$

2

6. $\dfrac{\csc \theta}{\sec \theta} = \cot \theta$

7. $\dfrac{\sin^2 \theta}{1 - \sin^2 \theta} = \tan^2 \theta$

8. $\dfrac{\cos^2 \theta}{1 - \sin \theta} = 1 + \sin \theta$

Lesson 14-4

NAME _____ DATE _____ PERIOD _____

14-4 Practice

Verifying Trigonometric Identities

Verify that each of the following is an identity.

1. $\dfrac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta} = \sec^2 \theta$

2. $\dfrac{\cos^2 \theta}{1 - \sin^2 \theta} = 1$

3. $(1 + \sin \theta)(1 - \sin \theta) = \cos^2 \theta$

4. $\tan^4 \theta + 2 \tan^2 \theta + 1 = \sec^4 \theta$

5. $\cos^2 \theta \cot^2 \theta = \cot^2 \theta - \cos^2 \theta$

6. $(\sin^2 \theta)(\csc^2 \theta + \sec^2 \theta) = \sec^2 \theta$

7. PROJECTILES The square of the initial velocity of an object launched from the ground is $v^2 = \dfrac{2gh}{\sin^2 \theta}$, where θ is the angle between the ground and the initial path, h is the maximum height reached, and g is the acceleration due to gravity. Verify the identity $\dfrac{2gh}{\sin^2 \theta} = \dfrac{2gh \sec^2 \theta}{\sec^2 \theta - 1}$.

8. LIGHT The intensity of a light source measured in candles is given by $I = ER^2 \sec \theta$, where E is the illuminance in foot candles on a surface, R is the distance in feet from the light source, and θ is the angle between the light beam and a line perpendicular to the surface. Verify the identity $ER^2(1 + \tan^2 \theta) \cos \theta = ER^2 \sec \theta$.

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14-4 Reading to Learn Mathematics

Verifying Trigonometric Identities

Pre-Activity **How can you verify trigonometric identities?**

Read the introduction to Lesson 14-4 at the top of page 782 in your textbook.

For $\theta = -\pi$, 0, or π, $\sin \theta = \sin 2\theta$. Does this mean that $\sin \theta = \sin 2\theta$ is an identity? Explain your reasoning.

Reading the Lesson

1. Determine whether each equation is an *identity* or *not an identity*.

a. $\dfrac{1}{\sin^2 \theta} - \dfrac{1}{\tan^2 \theta} = 1$

b. $\dfrac{\cos \theta}{\sin \theta \tan \theta}$

c. $\dfrac{\sin \theta}{\cos \theta} + \dfrac{\cos \theta}{\sin \theta} = \cos \theta \sin \theta$

d. $\cos^2 \theta (\tan^2 \theta + 1) = 1$

e. $\dfrac{\sin^2 \theta}{\cos^2 \theta} + \sin \theta \csc \theta = \sec^2 \theta$

f. $\dfrac{1}{1 - \sin \theta} + \dfrac{1}{1 + \sin \theta} = 2 \cos^2 \theta$

g. $\tan^2 \theta \cos^2 \theta = \dfrac{1}{\csc^2 \theta}$

h. $\dfrac{\sin \theta}{\sec \theta} = \dfrac{1}{\tan \theta} + \dfrac{1}{\cot \theta}$

2. Which of the following is *not* permitted when verifying an identity?

A. simplifying one side of the identity to match the other side

B. cross multiplying if the identity is a proportion

C. simplifying each side of the identity separately to get the same expression on both sides

Helping You Remember

3. Many students have trouble knowing where to start in verifying a trigonometric identity. What is a simple rule that you can remember that you can always use if you don't see a quicker approach?

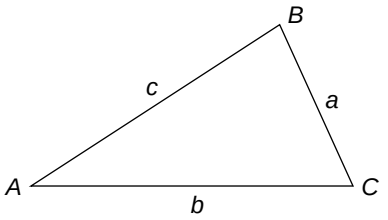
Lesson 14-4

NAME _____ DATE _____ PERIOD _____

14-4 Enrichment

Heron's Formula

Heron's formula can be used to find the area of a triangle if you know the lengths of the three sides. Consider any triangle ABC. Let K represent the area of $\triangle ABC$. Then



$$K = \frac{1}{2}bc \sin A$$

$$K^2 = \frac{b^2c^2 \sin^2 A}{4} \quad \text{Square both sides.}$$

$$= \frac{b^2c^2(1 - \cos^2 A)}{4}$$

$$= \frac{b^2c^2(1 + \cos A)(1 - \cos A)}{4}$$

$$= \frac{b^2c^2}{4}\left(1 + \frac{b^2 + c^2 - a^2}{2bc}\right)\left(1 - \frac{b^2 + c^2 - a^2}{2bc}\right) \quad \text{Use the law of cosines.}$$

$$= \frac{b + c + a}{2} \cdot \frac{b + c - a}{2} \cdot \frac{a + b - c}{2} \cdot \frac{a - b + c}{2} \quad \text{Simplify.}$$

Let $s = \frac{a + b + c}{2}$. Then $s - a = \frac{b + c - a}{2}$, $s - b = \frac{a + c - b}{2}$, $s - c = \frac{a + b - c}{2}$.

$$K^2 = s(s - a)(s - b)(s - c) \quad \text{Substitute.}$$

$$K = \sqrt{s(s - a)(s - b)(s - c)}$$

Heron's Formula	The area of $\triangle ABC$ is $\sqrt{s(s - a)(s - b)(s - c)}$, where $s = \frac{a + b + c}{2}$.

Use Heron's formula to find the area of $\triangle ABC$.

1. $a = 3, b = 4.4, c = 7$

2. $a = 8.2, b = 10.3, c = 9.5$

3. $a = 31.3, b = 92.0, c = 67.9$

4. $a = 0.54, b = 1.32, c = 0.78$

5. $a = 321, b = 178, c = 298$

6. $a = 0.05, b = 0.08, c = 0.04$

7. $a = 21.5, b = 33.0, c = 41.7$

8. $a = 2.08, b = 9.13, c = 8.99$

14-5 Study Guide and Intervention

Sum and Difference of Angles Formulas

Sum and Difference Formulas The following formulas are useful for evaluating an expression like sin 15° from the known values of sine and cosine of 60° and 45°.

Sum and Difference of Angles	The following identities hold true for all values of α and β. $\cos(\alpha \pm \beta) = \cos\alpha \cdot \cos\beta \mp \sin\alpha \cdot \sin\beta$ $\sin(\alpha \pm \beta) = \sin\alpha \cdot \cos\beta \pm \cos\alpha \cdot \sin\beta$

Example **Find the exact value of each expression.**

a. cos 345°

$$\begin{aligned}\cos 345° &= \cos(300° + 45°)\\ &= \cos 300° \cdot \cos 45° - \sin 300° \cdot \sin 45°\\ &= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{3}}{2}\right) \cdot \frac{\sqrt{2}}{2}\\ &= \frac{\sqrt{2} + \sqrt{6}}{4}\end{aligned}$$

b. sin (−105°)

$$\begin{aligned}\sin(-105°) &= \sin(45° - 150°)\\ &= \sin 45° \cdot \cos 150° - \cos 45° \cdot \sin 150°\\ &= \frac{\sqrt{2}}{2} \cdot \left(-\frac{\sqrt{3}}{2}\right) - \frac{\sqrt{2}}{2} \cdot \frac{1}{2}\\ &= -\frac{\sqrt{2} + \sqrt{6}}{4}\end{aligned}$$

Exercises

Find the exact value of each expression.

1. sin 105°

2. cos 285°

3. cos (−75°)

4. cos (−165°)

5. sin 195°

6. cos 420°

7. sin (−75°)

8. cos 135°

9. cos (−15°)

10. sin 345°

11. cos (−105°)

12. sin 495°

Lesson 14-5

14-5 Study Guide and Intervention *(continued)*

Sum and Difference of Angles Formulas

Verify Identities You can also use the sum and difference of angles formulas to verify identities.

Example 1 **Verify that $\cos\left(\theta + \frac{3\pi}{2}\right) = \sin\theta$ is an identity.**

$\cos\left(\theta + \frac{3\pi}{2}\right) \stackrel{?}{=} \sin\theta$	Original equation
$\cos\theta \cdot \cos\frac{3\pi}{2} - \sin\theta \cdot \sin\frac{3\pi}{2} \stackrel{?}{=} \sin\theta$	Sum of Angles Formula
$\cos\theta \cdot 0 - \sin\theta \cdot (-1) \stackrel{?}{=} \sin\theta$	Evaluate each expression.
$\sin\theta = \sin\theta$	Simplify.

Example 2 **Verify that $\sin\left(\theta - \frac{\pi}{2}\right) + \cos(\theta + \pi) = -2\cos\theta$ is an identity.**

$\sin\left(\theta - \frac{\pi}{2}\right) + \cos(\theta + \pi) \stackrel{?}{=} -2\cos\theta$	Original equation
$\sin\theta \cdot \cos\frac{\pi}{2} - \cos\theta \cdot \sin\frac{\pi}{2} + \cos\theta \cdot \cos\pi - \sin\theta \cdot \sin\pi \stackrel{?}{=} -2\cos\theta$	Sum and Difference of Angles Formulas
$\sin\theta \cdot 0 - \cos\theta \cdot 1 + \cos\theta \cdot (-1) - \sin\theta \cdot 0 \stackrel{?}{=} -2\cos\theta$	Evaluate each expression.
$-2\cos\theta = -2\cos\theta$	Simplify.

Exercises

Verify that each of the following is an identity.

1. $\sin(90° + \theta) = \cos\theta$

2. $\cos(270° + \theta) = \sin\theta$

3. $\sin\left(\frac{2\pi}{3} - \theta\right) + \cos\left(\theta - \frac{5\pi}{6}\right) = \sin\theta$

4. $\cos\left(\frac{3\pi}{4} + \theta\right) - \sin\left(\theta - \frac{\pi}{4}\right) = -\sqrt{2}\sin\theta$

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14-5 Skills Practice

Sum and Difference of Angles Formulas

Find the exact value of each expression.

1. $\sin 330°$

2. $\cos (-165°)$

3. $\sin (-225°)$

4. $\cos 135°$

5. $\sin (-45)°$

6. $\cos 210°$

7. $\cos (-135°)$

8. $\sin 75°$

9. $\sin (-195°)$

Verify that each of the following is an identity.

10. $\sin (90° + \theta) = \cos \theta$

11. $\sin (180° + \theta) = -\sin \theta$

12. $\cos (270° - \theta) = -\sin \theta$

13. $\cos (\theta - 90°) = \sin \theta$

14. $\sin \left(\theta - \frac{\pi}{2}\right) = -\cos \theta$

15. $\cos (\pi + \theta) = -\cos \theta$

Lesson 14-5

14-5 Practice

Sum and Difference of Angles Formulas

Find the exact value of each expression.

1. $\cos 75°$

2. $\cos 375°$

3. $\sin (-165°)$

4. $\sin (-105°)$

5. $\sin 150°$

6. $\cos 240°$

7. $\sin 225°$

8. $\sin (-75°)$

9. $\sin 195°$

Verify that each of the following is an identity.

10. $\cos (180° - \theta) = -\cos \theta$

11. $\sin (360° + \theta) = \sin \theta$

12. $\sin (45° + \theta) - \sin (45° - \theta) = \sqrt{2} \sin \theta$

13. $\cos \left(x - \frac{\pi}{6}\right) + \sin \left(x - \frac{\pi}{3}\right) = \sin x$

14. SOLAR ENERGY On March 21, the maximum amount of solar energy that falls on a square foot of ground at a certain location is given by $E \sin (90° - \phi)$, where ϕ is the latitude of the location and E is a constant. Use the difference of angles formula to find the amount of solar energy, in terms of $\cos \phi$, for a location that has a latitude of ϕ.

ELECTRICITY In Exercises 15 and 16, use the following information.

In a certain circuit carrying alternating current, the formula $i = 2 \sin (120t)$ can be used to find the current i in amperes after t seconds.

15. Rewrite the formula using the sum of two angles.

16. Use the sum of angles formula to find the exact current at $t = 1$ second.

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14-5 Reading to Learn Mathematics

Sum and Difference of Angles Formulas

Pre-Activity **How are the sum and difference formulas used to describe communication interference?**

Read the introduction to Lesson 14-5 at the top of page 786 in your textbook.

Consider the functions $y = \sin x$ and $y = 2 \sin x$. Do the graphs of these two functions have *constructive* interference or *destructive* interference?

Reading the Lesson

1. Match each expression from the list on the left with an expression from the list on the right that is equal to it for all values of the variables. (Some of the expressions from the list on the right may be used more than once or not at all.)

a. $\sin(\alpha - \beta)$	**i.** $\sin \beta$
b. $\cos(\alpha + \beta)$	**ii.** $\sin \alpha \cos \beta + \cos \alpha \sin \beta$
c. $\sin(180° + \beta)$	**iii.** $-\cos \beta$
d. $\sin(180° - \beta)$	**iv.** $\cos \alpha \cos \beta + \sin \alpha \sin \beta$
e. $\cos(180° + \beta)$	**v.** $\sin \alpha \cos \beta - \cos \alpha \sin \beta$
f. $\sin(\alpha + \beta)$	**vi.** $\cos \alpha \cos \beta - \sin \alpha \sin \beta$
g. $\cos(90° - \beta)$	**vii.** $-\sin \beta$
h. $\cos(\alpha - \beta)$	**viii.** $\cos \beta$

2. Which expressions are equal to $\sin 15°$? (There may be more than one correct choice.)

A. $\sin 45° \cos 30° + \cos 45° \sin 30°$

B. $\sin 45° \cos 30° - \cos 45° \sin 30°$

C. $\sin 60° \cos 45° - \cos 60° \sin 45°$

D. $\cos 60° \cos 45° - \sin 60° \sin 45°$

Helping You Remember

3. Some students have trouble remembering which signs to use on the right-hand sides of the sum and difference of angle formulas. What is an easy way to remember this?

Lesson 14-5

14-5 Enrichment

Identities for the Products of Sines and Cosines

By adding the identities for the sines of the sum and difference of the measures of two angles, a new identity is obtained.

$$\begin{aligned} \sin(\alpha + \beta) &= \sin\alpha\cos\beta + \cos\alpha\sin\beta \\ \sin(\alpha - \beta) &= \sin\alpha\cos\beta - \cos\alpha\sin\beta \\ \hline (i)\quad \sin(\alpha + \beta) + \sin(\alpha - \beta) &= 2\sin\alpha\cos\beta \end{aligned}$$

This new identity is useful for expressing certain products as sums.

Example **Write $\sin 3\theta \cos\theta$ as a sum.**

In the identity let $\alpha = 3\theta$ and $\beta = \theta$ so that
$2\sin 3\theta\cos\theta = \sin(3\theta + \theta) + \sin(3\theta - \theta)$. Thus,
$\sin 3\theta\cos\theta = \frac{1}{2}\sin 4\theta + \frac{1}{2}\sin 2\theta$.

By subtracting the identities for $\sin(\alpha + \beta)$ and $\sin(\alpha - \beta)$, a similar identity for expressing a product as a difference is obtained.

(ii) $\sin(\alpha + \beta) - \sin(\alpha - \beta) = 2\cos\alpha\sin\beta$

Solve.

1. Use the identities for $\cos(\alpha + \beta)$ and $\cos(\alpha - \beta)$ to find identities for expressing the products $2\cos\alpha\cos\beta$ and $2\sin\alpha\sin\beta$ as a sum or difference.

2. Find the value of $\sin 105° \cos 75°$ without using tables.

3. Express $\cos\theta\sin\frac{\theta}{2}$ as a difference.

14-6 Study Guide and Intervention

Double-Angle and Half-Angle Formulas

Double-Angle Formulas

Double-Angle Formulas	The following identities hold true for all values of θ. $\sin 2\theta = 2 \sin \theta \cdot \cos \theta$ $\quad$ $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ $\cos 2\theta = 1 - 2 \sin^2 \theta$ $\cos 2\theta = 2 \cos^2 \theta - 1$

Example **Find the exact values of $\sin 2\theta$ and $\cos 2\theta$ if $\sin \theta = -\frac{9}{10}$ and $180° < \theta < 270°$.**

First, find the value of $\cos \theta$.

$\cos^2 \theta = 1 - \sin^2 \theta$ $\quad$ $\cos^2 \theta + \sin^2 \theta = 1$

$\cos^2 \theta = 1 - \left(-\frac{9}{10}\right)^2$ $\quad$ $\sin \theta = -\frac{9}{10}$

$\cos^2 \theta = \frac{19}{100}$

$\cos \theta = \pm\frac{\sqrt{19}}{10}$

Since θ is in the third quadrant, $\cos \theta$ is negative. Thus $\cos \theta = -\frac{\sqrt{19}}{10}$.

To find $\sin 2\theta$, use the identity $\sin 2\theta = 2 \sin \theta \cdot \cos \theta$.

$$\begin{aligned}\sin 2\theta &= 2 \sin \theta \cdot \cos \theta \\ &= 2\left(-\frac{9}{10}\right)\left(-\frac{\sqrt{19}}{10}\right) \\ &= \frac{9\sqrt{19}}{50}\end{aligned}$$

The value of $\sin 2\theta$ is $\frac{9\sqrt{19}}{50}$.

To find $\cos 2\theta$, use the identity $\cos 2\theta = 1 - 2 \sin^2 \theta$.

$$\begin{aligned}\cos 2\theta &= 1 - 2 \sin^2 \theta \\ &= 1 - 2\left(-\frac{9}{10}\right)^2 \\ &= -\frac{31}{50}.\end{aligned}$$

The value of $\cos 2\theta$ is $-\frac{31}{50}$.

Exercises

Find the exact values of $\sin 2\theta$ and $\cos 2\theta$ for each of the following.

1. $\sin \theta = \frac{1}{4}$, $0° < \theta < 90°$

2. $\sin \theta = -\frac{1}{8}$, $270° < \theta < 360°$

3. $\cos \theta = -\frac{3}{5}$, $180° < \theta < 270°$

4. $\cos \theta = -\frac{4}{5}$, $90° < \theta < 180°$

5. $\sin \theta = -\frac{3}{5}$, $270° < \theta < 360°$

6. $\cos \theta = -\frac{2}{3}$, $90° < \theta < 180°$

14-6 Study Guide and Intervention *(continued)*

Double-Angle and Half-Angle Formulas

Half-Angle Formulas

Half-Angle Formulas	The following identities hold true for all values of α. $\sin \frac{\alpha}{2} = \pm\sqrt{\frac{1 - \cos \alpha}{2}}$ $\cos \frac{\alpha}{2} = \pm\sqrt{\frac{1 + \cos \alpha}{2}}$

Example **Find the exact value of $\sin \frac{\alpha}{2}$ if $\sin \alpha = \frac{2}{3}$ and $90° < \alpha < 180°$.**

First find $\cos \alpha$.

$\cos^2 \alpha = 1 - \sin^2 \alpha$ $\quad$ $\cos^2 \alpha + \sin^2 \alpha = 1$

$\cos^2 \alpha = 1 - \left(\frac{2}{3}\right)^2$ $\quad$ $\sin \alpha = \frac{2}{3}$

$\cos^2 \alpha = \frac{5}{9}$ $\quad$ Simplify.

$\cos \alpha = \pm\frac{\sqrt{5}}{3}$ $\quad$ Take the square root of each side.

Since α is in the second quadrant, $\cos \alpha = -\frac{\sqrt{5}}{3}$.

$\sin \frac{\alpha}{2} = \pm\sqrt{\frac{1 - \cos \alpha}{2}}$ $\quad$ Half-Angle formula

$= \pm\sqrt{\frac{1 - \left(-\frac{\sqrt{5}}{3}\right)}{2}}$ $\quad$ $\cos \alpha = -\frac{\sqrt{5}}{3}$

$= \pm\sqrt{\frac{3 + \sqrt{5}}{6}}$ $\quad$ Simplify.

$= \pm\frac{\sqrt{18 + 6\sqrt{5}}}{6}$ $\quad$ Rationalize.

Since α is between 90° and 180°, $\frac{\alpha}{2}$ is between 45° and 90°. Thus $\sin \frac{\alpha}{2}$ is positive and equals $\frac{\sqrt{18 + 6\sqrt{5}}}{6}$.

Exercises

Find the exact value of $\sin \frac{\alpha}{2}$ and $\cos \frac{\alpha}{2}$ for each of the following.

1. $\cos \alpha = -\frac{3}{5}$, $180° < \alpha < 270°$

2. $\cos \alpha = -\frac{4}{5}$, $90° < \alpha < 180°$

3. $\sin \alpha = -\frac{3}{5}$, $270° < \alpha < 360°$

4. $\cos \alpha = -\frac{2}{3}$, $90° < \alpha < 180°$

Find the exact value of each expression by using the half-angle formulas.

5. $\cos 22\frac{1}{2}°$

6. $\sin 67.5°$

7. $\cos \frac{7\pi}{8}$

14-6 Skills Practice

Double-Angle and Half-Angle Formulas

Find the exact values of $\sin 2\theta$, $\cos 2\theta$, $\sin \frac{\theta}{2}$, and $\cos \frac{\theta}{2}$ for each of the following.

1. $\cos \theta = \frac{7}{25}$, $0° < \theta < 90°$

2. $\sin \theta = -\frac{4}{5}$, $180° < \theta < 270°$

3. $\sin \theta = \frac{40}{41}$, $90° < \theta < 180°$

4. $\cos \theta = \frac{3}{7}$, $270° < \theta < 360°$

5. $\cos \theta = -\frac{3}{5}$, $90° < \theta < 180°$

6. $\sin \theta = \frac{5}{13}$, $0° < \theta < 90°$

Find the exact value of each expression by using the half-angle formulas.

7. $\cos 22\frac{1}{2}°$

8. $\sin 165°$

9. $\cos 105°$

10. $\sin \frac{\pi}{8}$

11. $\sin \frac{15\pi}{8}$

12. $\cos 75°$

Verify that each of the following is an identity.

13. $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$

14. $\tan \theta + \cot \theta = 2 \csc 2\theta$

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14-6 Practice

Double-Angle and Half-Angle Formulas

Find the exact values of $\sin 2\theta$, $\cos 2\theta$, $\sin \frac{\theta}{2}$, and $\cos \frac{\theta}{2}$ for each of the following.

1. $\cos \theta = \frac{5}{13}$, $0° < \theta < 90°$

2. $\sin \theta = \frac{8}{17}$, $90° < \theta < 180°$

3. $\cos \theta = \frac{1}{4}$, $270° < \theta < 360°$

4. $\sin \theta = -\frac{2}{3}$, $180° < \theta < 270°$

Find the exact value of each expression by using the half-angle formulas.

5. $\tan 105°$

6. $\tan 15°$

7. $\cos 67.5°$

8. $\sin \left(-\frac{\pi}{8}\right)$

Verify that each of the following is an identity.

9. $\sin^2 \frac{\theta}{2} = \frac{\tan \theta - \sin \theta}{2 \tan \theta}$

10. $\sin 4\theta = 4 \cos 2\theta \sin \theta \cos \theta$

11. AERIAL PHOTOGRAPHY In aerial photography, there is a reduction in film exposure for any point X not directly below the camera. The reduction E_θ is given by $E_\theta = E_0 \cos^4 \theta$, where θ is the angle between the perpendicular line from the camera to the ground and the line from the camera to point X, and E_0 is the exposure for the point directly below the camera. Using the identity $2 \sin^2 \theta = 1 - \cos 2\theta$, verify that $E_0 \cos^4 \theta = E_0\left(\frac{1}{2} + \frac{\cos 2\theta}{2}\right)^2$.

12. IMAGING A scanner takes thermal images from altitudes of 300 to 12,000 meters. The width W of the swath covered by the image is given by $W = 2H' \tan \theta$, where H' is the height and θ is half the scanner's field of view. Verify that $\frac{2H' \sin 2\theta}{1 + \cos 2\theta} = 2H' \tan \theta$.

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14-6 Reading to Learn Mathematics

Double-Angle and Half-Angle Formulas

Lesson 14-6

Pre-Activity **How can trigonometric functions be used to describe music?**

Read the introduction to Lesson 14-6 at the top of page 791 in your textbook.

Suppose that the equation for the second harmonic is $y = \sin a\theta$. Then what would be the equations for the fundamental tone (first harmonic), third harmonic, fourth harmonic, and fifth harmonic?

Reading the Lesson

1. Match each expression from the list on the left with *all* expressions from the list on the right that are equal to it for all values of β.

a. $\sin \frac{\beta}{2}$	**i.** $2 \sin \beta \cos \beta$
b. $\cos 2\beta$	**ii.** $1 - 2 \sin^2 \beta$
c. $\cos \frac{\beta}{2}$	**iii.** $\cos^2 \beta - \sin^2 \beta$
d. $\sin 2\beta$	**iv.** $\pm\sqrt{\frac{1 + \cos \beta}{2}}$
	v. $\pm\sqrt{\frac{1 - \cos \beta}{2}}$

2. Determine whether you would use the *positive* or *negative* square root in the half-angle identities for $\sin \frac{\alpha}{2}$ and $\cos \frac{\alpha}{2}$ in each of the following situations. (Do not actually calculate $\sin \frac{\alpha}{2}$ and $\cos \frac{\alpha}{2}$.)

a. $\sin \frac{\alpha}{2}$, if $\cos \alpha = \frac{2}{5}$ and α is in Quadrant I

b. $\cos \frac{\alpha}{2}$, if $\cos \alpha = -0.9$ and α is in Quadrant II

c. $\cos \frac{\alpha}{2}$, if $\sin \alpha = -0.75$ and α is in Quadrant III

d. $\sin \frac{\alpha}{2}$, if $\sin \alpha = -0.8$ and α is in Quadrant IV

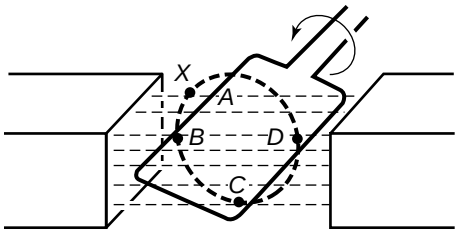
Helping You Remember

3. Many students find it difficult to remember a large number of identities. How can you obtain all three of the identities for $\cos 2\theta$ by remembering only one of them and using a Pythagorean identity?

14-6 Enrichment

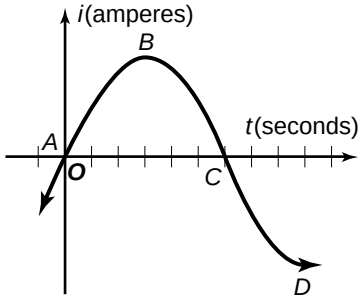
Alternating Current

The figure at the right represents an alternating current generator. A rectangular coil of wire is suspended between the poles of a magnet. As the coil of wire is rotated, it passes through the magnetic field and generates current.



As point X on the coil passes through the points A and C, its motion is along the direction of the magnetic field between the poles. Therefore, no current is generated. However, through points $B$and D, the motion of X is perpendicular to the magnetic field. This induces maximum current in the coil. Between A and B, B and C, C and D, and D and A, the current in the coil will have an intermediate value. Thus, the graph of the current of an alternating current generator is closely related to the sine curve.

The maximum current may have a positive or negative value.



The actual current, i, in a household current is given by $i = I_M \sin(120\pi t + \alpha)$ where I_M is the maximum value of the current, t is the elapsed time in seconds, and α is the angle determined by the position of the coil at time t_n.

Example **If $\alpha = \frac{\pi}{2}$, find a value of t for which $i = 0$.**

If $i = 0$, then $I_M \sin(120\pi t + \alpha) = 0$. $\quad i = I_M \sin(120\pi t + \alpha)$

Since $I_M \neq 0$, $\sin(120\pi t + \alpha) = 0$. $\quad$ If $ab = 0$ and $a \neq 0$, then $b = 0$.

Let $120\pi t + \alpha = s$. Thus, $\sin s = 0$.

$s = \pi$ because $\sin \pi = 0$.

$120\pi t + \alpha = \pi$ $\quad$ Substitute $120\pi t + \alpha$ for s.

$120\pi t + \frac{\pi}{2} = \pi$ $\quad$ Substitute $\frac{\pi}{2}$ for α.

$= \frac{1}{240}$ $\quad$ Solve for t.

This solution is the first positive value of t that satisfies the problem.

Using the equation for the actual current in a household circuit, $i = I_M \sin(120\pi t + \alpha)$, solve each problem. For each problem, find the first positive value of t.

1. If $\alpha = 0$, find a value of t for which $i = 0$.

2. If $\alpha = 0$, find a value of t for which $i = +I_M$.

3. If $\alpha = \frac{\pi}{2}$, find a value of t for which $i = -I_M$.

4. If $\alpha = \frac{\pi}{4}$, find a value of t for which $i = 0$.

14-7 Study Guide and Intervention

Solving Trigonometric Equations

Solve Trigonometric Equations You can use trigonometric identities to solve trigonometric equations, which are true for only certain values of the variable.

Example 1 **Find all solutions of $4 \sin^2 \theta - 1 = 0$ for the interval $0° < \theta < 360°$.**

$$4 \sin^2 \theta - 1 = 0$$
$$4 \sin^2 \theta = 1$$
$$\sin^2 \theta = \frac{1}{4}$$
$$\sin \theta = \pm\frac{1}{2}$$
$$\theta = 30°, 150°, 210°, 330°$$

Example 2 **Solve $\sin 2\theta + \cos \theta = 0$ for all values of θ. Give your answer in both radians and degrees.**

$$\sin 2\theta + \cos \theta = 0$$
$$2 \sin \theta \cos \theta + \cos \theta = 0$$
$$\cos \theta (2 \sin \theta + 1) = 0$$

$\cos \theta = 0$ or $2 \sin \theta + 1 = 0$

$$\sin \theta = -\frac{1}{2}$$

$\theta = 90° + k \cdot 180°$; $\theta = \frac{\pi}{2} + k \cdot \pi$

$\theta = 210° + k \cdot 360°$, $330° + k \cdot 360°$; $\theta = \frac{7\pi}{6} + k \cdot 2\pi$, $\frac{11\pi}{6} + k \cdot 2\pi$

Exercises

Find all solutions of each equation for the given interval.

1. $2 \cos^2 \theta + \cos \theta = 1, 0 \le \theta < 2\pi$

2. $\sin^2 \theta \cos^2 \theta = 0, 0 \le \theta < 2\pi$

3. $\cos 2\theta = \frac{\sqrt{3}}{2}, 0° \le \theta < 360°$

4. $2 \sin \theta - \sqrt{3} = 0, 0 \le \theta < 2\pi$

Solve each equation for all values of θ if θ is measured in radians.

5. $4 \sin^2 \theta - 3 = 0$

6. $2 \cos \theta \sin \theta + \cos \theta = 0$

Solve each equation for all values of θ if θ is measured in degrees.

7. $\cos 2\theta + \sin^2 \theta = \frac{1}{}$

8. $\tan 2\theta = -1$

14-7 Study Guide and Intervention *(continued)*

Solving Trigonometric Equations

Use Trigonometric Equations

Example **LIGHT Snell's law says that $\sin \alpha = 1.33 \sin \beta$, where α is the angle at which a beam of light enters water and β is the angle at which the beam travels through the water. If a beam of light enters water at 42°, at what angle does the light travel through the water?**

$\sin \alpha = 1.33 \sin \beta$	Original equation
$\sin 42° = 1.33 \sin \beta$	$\alpha = 42°$
$\sin \beta = \frac{\sin 42°}{1.33}$	Divide each side by 1.33.
$\sin \beta \approx 0.5031$	Use a calculator.
$\beta \approx 30.2°$	Take the arcsin of each side.

The light travels through the water at an angle of approximately 30.2°.

Exercises

1. A 6-foot pipe is propped on a 3-foot tall packing crate that sits on level ground. One foot of the pipe extends above the top of the crate and the other end rests on the ground. What angle does the pipe form with the ground?

2. At 1:00 P.M. one afternoon a 180-foot statue casts a shadow that is 85 feet long. Write an equation to find the angle of elevation of the Sun at that time. Find the angle of elevation.

3. A conveyor belt is set up to carry packages from the ground into a window 28 feet above the ground. The angle that the conveyor belt forms with the ground is 35°. How long is the conveyor belt from the ground to the window sill?

SPORTS The distance a golf ball travels can be found using the formula $d = \frac{v_0^2}{g} \sin 2\theta$, where v_0 is the initial velocity of the ball, g is the acceleration due to gravity (which is 32 feet per second squared), and θ is the angle that the path of the ball makes with the ground.

4. How far will a ball travel hit 90 feet per second at an angle of 55°?

5. If a ball that traveled 300 feet had an initial velocity of 110 feet per second, what angle did the path of the ball make with the ground?

6. Some children set up a teepee in the woods. The poles are 7 feet long from their intersection to their bases, and the children want the distance between the poles to be 4 feet at the base. How wide must the angle be between the poles?

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14-7 Skills Practice

Solving Trigonometric Equations

Find all solutions of each equation for the given interval.

1. $\sin \theta = \frac{\sqrt{2}}{2}$, $0° \le \theta < 360°$

2. $2 \cos \theta = -\sqrt{3}$, $90° < \theta < 180°$

3. $\tan^2 \theta = 1$, $180° < \theta < 360°$

4. $2 \sin \theta = 1$, $0 \le \theta < \pi$

5. $\sin^2 \theta + \sin \theta = 0$, $\pi \le \theta < 2\pi$

6. $2 \cos^2 \theta + \cos \theta = 0$, $0 \le \theta < \pi$

Solve each equation for all values of θ if θ is measured in radians.

7. $2 \cos^2 \theta - \cos \theta = 1$

8. $\sin^2 \theta - 2 \sin \theta + 1 = 0$

9. $\sin \theta + \sin \theta \cos \theta = 0$

10. $\sin^2 \theta = 1$

11. $4 \cos \theta = -1 + 2 \cos \theta$

12. $\tan \theta \cos \theta = \frac{1}{2}$

Solve each equation for all values of θ if θ is measured in degrees.

13. $2 \sin \theta + 1 = 0$

14. $2 \cos \theta + \sqrt{3} = 0$

15. $\sqrt{2} \sin \theta + 1 = 0$

16. $2 \cos^2 \theta = 1$

17. $4 \sin^2 \theta = 3$

18. $\cos 2\theta = -1$

Solve each equation for all values of θ.

19. $3 \cos^2 \theta - \sin^2 \theta = 0$

20. $\sin \theta + \sin 2\theta = 0$

21. $2 \sin^2 \theta = \sin \theta + 1$

22. $\cos \theta + \sec \theta = 2$

Lesson 14-7

NAME _____ DATE _____ PERIOD _____

14-7 Practice

Solving Trigonometric Equations

Find all solutions of each equation for the given interval.

1. $\sin 2\theta = \cos \theta$, $90° \le \theta < 180°$

2. $\sqrt{2} \cos \theta = \sin 2\theta$, $0° \le \theta < 360°$

3. $\cos 4\theta = \cos 2\theta$, $180° \le \theta < 360°$

4. $\cos \theta + \cos (90 - \theta) = 0$, $0 \le \theta < 2\pi$

5. $2 + \cos \theta = 2 \sin^2 \theta$, $\pi \le \theta \le \frac{3\pi}{2}$

6. $\tan^2 \theta + \sec \theta = 1$, $\frac{\pi}{2} \le \theta < \pi$

Solve each equation for all values of θ if θ is measured in radians.

7. $\cos^2 \theta = \sin^2 \theta$

8. $\cot \theta = \cot^3 \theta$

9. $\sqrt{2} \sin^3 \theta = \sin^2 \theta$

10. $\cos^2 \theta \sin \theta = \sin \theta$

11. $2 \cos 2\theta = 1 - 2 \sin^2 \theta$

12. $\sec^2 \theta = 2$

Solve each equation for all values of θ if θ is measured in degrees.

13. $\sin^2 \theta \cos \theta = \cos \theta$

14. $\csc^2 \theta - 3 \csc \theta + 2 = 0$

15. $\frac{3}{1 + \cos \theta} = 4(1 - \cos \theta)$

16. $\sqrt{2} \cos^2 \theta = \cos^2 \theta$

Solve each equation for all values of θ.

17. $4 \sin^2 \theta = 3$

18. $4 \sin^2 \theta - 1 = 0$

19. $2 \sin^2 \theta - 3 \sin \theta = -1$

20. $\cos 2\theta + \sin \theta - 1 = 0$

21. WAVES Waves are causing a buoy to float in a regular pattern in the water. The vertical position of the buoy can be described by the equation $h = 2 \sin x$. Write an expression that describes the position of the buoy when its height is at its midline.

22. ELECTRICITY The electric current in a certain circuit with an alternating current can be described by the formula $i = 3 \sin 240t$, where i is the current in amperes and t is the time in seconds. Write an expression that describes the times at which there is no current.

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14-7 Reading to Learn Mathematics

Solving Trigonometric Equations

Pre-Activity **How can trigonometric equations be used to predict temperature?**

Read the introduction to Lesson 14-7 at the top of page 799 in your textbook.

Describe how you could use a graphing calculator to determine the months in which the average daily high temperature is above 80°F. (Assume that $x = 1$ represents January.) Specify the graphing window that you would use.

Lesson 14-7

Reading the Lesson

1. Identify which equations have no solution.

A. $\sin \theta = 1$ **B.** $\tan \theta = 0.001$ **C.** $\sec \theta = \frac{1}{2}$

D. $\csc \theta = -3$ **E.** $\cos \theta = 1.01$ **F.** $\cot \theta = -1000$

G. $\cos \theta + 2 = -1$ **H.** $\sec \theta - 1.5 = 0$ **I.** $\sin \theta - 0.009 = 0.99$

2. Use a trigonometric identity to write the first step in the solution of each trigonometric equation. (Do not complete the solution.)

a. $\tan \theta = \cos^2 \theta + \sin^2 \theta, 0 \leq \theta < 2\pi$

b. $\sin^2 \theta - 2 \sin \theta + 1 = 0, 0° \leq \theta < 360°$

c. $\cos 2\theta = \sin \theta, 0° \leq \theta < 360°$

d. $\sin 2\theta = \cos \theta, 0 \leq \theta < 2\pi$

e. $2 \cos 2\theta + 3 \cos \theta = -1, 0° \leq \theta < 360°$

f. $3 \tan^2 \theta + 5 \tan \theta - 2 = 0$

Helping You Remember

3. A good way to remember something is to explain it to someone else. How would you explain to a friend the difference between verifying a trigonometric identity and solving a trigonometric equation.

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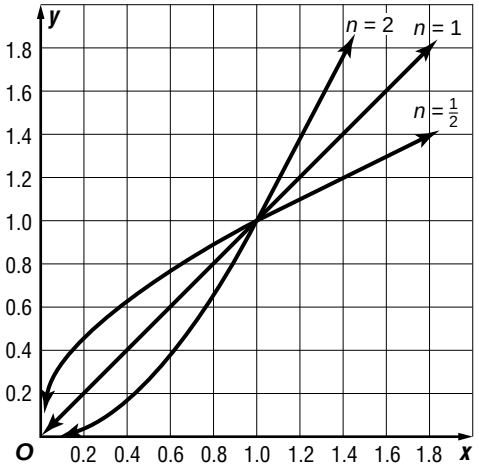
14-7 Enrichment

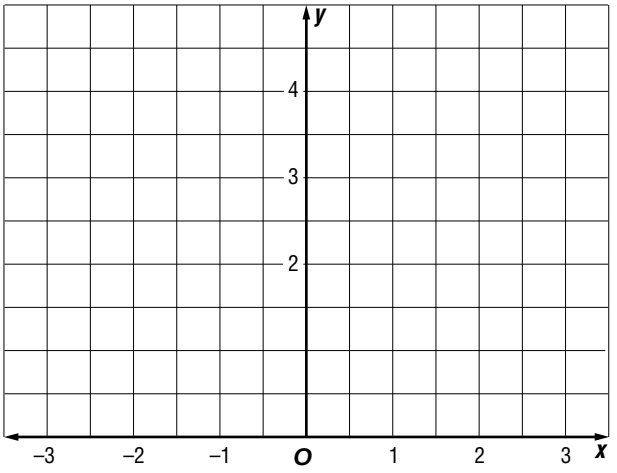
Families of Curves

Use these graphs for the problems below.

The Family $y \quad x^n$



The Family $y \quad e^{mx}$



1. Use the graph on the left to describe the relationship among the curves $y = x^{\frac{1}{2}}$, $y = x^1$, and $y = x^2$.

2. Graph $y = x^n$ for $n = \frac{1}{10}, \frac{1}{4}$, 4, and 10 on the grid with $y = x^{\frac{1}{2}}$, $y = x^1$, and $y = x^2$.

3. Which two regions in the first quadrant contain no points of the graphs of the family for $y = x^n$?

4. On the right grid, graph the members of the family $y = e^{mx}$ for which $m = 1$ and $m = -1$.

5. Describe the relationship among these two curves and the y-axis.

6. Graph $y = e^{mx}$ for $m = 0, \pm\frac{1}{4}, \pm\frac{1}{2}, \pm 2$, and ± 4.

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14 Chapter 14 Test, Form 1

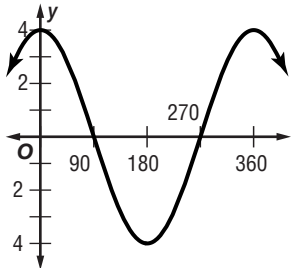
SCORE _____

Write the letter for the correct answer in the blank at the right of each question.

1. Which equation is graphed?

A. $y = 4 \sin \theta$ **B.** $y = 4 \cos \theta$

C. $y = \sin 4\theta$ **D.** $y = \cos 4\theta$



1. _____

2. Find the amplitude of $y = 6 \sin \theta$.

A. 6 **B.** π **C.** -6 **D.** 2π **2.** _____

3. Find the period of $y = 5 \cos \theta$.

A. -5 **B.** 5 **C.** π **D.** 2π **3.** _____

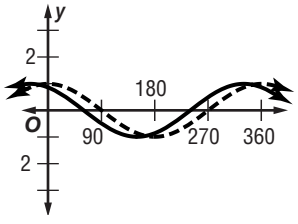
4. Which equation is graphed?

A. $y = \sin (\theta + 30°)$

B. $y = \sin (\theta - 30°)$

C. $y = \cos (\theta + 30°)$

D. $y = \cos (\theta - 30°)$

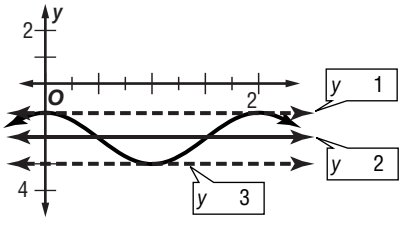


4. _____

5. Which equation is graphed?

A. $y = \cos \theta - 2$ **B.** $y = \cos \theta + 2$

C. $y = \sin \theta + 2$ **D.** $y = \sin \theta - 2$



5. _____

6. Find $\sin \theta$ if $\cos \theta = \frac{1}{2}$ and $0° \le \theta < 90°$.

A. $\frac{\sqrt{3}}{2}$ **B.** $-\frac{\sqrt{3}}{2}$ **C.** $\frac{3}{4}$ **D.** $\frac{1}{2}$ **6.** _____

7. Find $\cot \theta$ if $\tan \theta = \frac{1}{3}$ and $0° \le \theta < 90°$.

A. 4 **B.** 3 **C.** -3 **D.** $-\frac{1}{3}$ **7.** _____

8. Simplify $\sin \theta \csc \theta$.

A. $\sin^2 \theta$ **B.** -1 **C.** $\tan \theta$ **D.** 1 **8.** _____

9. Simplify $\tan \theta \cos \theta$.

A. $\frac{\cos^2 \theta}{\sin \theta}$ **B.** $\cot \theta$ **C.** $\sin \theta$ **D.** $1 - \sec^2 \theta$ **9.** _____

Assessment

14 Chapter 14 Test, Form 1 *(continued)*

10. Simplify $\cot \theta \sec \theta$.

A. $\frac{\cos \theta}{\sin^2 \theta}$ **B.** $\sin \theta$ **C.** $\csc \theta$ **D.** $\sec^2 \theta$ **10.** _____

11. Which expression is equivalent to $\frac{\sin^2 \theta + \cos^2 \theta}{\tan^2 \theta}$?

A. $\cot^2 \theta$ **B.** $\cos^2 \theta + \cot^2 \theta$ **C.** $\cos^2 \theta + \cos^4 \theta$ **D.** $\csc^2 \theta$ **11.** _____

12. Which expression is equivalent to $\csc \theta(\csc \theta - \sin \theta)$?

A. $\sec^2 \theta - 1$ **B.** $\cot^2 \theta$ **C.** $\tan^2 \theta$ **D.** 1 **12.** _____

13. Find the exact value of $\cos 135°$.

A. $\frac{\sqrt{2}}{2}$ **B.** $\frac{1}{2}$ **C.** $-\frac{1}{2}$ **D.** $-\frac{\sqrt{2}}{2}$ **13.** _____

14. Find the exact value of $\sin 105°$.

A. $-\frac{\sqrt{2}}{2}$ **B.** 0 **C.** $\frac{\sqrt{2}+\sqrt{6}}{4}$ **D.** $\frac{\sqrt{2}-\sqrt{6}}{4}$ **14.** _____

15. Which expression is equivalent to $\sin (90° - \theta)$?

A. $\sin \theta$ **B.** $-\sin \theta$ **C.** $-\cos \theta$ **D.** $\cos \theta$ **15.** _____

16. Find the exact value of $\cos 2\theta$ if $\cos \theta = \frac{5}{13}$ and $0° \le \theta < 90°$.

A. $\frac{25}{169}$ **B.** $\frac{120}{169}$ **C.** $-\frac{119}{169}$ **D.** $\frac{119}{169}$ **16.** _____

17. Find the exact value of $\sin 2\theta$ if $\sin \theta = \frac{4}{5}$ and $0° \le \theta < 90°$.

A. $\frac{24}{25}$ **B.** $\frac{12}{25}$ **C.** $\frac{24}{5}$ **D.** $-\frac{7}{25}$ **17.** _____

18. Find the exact value of $\cos 22\frac{1}{2}°$ by using a half-angle formula.

A. $\sqrt{2-\sqrt{2}}$ **B.** $\sqrt{2+\sqrt{2}}$ **C.** $-\sqrt{2+\sqrt{2}}$ **D.** $-\sqrt{2-\sqrt{2}}$ **18.** _____

19. Which is *not* a solution of $\sin 2\theta = 1$?

A. 90° **B.** 45° **C.** 225° **D.** −135° **19.** _____

20. **LIGHT** The length of the shadow S given by a tower that is 100 meters high is $S = \frac{100}{\tan \theta}$, where θ is the angle of inclination of the Sun. If the angle of inclination is 45°, find the length of the shadow.

A. 162 m **B.** 62 m **C.** 100 m **D.** 84 m **20.** _____

Bonus Verify that $\frac{1 + \tan \theta}{\cos \theta} = \sec \theta + \sin \theta \sec^2 \theta$ is an identity. **B:** _____

NAME _____ DATE _____ PERIOD _____

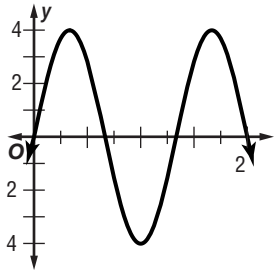
14

Chapter 14 Test, Form 2A

SCORE _____

Write the letter for the correct answer in the blank at the right of each question.

1. Which equation is graphed?



A. $y = 4 \sin \frac{3}{2}\theta$ **B.** $y = 4 \cos \frac{3}{2}\theta$

C. $y = 4 \sin \frac{2}{3}\theta$ **D.** $y = 4 \cos \frac{2}{3}\theta$ **1.** _____

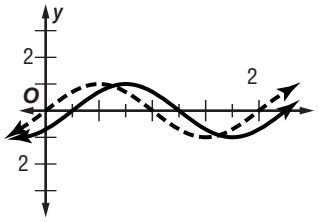
2. Find the amplitude of $y = 8 \sin 2\theta$.

A. 2 **B.** π **C.** 8 **D.** 4 **2.** _____

3. Find the period of $y = \tan 3\theta$.

A. $\frac{2\pi}{3}$ **B.** $\frac{\pi}{3}$ **C.** 3π **D.** 6π **3.** _____

4. Which equation is graphed?



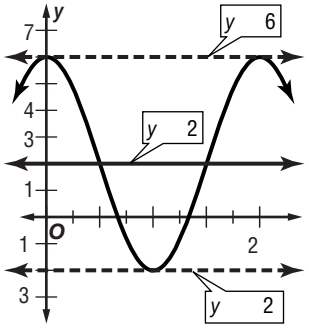
A. $y = \sin (\theta - \frac{\pi}{4})$ **B.** $y = \sin \left(\theta + \frac{\pi}{4}\right)$

C. $y = \cos \left(\theta - \frac{\pi}{4}\right)$ **D.** $y = \cos \left(\theta + \frac{\pi}{4}\right)$ **4.** _____

5. Find the phase shift of $y = \cos \left(\theta + \frac{2\pi}{5}\right)$.

A. $\frac{\pi}{5}$ **B.** $\frac{2\pi}{5}$ **C.** $-\frac{\pi}{5}$ **D.** $-\frac{2\pi}{5}$ **5.** _____

6. Which equation is graphed?



A. $y = 4 \sin \theta + 2$

B. $y = 4 \sin \theta - 2$

C. $y = 4 \cos \theta + 2$

D. $y = 4 \cos \theta - 2$ **6.** _____

7. Find the vertical shift of $y = 3 \csc \theta - 5$.

A. -3 **B.** -5 **C.** 5 **D.** 3 **7.** _____

8. Find $\csc \theta$ if $\cot \theta = \frac{1}{3}$ and $90° < \theta < 180°$.

A. $-\frac{2\sqrt{2}}{3}$ **B.** $\frac{2\sqrt{2}}{3}$ **C.** $\frac{\sqrt{10}}{3}$ **D.** $-\frac{\sqrt{10}}{3}$ **8.** _____

9. Find $\sin \theta$ if $\cos \theta = -\frac{2}{3}$ and $90° < \theta < 180°$.

A. $-\frac{\sqrt{5}}{3}$ **B.** $\frac{\sqrt{5}}{3}$ **C.** $-\frac{\sqrt{13}}{3}$ **D.** $\frac{\sqrt{13}}{3}$ **9.** _____

Assessment

14 Chapter 14 Test, Form 2A *(continued)*

10. Simplify $\frac{1 - \cos^2 \theta}{\tan^2 \theta}$.

A. $-\cos^2 \theta$ **B.** $\sec^2 \theta$ **C.** $\cos^2 \theta$ **D.** $\sin^2 \theta$ **10.** _____

11. Simplify $-5(\cot^2 \theta - \csc^2 \theta)$.

A. 5 **B.** -5 **C.** $-5 \csc^2 \theta$ **D.** $5 \sec^2 \theta$ **11.** _____

12. Which expression is *not* equivalent to 1?

A. $\sin^2 \theta + \cot^2 \theta \sin^2 \theta$ **B.** $\frac{\sin^2 \theta}{1 - \cos \theta} - \cos \theta$

C. $\sec^2 \theta + \tan^2 \theta$ **D.** $\frac{\cot^2 \theta \sin^2 \theta}{\cos^2 \theta}$ **12.** _____

13. Which expression is equivalent to $\tan \theta - \frac{\sec \theta}{\sin \theta}$?

A. $-\cot \theta$ **B.** $\cot \theta$ **C.** $\tan \theta - \cot \theta$ **D.** $\tan \theta - \sec^2 \theta$ **13.** _____

14. Find the exact value of $\cos 375°$.

A. $\frac{\sqrt{6} - \sqrt{2}}{4}$ **B.** $\frac{\sqrt{6} + \sqrt{2}}{4}$ **C.** $\frac{\sqrt{2} - \sqrt{6}}{4}$ **D.** $-\frac{\sqrt{2} - \sqrt{6}}{4}$ **14.** _____

15. Which expression is equivalent to $\cos \left(\theta + \frac{\pi}{2}\right)$?

A. $\cos \theta$ **B.** $-\cos \theta$ **C.** $\sin \theta$ **D.** $-\sin \theta$ **15.** _____

16. Find the exact value of $\sin 2\theta$ if $\cos \theta = -\frac{\sqrt{5}}{3}$ and $180° < \theta < 270°$.

A. $-\frac{1}{9}$ **B.** $-\frac{4\sqrt{5}}{9}$ **C.** $\frac{1}{9}$ **D.** $\frac{4\sqrt{5}}{9}$ **16.** _____

17. Find the exact value of $\sin \frac{\theta}{2}$ if $\cos \theta = \frac{2}{3}$ and $270° < \theta < 360°$.

A. $\frac{1}{3}$ **B.** $-\frac{1}{3}$ **C.** $\frac{\sqrt{6}}{6}$ **D.** $-\frac{\sqrt{6}}{6}$ **17.** _____

18. Find the exact value of $\cos 105°$ by using a half-angle formula.

A. $\sqrt{2 - \sqrt{3}}$ **B.** $-\sqrt{2 - \sqrt{3}}$ **C.** $-\sqrt{2 + \sqrt{3}}$ **D.** $\sqrt{2 + \sqrt{3}}$ **18.** _____

19. Find the solutions of $\sin 2\theta = \cos \theta$ if $0° \leq \theta < 180°$.

A. 30°, 90° **B.** 30°, 150° **C.** 30°, 90°, 150° **D.** 0°, 90°, 150° **19.** _____

20. **BIOLOGY** An insect population P in a certain area fluctuates with the seasons. It is estimated that $P = 17{,}000 + 4500 \sin \frac{\pi t}{52}$, where t is given in weeks. Determine the number of weeks it would take for the population to initially reach 20,000.

A. 12 weeks **B.** 692 weeks **C.** 38 weeks **D.** 42 weeks **20.** _____

Bonus Verify that $\frac{1 + \cot \theta}{\csc \theta} = \sin \theta + \cos \theta$ is an identity. **B:** _____

NAME _____ DATE _____ PERIOD _____

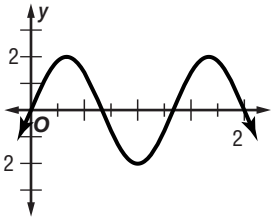
14

Chapter 14 Test, Form 2B

SCORE _____

Write the letter for the correct answer in the blank at the right of each question.

1. Which equation is graphed?



A. $y = 3 \sin \frac{2}{3}\theta$ **B.** $y = 3 \cos \frac{2}{3}\theta$

C. $y = 2 \sin \frac{3}{2}\theta$ **D.** $y = 2 \cos \frac{3}{2}\theta$ **1.** _____

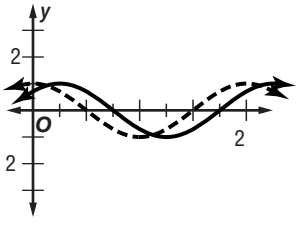
2. Find the amplitude of $y = 6 \cos 4\theta$.

A. $\frac{3}{2}$ **B.** 6 **C.** 4 **D.** $\frac{\pi}{2}$ **2.** _____

3. Find the period of $y = \tan 5\theta$.

A. 10π **B.** $\frac{2\pi}{5}$ **C.** 5π **D.** $\frac{\pi}{5}$ **3.** _____

4. Which equation is graphed?



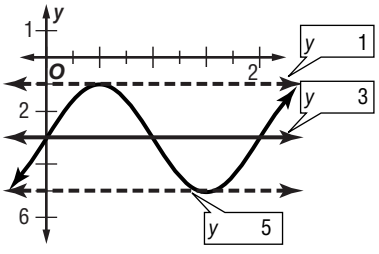
A. $y = \sin\left(\theta - \frac{\pi}{4}\right)$ **B.** $y = \cos\left(\theta - \frac{\pi}{4}\right)$

C. $y = \sin\left(\theta + \frac{\pi}{4}\right)$ **D.** $y = \cos\left(\theta + \frac{\pi}{4}\right)$ **4.** _____

5. Find the phase shift of $y = \sin\left(\theta - \frac{3\pi}{4}\right)$.

A. $\frac{3\pi}{4}$ **B.** $-\frac{3\pi}{4}$ **C.** $\frac{4\pi}{3}$ **D.** $-\frac{4\pi}{3}$ **5.** _____

6. Which equation is graphed?



A. $y = 2 \sin \theta - 3$ **B.** $y = 2 \sin \theta + 3$

C. $y = 3 \cos \theta - 2$ **D.** $y = 3 \cos \theta + 2$ **6.** _____

7. Find the vertical shift of $y = -4 \sec \theta + 7$.

A. -4 **B.** -7

C. 7 **D.** 4 **7.** _____

8. Find $\sec \theta$ if $\tan \theta = \frac{1}{4}$ and $180° < \theta < 270°$.

A. $\frac{\sqrt{15}}{4}$ **B.** $-\frac{\sqrt{15}}{4}$ **C.** $\frac{\sqrt{17}}{4}$ **D.** $-\frac{\sqrt{17}}{4}$ **8.** _____

9. Find $\cos \theta$ if $\sin \theta = \frac{3}{5}$ and $90° < \theta < 180°$.

A. $\frac{4}{5}$ **B.** $-\frac{4}{5}$ **C.** $\frac{\sqrt{34}}{5}$ **D.** $-\frac{\sqrt{34}}{5}$ **9.** _____

10. Simplify $\frac{1 - \csc^2 \theta}{\cot^2 \theta}$.

A. -1 **B.** 1 **C.** $\tan^2 \theta$ **D.** $\frac{1}{\sin^4 \theta}$ **10.** _____

Assessment

14 Chapter 14 Test, Form 2B *(continued)*

11. Simplify $-4(\sec^2 \theta - \tan^2 \theta)$.

A. $-4 \tan^2 \theta$ **B.** $4 \tan^2 \theta$ **C.** 4 **D.** -4 **11.** _____

12. Which expression is equivalent to 1?

A. $\frac{1 + \sin \theta}{\sin \theta}$ **B.** $\frac{1}{\sec^2 \theta} + \frac{1}{\csc^2 \theta}$

C. $\tan^2 \theta - \sec^2 \theta$ **D.** $\frac{\cot \theta \csc \theta}{\sec \theta}$ **12.** _____

13. Which expression is equivalent to $\frac{\sin \theta}{1 + \cos \theta} + \frac{\sin \theta}{1 - \cos \theta}$?

A. $\frac{2 \sin \theta}{1 + \cos^2 \theta}$ **B.** $2 \sin \theta$ **C.** $2 \csc \theta$ **D.** $-2 \csc \theta$ **13.** _____

14. Find the exact value of $\sin(-15°)$.

A. $\frac{\sqrt{6} - \sqrt{2}}{4}$ **B.** $\frac{\sqrt{6} + \sqrt{2}}{4}$ **C.** $\frac{-\sqrt{6} + \sqrt{2}}{4}$ **D.** $\frac{-\sqrt{6} - \sqrt{2}}{4}$ **14.** _____

15. Which expression is equivalent to $\sin\left(\theta + \frac{\pi}{2}\right)$?

A. $\cos\left(\theta - \frac{\pi}{2}\right)$ **B.** $-\cos \theta$ **C.** $-\sin \theta$ **D.** $\cos \theta$ **15.** _____

16. Find the exact value of $\cos 2\theta$ if $\sin \theta = -\frac{2}{3}$ and $180° < \theta < 270°$.

A. $\frac{1}{9}$ **B.** $-\frac{4\sqrt{5}}{9}$ **C.** $-\frac{1}{9}$ **D.** $\frac{4\sqrt{5}}{9}$ **16.** _____

17. Find the exact value of $\cos \frac{\theta}{2}$ if $\sin \theta = \frac{1}{4}$ and $0° \le \theta < 90°$.

A. $\frac{\sqrt{15}}{4}$ **B.** $-\frac{\sqrt{15}}{4}$ **C.** $\sqrt{8 + 2\sqrt{15}}$ **D.** $\sqrt{4 + \sqrt{15}}$ **17.** _____

18. Find the exact value of $\sin 105°$ by using a half-angle formula.

A. $\sqrt{2 - \sqrt{3}}$ **B.** $\sqrt{2 + \sqrt{3}}$ **C.** $-\sqrt{2 + \sqrt{3}}$ **D.** $-\sqrt{2 - \sqrt{3}}$ **18.** _____

19. Find the solutions of $3 \sin \theta = 2 \cos^2 \theta$ if $0° \le \theta < 360°$.

A. 30°, 150° **B.** 30°, 120° **C.** 30°, 330° **D.** 150°, 330° **19.** _____

20. **BIOLOGY** An insect population P in a certain area fluctuates with the seasons. It is estimated that $P = 15{,}000 + 2500 \sin \frac{\pi t}{52}$, where t is given in weeks. Determine the number of weeks it would take for the population to initially reach 16,000.

A. 21 weeks **B.** 24 weeks **C.** 109 weeks **D.** 7 weeks **20.** _____

Bonus Verify that $1 + \csc^2 \theta \tan^2 \theta = 2 + \tan^2 \theta$ is an identity. **B:** _____

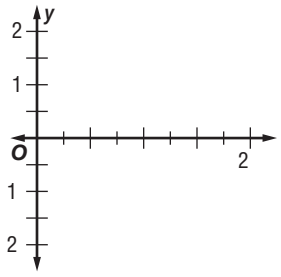
NAME _____ DATE _____ PERIOD _____

14 Chapter 14 Test, Form 2C

SCORE _____

1. Graph the function $y = \frac{3}{2}\cos 2\theta$.

1.



For Questions 2 and 3, find the amplitude, if it exists, and period of each function.

2. $y = 3 \sin 4\theta$

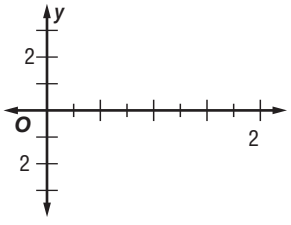
2. _____

3. $y = \frac{1}{2}\tan \frac{1}{5}\theta$

3. _____

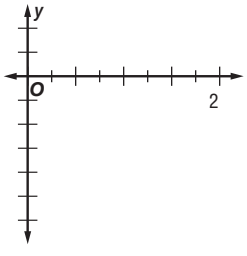
4. State the phase shift of $y = \cos\left(\theta + \frac{2\pi}{3}\right)$. Then graph the function.

4. _____



5. State the vertical shift and the equation of the midline for $y = 3 \cos \theta - 2$. Then graph the function.

5. _____



6. Find $\sec \theta$ if $\sin \theta = \frac{3}{5}$ and $0° \leq \theta < 90°$.

6. _____

7. Find $\cot \theta$ if $\csc \theta = -\frac{5}{2}$ and $270° < \theta < 360°$.

7. _____

8. Simplify $\frac{\cos \theta \cdot \csc \theta}{\cot \theta}$.

8. _____

9. Simplify $\frac{1 - \cos^2 \theta}{\cos^2 \theta}$.

9. _____

Assessment

14 Chapter 14 Test, Form 2C *(continued)*

10. Verify that $(\cos \theta + \sin \theta)^2 - 2 \cos \theta \sin \theta = 1$ is an identity. **10.** _____

11. Verify that $\frac{1 + \cot \theta}{\csc \theta} = \sin \theta + \cos \theta$ is an identity. **11.** _____

12. Find the exact value of $\sin (-195°)$. **12.** _____

13. Find the exact value of $\cos 255°$. **13.** _____

14. Verify that $\sin \left(\theta - \frac{\pi}{2}\right) = -\cos \theta$ is an identity. **14.** _____

15. Find the exact value of $\sin 2\theta$ if $\cos \theta = \frac{1}{4}$ and $270° < \theta < 360°$. **15.** _____

16. Find the exact value of $\cos \frac{\theta}{2}$ if $\sin \theta = \frac{1}{3}$ and $90° < \theta < 180°$. **16.** _____

17. Find the exact value of $\sin 195°$ by using a half-angle formula. **17.** _____

18. Verify that $\sin 2\theta = \frac{2 \cot \theta}{\csc^2 \theta}$ is an identity. **18.** _____

19. Solve $\cos 2\theta + \cos \theta = 0$ for all values of θ if θ is measured in degrees. **19.** _____

20. **BUSINESS** The profit P for a product whose sales fluctuate with the seasons is estimated to be $P = 14 + 5 \sin \frac{\pi t}{52}$, where t is given in weeks and P is in thousands of dollars. Determine the number of weeks it would take for the profit to initially reach \$18,000. **20.** _____

Bonus Find $\cos 2\theta$ if $\sin \frac{\theta}{2} = \frac{\sqrt{2 + \sqrt{3}}}{2}$. **B:** _____

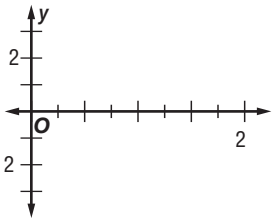
NAME _____ DATE _____ PERIOD _____

14 Chapter 14 Test, Form 2D

SCORE _____

1. Graph $y = \frac{5}{2} \sin 2\theta$.

1.



For Questions 2 and 3, find the amplitude, if it exists, and period of each function.

2. $y = 2 \sin 3\theta$

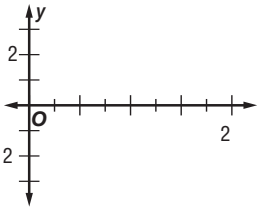
2. _____

3. $y = \frac{1}{3} \tan \frac{1}{4}\theta$

3. _____

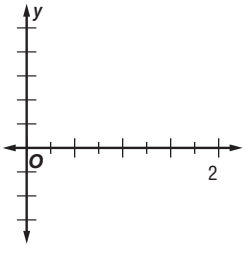
4. State the phase shift of $y = \sin\left(\theta - \frac{2\pi}{3}\right)$. Then graph the function.

4. _____



5. State the vertical shift and the equation of the midline for $y = 3 \cos \theta + 1$. Then graph the function.

5. _____



6. Find $\csc \theta$ if $\cos \theta = -\frac{1}{3}$ and $90° < \theta < 180°$.

6. _____

7. Find $\tan \theta$ if $\sec \theta = \frac{5}{2}$ and $270° \leq \theta < 360°$.

7. _____

8. Simplify $\frac{\csc \theta \tan \theta}{\sec \theta}$.

8. _____

9. Simplify $\frac{1 - \sec^2 \theta}{\sin^2 \theta}$.

9. _____

Assessment

14 Chapter 14 Test, Form 2D *(continued)*

10. Verify that $\cos^2 \theta \sec^2 \theta - \cos^2 \theta - \sin^2 \theta = 0$ is an identity. **10.** _____

11. Verify that $\frac{\tan \theta + \sec \theta}{\cot \theta \sec \theta} = \frac{\sin \theta + 1}{\cot \theta}$ is an identity. **11.** _____

12. Find the exact value of $\sin 165°$. **12.** _____

13. Find the exact value of $\cos (-345°)$. **13.** _____

14. Verify that $\cos \left(\theta - \frac{\pi}{2}\right) = \sin \theta$ is an identity. **14.** _____

15. Find the exact value of $\cos 2\theta$ if $\cos \theta = \frac{1}{4}$ and $270° < \theta < 360°$. **15.** _____

16. Find the exact value of $\sin \frac{\theta}{2}$ if $\sin \theta = \frac{1}{3}$ and $90° < \theta < 180°$. **16.** _____

17. Find the exact value of $\cos 195°$ by using a half-angle formula. **17.** _____

18. Verify that $\cos 2\theta = \sin^2 \theta(2 \cot^2 \theta - \csc^2 \theta)$ is an identity. **18.** _____

19. Solve $\sin 2\theta - \sin \theta = 0$ for all values of θ if θ is measured in degrees. **19.** _____

20. **BUSINESS** The profit P for a product whose sales fluctuate with the seasons is estimated to be $P = 16 + 7 \sin \frac{\pi t}{52}$, where t is given in weeks and P is in thousands of dollars. Determine the number of weeks it would take for the profit to initially reach \$20,000. **20.** _____

Bonus Find $\cos 2\theta$ if $\cos \frac{\theta}{2} = \frac{\sqrt{2 - \sqrt{2}}}{}$. **B:** _____

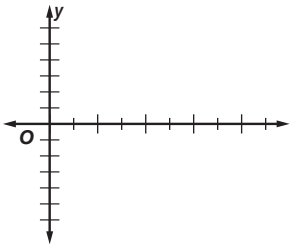
NAME _____ DATE _____ PERIOD _____

14 Chapter 14 Test, Form 3

SCORE _____

1. Graph $\frac{1}{2}y = \frac{3}{4}\csc\frac{1}{2}\theta$.

1.



Find the amplitude, if it exists, and period of each function.

2. $5y = \frac{2}{3}\cos 4\theta$

3. $-\frac{1}{4}y = -\frac{3}{8}\tan\frac{1}{5}\theta$

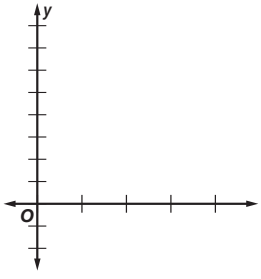
2. _____

3. _____

For Questions 4 and 5, state the vertical shift, amplitude, period, and phase shift of each function. Then graph the function.

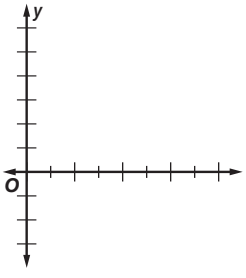
4. $y = 2\tan(2\theta - 90°) + 3$

4. _____



5. $y = \frac{3}{2} + 3\cos\left[2\left(\theta + \frac{\pi}{4}\right)\right]$

5. _____



6. Find $\sec\theta$ if $\sin\theta = \frac{1}{4}$ and $90° < \theta < 180°$.

6. _____

7. Find $\tan\theta$ if $\sec\theta = \frac{4}{3}$ and $270° < \theta < 360°$.

7. _____

8. Simplify $\frac{\cot^2\theta - \cos^2\theta}{\cot^2\theta\cos^2\theta}$.

8. _____

9. Verify that $\frac{\csc^2\theta - 1}{\cot\theta\sin\theta} = \cot\theta\csc\theta$ is an identity.

9. _____

Assessment

14 Chapter 14 Test, Form 3 *(continued)*

10. Verify that $1 - \cot^4 \theta = \dfrac{2 \sin^2 \theta - 1}{\sin^4 \theta}$ is an identity. **10.** _____

11. Find the exact value of $\cos 75° - \cos 15°$. **11.** _____

12. Find the exact value of $\sin 105° + \sin 225°$. **12.** _____

13. Verify that $\sin\left(\theta + \dfrac{\pi}{4}\right) - \cos\left(\theta - \dfrac{3\pi}{4}\right) = \sqrt{2} \cos \theta$ is an identity. **13.** _____

14. Verify that $\dfrac{[\sin (\alpha + \beta)]^2}{\sin \alpha \cos \alpha \sin \beta \cos \beta} = 2 + \tan \alpha \cot \beta + \cot \alpha \tan \beta$ is an identity. **14.** _____

15. Find the exact value of $\sin 2\theta$ if $\cos \theta = \dfrac{3}{8}$ and $270° < \theta < 360°$. **15.** _____

16. Find the exact value of $\cos \dfrac{\theta}{2}$ if $\sin \theta = -\dfrac{13}{16}$ and $180° < \theta < 270°$. **16.** _____

17. Find the exact value of $\cos \dfrac{17\pi}{12}$ by using half-angle formulas. **17.** _____

18. Verify that $\sin^2 \dfrac{\theta}{2} = \dfrac{\sin^2 \theta + \cos \theta - 1}{2 \cos \theta}$ is an identity. **18.** _____

19. Solve $\sin \dfrac{\theta}{2} - \cos \theta = 0$ for all values of θ if θ is measured in radians. **19.** _____

20. **WAVES** For a short time after a wave is created by a boat, its height can be modeled by $y = \dfrac{1}{2}h + \dfrac{1}{2}h \sin \dfrac{2\pi t}{P}$, where h is the maximum height of the wave in feet, P is the period in seconds, and t is the propagation of the wave in seconds. If a wave has a maximum height of 3.2 feet and a period of 2.5 seconds, how long after its creation will the wave initially reach a height of 3 feet? Round to the nearest hundredth. **20.** _____

Bonus Find the exact value of $\dfrac{\sin 2\theta - \cos 2\theta}{\sin \dfrac{\theta}{2}}$ if $\sin \theta = -\dfrac{3}{5}$ and $180° < \theta < 270°$. **B:** _____

NAME _____ DATE _____ PERIOD _____

14 Chapter 14 Open-Ended Assessment

SCORE _____

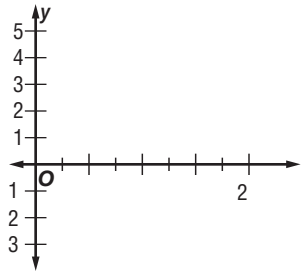
Demonstrate your knowledge by giving a clear, concise solution to each problem. Be sure to include all relevant drawings and justify your answers. You may show your solutions in more than one way or investigate beyond the requirements of the problem.

1. Ms. Rollins divided her students into four groups, asking each to solve the equation $\sin\theta \cot\theta = \cos^2\theta$. The answers given were:

Group A: $0° + k \cdot 360°$, $90° + k \cdot 360°$, $270° + k \cdot 360°$
Group B: $0° + k \cdot 360°$, $90° + k \cdot 180°$
Group C: $90° + k \cdot 180°$
Group D: $90° + k \cdot 360°$, $270° + k \cdot 360°$

Do any of the groups have the correct solution? Explain your reasoning.

2. Write a trigonometric function that has no amplitude, a period of $\frac{\pi}{2}$, a phase shift to the left, and a vertical shift upward. Then graph your function for $0 \le \theta < 2\pi$.



3. Show two different methods of verifying that $\frac{1}{1 - \sin^2\theta} = \tan^2\theta + 1$ is a trigonometric identity.

4. Select a quadrant, other than Quadrant I, and values for p and q so that $\sin\theta = \frac{p}{q}$. Use your values of p and q to find the exact values of $\cos\theta$, $\tan\theta$, $\csc\theta$, $\sec\theta$, $\cot\theta$, $\sin 2\theta$, $\cos 2\theta$, $\sin\frac{\theta}{2}$, and $\cos\frac{\theta}{2}$.

5. Show how to find the exact value of $\sin 240°$ by each method indicated.

a. using a sum of angles formula
b. using a difference of angles formula
c. using a double-angle formula
d. using a half-angle formula

Assessment

NAME _____ DATE _____ PERIOD _____

14

Chapter 14 Vocabulary Test/Review

SCORE _____

amplitude	half-angle formula	phase shift	trigonometric identity
double-angle formula	midline	trigonometric equation	vertical shift

Tell whether each sentence is *true* or *false*. If false, replace the underlined word or words to make a true sentence.

1. For the graph of $y = 3 \sin \left(x - \frac{\pi}{2}\right)$, the <u>vertical shift</u> is 3. **1.** _____

2. For the graph of $y = 2 \cos (x + 45°) + 5$, the <u>phase shift</u> is 5. **2.** _____

3. For the graph of $y = 3 \sin \left(x - \frac{\pi}{6}\right) - 2$, the line $y = -2$ is the <u>amplitude</u>. **3.** _____

4. $\sin^2 \theta + \cos^2 \theta = 1$ is a(n) <u>trigonometric identity</u>. **4.** _____

5. The exact value of sin 15° can be found by using a(n) <u>phase shift</u>. **5.** _____

6. $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ is a(n) <u>double-angle formula</u>. **6.** _____

7. $2 \cos^2 \theta - \cos \theta - 1 = 0$ is a(n) <u>trigonometric equation</u>. **7.** _____

In your own words—
Define the term.

8. phase shift

NAME _____ DATE _____ PERIOD _____

14

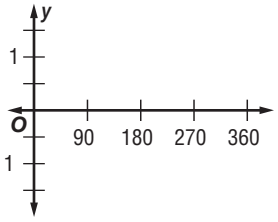
Chapter 14 Quiz

(Lessons 14–1 and 14–2)

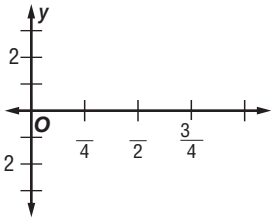
SCORE _____

For Questions 1 and 2, find the amplitude, if it exists, and period of each function. Then graph the function.

1. $y = \frac{1}{2}\cos\theta$ **1.** _____



2. $y = \tan 2\theta$ **2.** _____



3. State the phase shift of $y = \sin\left(\theta + \frac{\pi}{4}\right)$. **3.** _____

4. State the vertical shift and the equation of the midline for $y = 4\cos\theta + 2$. **4.** _____

Assessment

NAME _____ DATE _____ PERIOD _____

14

Chapter 14 Quiz

(Lessons 14–3 and 14–4)

SCORE _____

For Questions 1 and 2, find the value of each expression.

1. $\cos\theta$, if $\sin\theta = \frac{1}{2}$; $90° \le \theta < 180°$ **1.** _____

2. $\cot\theta$, if $\tan\theta = 2$; $180° < \theta < 270°$ **2.** _____

3. Simplify $4(\tan^2\theta - \sec^2\theta)$. **3.** _____

4. Simplify $\frac{1 + \tan^2\theta}{\csc^2\theta}$. **4.** _____

5. Standardized Test Practice $\frac{\sec\theta - 1}{\tan^2\theta} =$ **5.** _____

A. $\frac{\cos\theta}{\cos\theta + 1}$ **B.** $\frac{\sin\theta}{\sin\theta + 1}$ **C.** $\frac{\sin^2\theta}{\sin\theta + 1}$ **D.** 1

NAME _____ DATE _____ PERIOD _____

14

Chapter 14 Quiz

SCORE _____

(Lessons 14–5 and 14–6)

Find the exact value of each expression.

1. $\sin 75°$ **2.** $\cos(-225°)$

3. $\tan 210°$

Verify that each is an identity.

4. $\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$

5. $\cos(180° + \theta) = -\cos\theta$

For Questions 6–8, find the exact value for each.

6. $\cos 2\theta$, if $\cos\theta = -\frac{2}{5}$; $90° < \theta < 180°$

7. $\sin 2\theta$, if $\sin\theta = -\frac{4}{9}$; $270° < \theta < 360°$

8. $\cos\frac{\theta}{2}$, if $\sin\theta = -\frac{2}{5}$; $180° < \theta < 270°$

9. Find the exact value of $\cos 112\frac{1}{2}°$ by using a half-angle formula.

10. Verify that $\cos 2\theta = 1 - \sin 2\theta \tan\theta$ is an identity.

NAME _____ DATE _____ PERIOD _____

14

Chapter 14 Quiz

SCORE _____

(Lesson 14–7)

1. Find all solutions for $\sin\theta = \cos 2\theta$ if $0° \le \theta < 360°$.

2. Find all solutions for $4\cos^2\theta = 1$ if $0 \le \theta < 2\pi$.

3. Solve $\cos 2\theta = \cos\theta$ for all values of θ if θ is measured in degrees.

4. Solve $\cos 2\theta = 3\sin\theta - 1$ for all values of θ if θ is measured in radians.

5. **LIGHT** The length of the shadow s cast by a 40-foot tree depends on the angle of inclination of the sun, θ. Express s as a function of θ. Then find the angle of inclination that produces a shadow 30 feet long.

NAME _____ DATE _____ PERIOD _____

14

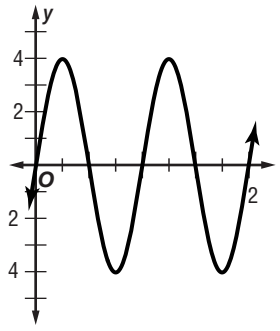
Chapter 14 Mid-Chapter Test

SCORE _____

(Lessons 14–1 through 14–4)

Part I ***For Questions 1–5, write the letter for the correct answer in the blank at the right of each question.***

Use the graph shown at the right.



1. Find the period of the function.

A. 4 **B.** 2π

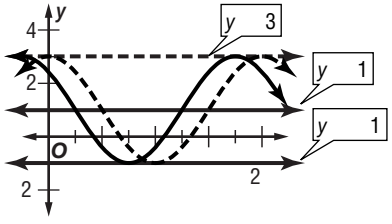
C. π **D.** 2 **1.** _____

2. Find the amplitude of the function.

A. 4 **B.** 8

C. π **D.** $\frac{\pi}{4}$ **2.** _____

For Questions 3 and 4, use the graph shown at the right.



3. Find the phase shift of the function.

A. $\frac{\pi}{4}$ **B.** $-\frac{\pi}{4}$

C. 1 **D.** 2 **3.** _____

4. Find the vertical shift of the function.

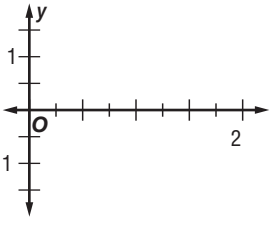
A. 1 **B.** 2 **C.** $\frac{\pi}{4}$ **D.** $-\frac{\pi}{4}$ **4.** _____

5. Which expression is equivalent to $\frac{1 + \sin^2 \theta \sec^2 \theta}{\sec^2 \theta} - \cos^2 \theta$?

A. 1 **B.** $\csc^2 \theta$ **C.** $\sin^2 \theta$ **D.** $2 \cos^2 \theta$ **5.** _____

Part II

6. Graph the function $y = \frac{1}{2} \cos 4\theta$. **6.**



7. Find the amplitude, if it exists, and period of the function $y = 2 \tan 4\theta$. **7.** _____

8. Find $\sin \theta$ if $\cos \theta = \frac{3}{4}$ and $0° \leq \theta < 90°$. **8.** _____

9. Simplify $\frac{\cos^2 \theta + \sin^2 \theta}{\sec \theta}$. **9.** _____

10. Simplify $\frac{\cot \theta \sec \theta}{\csc \theta}$. **10.** _____

11. Verify that $\frac{\csc^2 \theta - \cot^2 \theta}{\cot \theta} = \tan \theta$ is an identity. **11.** _____

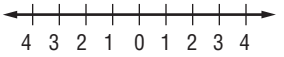
Assessment

NAME _____ DATE _____ PERIOD _____

14 Chapter 14 Cumulative Review

(Chapters 1–14)

1. Solve $5 + |2x - 1| \leq 10$ and graph its solution set. (Lesson 1-6) **1.** _____



2. Use long division to find $(x^3 - 4x^2 + 12x - 25) \div (x - 1)$. (Lesson 5-3) **2.** _____

3. Write $13n^4 = 52n^2$ in quadratic form, if possible. Then solve. (Lesson 7-3) **3.** _____

4. Express $\log_8 20$ in terms of common logarithms. Then approximate its value to four decimal places. (Lesson 10-4) **4.** _____

5. Find a_1 in a geometric series for which $S_n = 315$, $r = 2$, and $a_n = 168$. (Lesson 11-4) **5.** _____

6. From a group of 5 students and 3 faculty members, a committee of 3 is selected. Find the probability that all 3 are students or all 3 are faculty. (Lesson 12-5) **6.** _____

7. Six coins are tossed. Find P(at least 4 tails). (Lesson 12-8) **7.** _____

8. Find one angle with positive measure and one angle with negative measure coterminal with $-\frac{7\pi}{11}$. (Lesson 13-2) **8.** _____

9. Find the exact value of $\sin 120°$. (Lesson 13-3) **9.** _____

10. $P\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ is located on the unit circle. Find $\sin \theta$ and $\cos \theta$. (Lesson 13-6) **10.** _____

11. Find the amplitude, if it exists, and period of the function $y = 2 \cos \frac{1}{3}\theta$. (Lesson 14-1) **11.** _____

12. Find $\tan \theta$ if $\cos \theta = \frac{12}{13}$ and $270° < \theta < 360°$. (Lesson 14-3) **12.** _____

13. Find the exact value of $\sin \frac{\theta}{2}$ if $\sin \theta = -\frac{3}{7}$ and $180° < \theta < 270°$. (Lesson 14-6) **13.** _____

14. Solve $\cos^2 \theta \sin \theta = \sin \theta$ for all values of θ if θ is measured in radians. (Lesson 14-7) **14.** _____

NAME _____ DATE _____ PERIOD _____

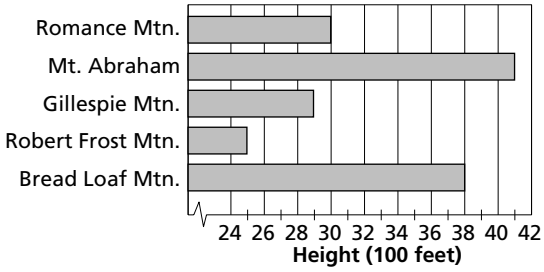
14 Standardized Test Practice

(Chapters 1–14)

Part 1: Multiple Choice

Instructions: Fill in the appropriate oval for the best answer.

For Questions 1 and 2, use the bar graph that shows the height, to the nearest hundred feet, of five mountains in Vermont's Green Mountain National Forest.



1. What is the difference in height between the highest and lowest of the given mountains?

A. 16 ft **B.** 160 ft
C. 1600 ft **D.** 16,000 ft

1. Ⓐ Ⓑ Ⓒ Ⓓ

2. What is the mean height of the given mountains?

E. 3200 ft **F.** 32.6 ft **G.** 3260 ft **H.** 320 ft

2. Ⓔ Ⓕ Ⓖ Ⓗ

3. If $\frac{x}{y} = 10$ and $yz = 12$, then $xz =$ _____.

A. $\frac{5}{6}$ **B.** $\frac{6}{5}$ **C.** 22 **D.** 120

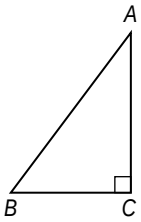
3. Ⓐ Ⓑ Ⓒ Ⓓ

4. A tank that holds 500 gallons of water is filled at a rate of 4.5 gallons per minute. How long, to the nearest minute, will it take the tank to fill if it already contains 325 gallons of water?

E. 788 min **F.** 39 min **G.** 111 min **H.** 4 min

4. Ⓔ Ⓕ Ⓖ Ⓗ

5. In the figure, the ratio of AC to CB is 12:5. If the area of triangle ABC is 120 cm^2, then $AB =$ _____.



A. 26 cm **B.** 10 cm
C. 104 cm **D.** 24 cm

5. Ⓐ Ⓑ Ⓒ Ⓓ

6. Line ℓ passes through the points $(3, -5)$ and $(-2, 10)$. Which point does *not* lie on line ℓ?

E. $(0, 4)$ **F.** $(-3, 13)$ **G.** $\left(\frac{1}{3}, 1\right)$ **H.** $(1, 1)$

6. Ⓔ Ⓕ Ⓖ Ⓗ

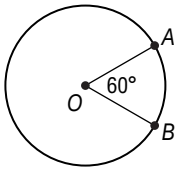
7. The number 5610 is divisible by which of the following?

I. 3 II. 6 III. 15

A. I only **B.** I and II only
C. I and III only **D.** I, II, and III

7. Ⓐ Ⓑ Ⓒ Ⓓ

8. In the figure, the length of arc AB is 8π. What is the length of a radius of circle O?



E. 24 **F.** 48
G. $2\sqrt{6}$ **H.** 24π

8. Ⓔ Ⓕ Ⓖ Ⓗ

Assessment

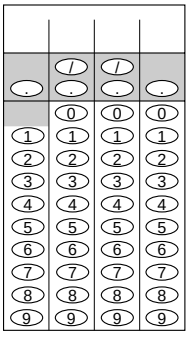
14 Standardized Test Practice *(continued)*

Part 2: Grid In

Instructions: Enter your answer by writing each digit of the answer in a column box and then shading in the appropriate oval that corresponds to that entry.

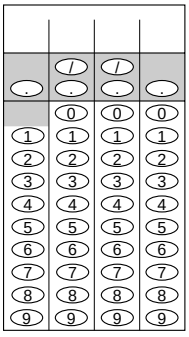
9. The probability of randomly selecting a white marble from a bag is $\frac{1}{10}$. The probability of randomly selecting a red marble is $\frac{3}{5}$. If the bag also contains 9 blue marbles, what is the total number of marbles in the bag?

9.

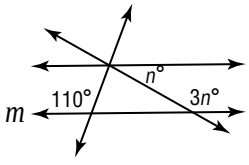


10. If the mean of x, $x + 2$, $3x - 2$, $x + 7$, $2x - 1$, $2x + 1$, and $x + 3$ is 14, what is the mode?

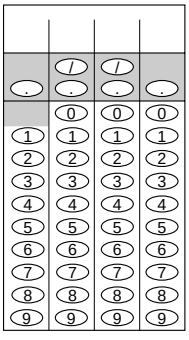
10.



11. Find the value of n in the figure if $\ell \parallel m$.

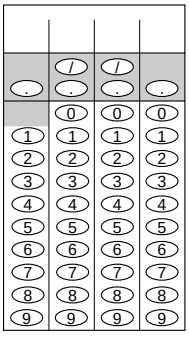


11.



12. Catherine purchased a hammer for \$12, a rake for \$17, and a shovel for \$26 at a local hardware store. If the state sales tax rate is 6%, how much change did Catherine receive from the \$60 she gave to the cashier?

12.

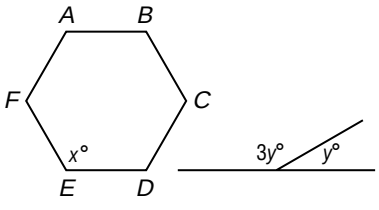


Part 3: Quantitative Comparison

Instructions: Compare the quantities in columns A and B. Shade in
(A) if the quantity in column A is greater;
(B) if the quantity in column B is greater;
(C) if the quantities are equal; or
(D) if the relationship cannot be determined from the information given.

	Column A	Column B	
13.	The 8th term of the sequence 16, 32, 48, 64, …	The 8th term of the sequence −1, 2, −4, 8, …	**13.** (A) (B) (C) (D)
14.	a where $3^{2a+1} > 81$	1.25	**14.** (A) (B) (C) (D)
15.	x	y	**15.** (A) (B) (C) (D)

15. Regular hexagon *ABCDEF*



NAME _____ DATE _____ PERIOD _____

Unit 5 Test

(Chapters 13–14)

SCORE _____

1. Solve $\triangle ABC$ if $C = 90°$, $B = 20°$, and $b = 10$. Round measures of sides to the nearest tenth and measures of angles to the nearest degree. **1.** _____

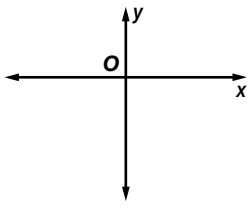
2. Rewrite $-25°$ in radian measure. **2.** _____

3. Rewrite $\frac{9\pi}{5}$ radians in degree measure. **3.** _____

4. Find one angle with positive measure and one angle with negative measure coterminal with $-310°$. **4.** _____

5. Find the exact values of the six trigonometric functions of θ if the terminal side of θ in standard position contains the point $(-5, -4)$. **5.** _____

6. Sketch the angle with measure $-\frac{2\pi}{3}$ radians. Then label its reference angle. **6.**



For Questions 7–10, find the exact value of each trigonometric function.

7. $\cot\left(-\frac{\pi}{6}\right)$ **8.** $\sin 405°$ **7.** _____

9. $\tan(-3\pi)$ **10.** $\sin 60° - \cos 60°$ **8.** _____

11. Find the area of $\triangle ABC$ if $A = 56°$, $b = 20$ feet, and $c = 12$ feet. Round to the nearest tenth. **9.** _____

12. In $\triangle ABC$, $A = 35°$, $a = 43$, and $c = 20$. Determine whether $\triangle ABC$ has *no* solution, *one* solution or *two* solutions. Then solve the triangle. Round to the nearest tenth. **10.** _____ **11.** _____ **12.** _____

For Questions 13 and 14, determine whether each triangle should be solved by beginning with the Law of Sines or Law of Cosines. Then solve each triangle. Round to the nearest tenth.

13. $a = 16,\ b = 13,\ c = 10$ **13.** _____

14. $A = 56°,\ B = 38°,\ a = 13$ **14.** _____

15. $P\left(-\frac{15}{17}, -\frac{8}{17}\right)$ is located on the unit circle. Find $\sin\theta$ and $\cos\theta$. **15.** _____

16. Solve $x = \text{Arctan}(-\sqrt{3})$. **16.** _____

17. Verify that $\frac{\sec\theta}{\sin\theta} \cdot \frac{\cot\theta}{\csc\theta} = \csc\theta$ is an identity. **17.** _____

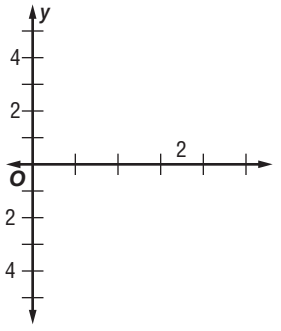
Assessment

Unit 5 Test *(continued)*

(Chapters 13–14)

18. Graph the function $y = 4 \sin 2\theta$.

18.



For Questions 19 and 20, find the amplitude, if it exists, and period of each function.

19. $y = \cos 3\theta$

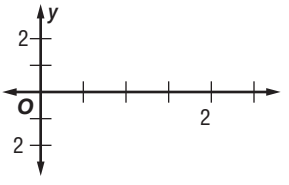
20. $y = \tan \frac{1}{4}\theta$

19. _____

20. _____

21. State the phase shift of $y = \cos\left(\theta - \frac{\pi}{3}\right)$. Then graph the function.

21. _____



22. State the vertical shift and the equation of the midline for $y = 4 \cos \theta - 1$.

22. _____

23. Find $\sec \theta$ if $\sin \theta = -\frac{4}{5}$ and $270° < \theta < 360$.

23. _____

24. Simplify $\left(\frac{1}{\sec \theta} + \frac{\sin^2 \theta}{\cos \theta}\right) \cos \theta$.

24. _____

For Questions 25 and 26, find the exact value of each expression.

25. $\cos 315°$

26. $\sin 195°$

25. _____

26. _____

27. Verify that $\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$ is an identity.

27. _____

For Questions 28 and 29, use the fact that $\cos \theta = \frac{1}{6}$ and $0° \le \theta < 90°$ to find the exact value of each expression.

28. $\sin 2\theta$

29. $\cos \frac{\theta}{2}$

28. _____

29. _____

30. The profit P for a product whose sales fluctuate with the seasons is estimated to be $P = 21 + 6 \sin \frac{\pi t}{52}$, where t is given in weeks and P is in thousands of dollars. Determine the number of weeks it would take for the profit to initially reach \$25,000.

30. _____

NAME _____ DATE _____ PERIOD _____

Second Semester Test

(Chapters 8–14)

SCORE _____

For Questions 1–20, write the letter for the correct answer in the blank at the right of each question.

1. Find the distance between (2, 5) and (−4, 1).

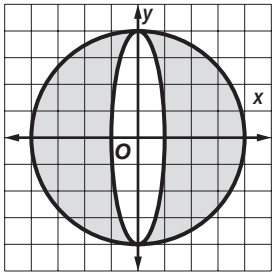
A. $\sqrt{34}$ **B.** $2\sqrt{13}$ **C.** $4\sqrt{2}$ **D.** $6\sqrt{2}$ **1.** _____

2. Write the equation $9x^2 = 4y^2 - 16y + 52$ in standard form.

A. $\frac{x^2}{4} - \frac{(y-2)^2}{9} = 1$ **B.** $\frac{x^2}{4} - \frac{(y+2)^2}{9} = 1$

C. $\frac{x^2}{4} + \frac{(y-2)^2}{9} = 1$ **D.** $\frac{x^2}{4} + \frac{(y+2)^2}{9} = 1$ **2.** _____

3. Which system of inequalities is graphed?



A. $x^2 + y^2 \le 16$, $x^2 + 16y^2 \ge 16$ **B.** $x^2 + y^2 \ge 16$, $16x^2 + y^2 \le 16$

C. $x^2 + y^2 \le 16$, $16x^2 + y^2 \ge 16$ **D.** $x^2 + y^2 \le 16$, $\frac{x^2}{16} + y^2 \ge 1$ **3.** _____

4. Simplify $\frac{5t^2 - 45}{4(t+3)^2} \cdot \frac{2t+6}{5t-15}$.

A. $\frac{(t-3)^2}{(t+3)^2}$ **B.** $\frac{t^2-9}{(t+3)(t-3)}$ **C.** $\frac{1}{2}$ **D.** $\frac{2}{5}$ **4.** _____

5. Determine the values of x for any holes in the graph of the rational function $f(x) = \frac{x-3}{x^2 + 2x - 15}$.

A. $x = -5, x = 3$ **B.** $x = -5$ **C.** $x = -3, x = 5$ **D.** $x = 3$ **5.** _____

6. Solve $\frac{1}{2m} - \frac{2}{5m} < \frac{1}{10}$.

A. $m < 0$ or $m > 1$ **B.** $m > 1$

C. $m < -1$ or $m > 0$ **D.** $m < -1$ or $m > 1$ **6.** _____

7. Solve $\log_{16} n = \frac{5}{4}$.

A. 32 **B.** 20 **C.** 8 **D.** 64 **7.** _____

8. Use $\log_5 2 \approx 0.4307$ and $\log_5 3 \approx 0.6826$ to approximate the value of $\log_5 24$.

A. 0.7625 **B.** 0.2760 **C.** 0.6812 **D.** 1.9747 **8.** _____

9. Write an equivalent logarithmic equation for $e^3 = 6x$.

A. $3 = 6 \ln x$ **B.** $3 = \ln 6x$ **C.** $6x = \ln 3$ **D.** $x = \ln 2$ **9.** _____

10. Evaluate $\sum_{k=7}^{12} (3k - 6)$.

A. 105 **B.** 165 **C.** 135 **D.** 162 **10.** _____

Assessment

Second Semester Test *(continued)*

(Chapters 8–14)

11. Find the next two terms of the geometric sequence 81, 54, 36, … .

A. 54, 81 **B.** 9, −18 **C.** 18, 0 **D.** 24, 16 **11.** _____

12. Find the fifth term of the sequence in which $a_1 = 12$ and $a_{n+1} = a_n + 2n$.

A. 24 **B.** 32 **C.** 42 **D.** 30 **12.** _____

13. A password has three letters followed by three digits. How many different passwords are possible?

A. 12,812,904 **B.** 13,824,000 **C.** 11,232,000 **D.** 17,576,000 **13.** _____

14. The odds that an event will occur are 5:3. What is the probability that the event will *not* occur?

A. $\frac{3}{8}$ **B.** $\frac{5}{8}$ **C.** $\frac{3}{5}$ **D.** $\frac{5}{2}$ **14.** _____

15. On a geometry test, $\frac{1}{5}$ of the students earned an A. Find the probability that 4 of 5 randomly-selected students earned an A.

A. $\frac{4}{3125}$ **B.** $\frac{4}{625}$ **C.** $\frac{1}{625}$ **D.** $\frac{1}{125}$ **15.** _____

16. In a survey of 550 residents, 42% favored the expansion of the town library. Find the margin of sampling error.

A. 8% **B.** 2% **C.** 4% **D.** 6% **16.** _____

17. In $\triangle ABC$, $a = 15$, $b = 25$, and $c = 30$. Find C.

A. 56° **B.** 30° **C.** 94° **D.** 98° **17.** _____

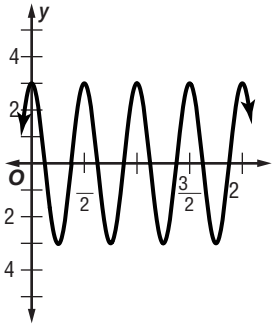
18. Find the exact value of $4(\cos 150°)(\tan 120°)$.

A. $\frac{\sqrt{3}}{3}$ **B.** $\sqrt{3}$ **C.** $2\sqrt{3}$ **D.** 6 **18.** _____

19. Which equation is graphed?

A. $y = 4 \cos 3\theta$ **B.** $y = 3 \cos 4\theta$

C. $y = 3 \sin 4\theta$ **D.** $y = 4 \sin 3\theta$



19. _____

20. Find $\csc \theta$ if $\cos \theta = -\frac{2}{7}$ and $90° < \theta < 180°$.

A. $\frac{7\sqrt{5}}{15}$ **B.** $\frac{3\sqrt{5}}{7}$ **C.** $-\frac{7\sqrt{5}}{15}$ **D.** $-\frac{3\sqrt{5}}{7}$ **20.** _____

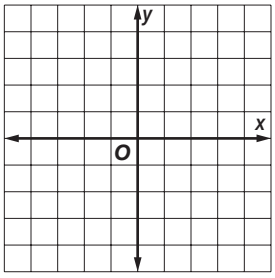
Second Semester Test *(continued)*

(Chapters 8–14)

21. Write an equation for the parabola with focus (2, 5) and directrix $y = 1$. **21.** _____

22. Write an equation for a circle with center at (10, −3) and radius $\frac{1}{5}$ unit. **22.** _____

23. Find the coordinates of the vertices and foci and the equations of the asymptotes for the hyperbola with equation $9y^2 - x^2 = 9$. Then graph the hyperbola. **23.** _____



24. Write the equation $x^2 + y^2 = -2x - 2y + 23$ in standard form. Then state whether the graph of the equation is a *parabola*, *circle*, *ellipse*, or *hyperbola*. **24.** _____

25. Find the LCM of $4t + 20$ and $6t + 30$. **25.** _____

26. State whether the equation $\frac{3.1}{p} = r$ represents a *direct*, *joint*, or *inverse* variation. Then name the constant of variation. **26.** _____

27. Solve $\frac{2}{t-3} = \frac{2t+1}{t^2+2t-15} + \frac{6}{t+5}$. **27.** _____

For Questions 28–30, solve each equation. **28.** _____

28. $\left(\frac{1}{25}\right)^m = 625^{m+2}$ **29.** $\ln(2x + 1) = 2$ **29.** _____

30. $4 \log_8 3 - \frac{1}{2} \log_8 9 = \log_8 x$ **30.** _____

31. Express $\log_7 32$ in terms of common logarithms. Then approximate its value to four decimal places. **31.** _____

32. The half-life of carbon-14 is 5760 years. A scientist unearthed a fossil whose bones contained only 2% as much carbon-14 as they would have contained when the animal was alive. Find the constant k for carbon-14 for t in years, and write the equation for modeling this exponential decay. Then determine how long ago the animal died. **32.** _____

33. Find the three arithmetic means between −2 and 10. **33.** _____

Assessment

Second Semester Test *(continued)*

(Chapters 8–14)

34. Find a_1 in a geometric series for which $S_n = 153$, $a_n = -3$, and $r = -\frac{1}{4}$. **34.** _____

35. Write $0.\overline{72}$ as a fraction. **35.** _____

36. Use Pascal's triangle to expand $(3x - y)^5$. **36.** _____

37. Find a counterexample to the statement $1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(5n - 1)}{4}$. **37.** _____

38. How many ways can you choose three books from a locker containing seven books? **38.** _____

39. Elias, Alisa, and Drew each roll a die. What is the probability that Elias rolls a 5, Alisa rolls an even number, and Drew does *not* roll a 1 or 2? **39.** _____

40. At a local gym with 800 members, 450 members take an aerobics class, 200 members do weight training, and 125 members do both weight training and take an aerobics class. What is the probability that a randomly-selected member takes an aerobics class or does weight training? **40.** _____

41. Determine whether the data {2, 1, 5, 9, 2, 3, 1, 7, 3, 2, 4, 8, 3, 6, 4, 3} appear to be *positively skewed*, *negatively skewed*, or *normally distributed*. **41.** _____

42. On a multiple-choice quiz with eight questions, each question has four answer choices. If Noreen randomly guesses at all eight questions, find P(more than 6 correct). **42.** _____

43. Find the exact values of the six trigonometric functions of θ if the terminal side of θ in standard position contains the point $(-8, -15)$. **43.** _____

44. Determine whether $\triangle ABC$ with $A = 35°$, $a = 20$, and $b = 13$ has *no* solution, *one* solution, or *two* solutions. Then, if possible, solve the triangle. Round measures of sides to the nearest tenth and measures of angles to the nearest degree. **44.** _____

45. Verify that $\frac{\sin\theta \cot\theta}{\csc\theta \tan\theta} = \cos^2\theta$ is an identity. **45.** _____

46. Find the exact value of $\cos 2\theta$ if $\sin\theta = -\frac{5}{6}$ and $180° < \theta < 270°$. **46.** _____

NAME _____ DATE _____ PERIOD _____

Final Test

(Chapters 1–14)

SCORE _____

For Questions 1–28, write the letter for the correct answer in the blank at the right of each question.

1. The five fastest roller coasters in the world are Fujiyama (Japan), Goliath (CA), Millennium Force (OH), Steel Dragon 2000 (Japan), and Superman the Escape (CA). The speeds, in miles per hour, of the first four coasters are 83, 85, 92, and 95, respectively. How fast can Superman the Escape travel if the average speed of all five coasters is no more than 91 miles per hour? **Source:** *World Almanac*

A. no more than 100 mph **B.** at least 93 mph
C. at least 100 mph **D.** no more than 93 mph **1.** _____

2. Write an equation of the line that passes through (9, 6) and is perpendicular to the line whose equation is $y = -\frac{1}{3}x + 7$.

A. $y = -\frac{1}{3}x + 9$ **B.** $y = -3x + 33$
C. $y = 3x - 21$ **D.** $y = \frac{1}{3}x + 3$ **2.** _____

3. Find x in the solution of the system $3x + y = 2$ and $2x - 3y = 16$.

A. 2 **B.** -4 **C.** $\frac{18}{11}$ **D.** $\frac{10}{11}$ **3.** _____

4. Find the coordinates of the vertices of the figure formed by $y \le x + 2$, $x + y \le 6$, and $y \ge -2$.

A. (0, 0), (2, 4), (8, −2) **B.** (−4, −2), (2, 4), (8, −2)
C. (−4, −2), (4, 2), (8, −2) **D.** (−2, −4), (2, 4), (8, −2) **4.** _____

5. Solve $\begin{bmatrix} 2x + 5y \\ x + 3y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ for y.

A. 1 **B.** 3 **C.** -3 **D.** -1 **5.** _____

6. The vertices of $\triangle ABC$ are $A(-3, -4)$, $B(-1, 3)$, and $C(3, -2)$. The triangle is rotated 90° counterclockwise. Use the rotation matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ to find the coordinates of C'.

A. (−3, 2) **B.** (4, −3) **C.** (−3, −1) **D.** (2, 3) **6.** _____

7. Simplify $\frac{y^2 + y - 20}{y^2 - 2y - 8}$. Assume that the denominator is not equal to 0.

A. $\frac{y - 5}{y - 2}$ **B.** $\frac{y + 5}{y + 2}$ **C.** $\frac{5}{2}$ **D.** $\frac{y - 10}{y - 4}$ **7.** _____

8. Simplify $\frac{1 + \boldsymbol{i}}{2 - \boldsymbol{i}}$.

A. $\frac{1}{3} + \frac{2}{3}\boldsymbol{i}$ **B.** $\frac{1}{5} + \frac{2}{5}\boldsymbol{i}$ **C.** $\frac{1}{3} + \boldsymbol{i}$ **D.** $\frac{1}{5} + \frac{3}{5}\boldsymbol{i}$ **8.** _____

Assessment

Final Test *(continued)*

(Chapters 1–14)

9. Solve $3x^2 - 8x + 4 = 0$ by graphing. If exact roots cannot be found, state the consecutive integers between which the roots are located.

A. 2; between 0 and 1 **B.** between 0 and 1; between 7 and 8
C. 1, 2 **D.** between 0 and 1; between 3 and 4 **9.** _____

10. Find the exact solutions to $6x^2 + 1 = -8x$ by using the Quadratic Formula.

A. $-4 \pm \sqrt{10}$ **B.** $\frac{-4 \pm \sqrt{22}}{6}$ **C.** $\frac{-2 \pm 2\sqrt{10}}{3}$ **D.** $\frac{-4 \pm \sqrt{10}}{6}$ **10.** _____

11. State the degree of $9 + 4x^2 - 6x^3 + x^4 - 7x$.

A. 9 **B.** 1 **C.** 4 **D.** 10 **11.** _____

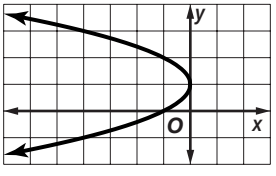
12. Which describes the number and type of roots of the equation $x^4 - 625 = 0$?

A. 1 real root, 1 imaginary root **B.** 2 real roots, 2 imaginary roots
C. 2 real roots **D.** 4 real roots **12.** _____

13. If $g(x) = 3x - 8$, find $g[g(-4)]$.

A. -68 **B.** 4 **C.** -20 **D.** 52 **13.** _____

14. Which equation is graphed?



A. $y = -x^2 + 2x - 1$
B. $x = -y^2 + 2y - 1$
C. $y = x^2 - 2x + 1$
D. $x = y^2 - 2y + 1$ **14.** _____

15. Write an equation for an ellipse if the endpoints of the major axis are at $(-8, 1)$ and $(8, 1)$ and the endpoints of the minor axis are at $(0, -1)$ and $(0, 3)$.

A. $\frac{x^2}{16} + \frac{(y-1)^2}{4} = 1$ **B.** $\frac{(x-1)^2}{64} + \frac{y^2}{4} = 1$
C. $\frac{(x-1)^2}{16} + \frac{y^2}{4} = 1$ **D.** $\frac{x^2}{64} + \frac{(y-1)^2}{4} = 1$ **15.** _____

16. Find the exact solution(s) of the system $\frac{x^2}{4} - y^2 = 1$ and $x = y^2 + 1$.

A. $(4, \sqrt{3}), (4, -\sqrt{3}), (-4, \sqrt{3}), (-4, -\sqrt{3})$
B. $(4, \sqrt{3}), (-4, \sqrt{3})$
C. $(2, 1), (2, -1), (4, \sqrt{3}), (4, -\sqrt{3})$
D. $(4, \sqrt{3}), (4, -\sqrt{3})$ **16.** _____

17. Simplify $\frac{4m}{5n^2} - \frac{n}{2m}$.

A. $\frac{8m^2 - 5n^3}{10n^2m}$ **B.** $\frac{8m^2 + 5n^3}{10n^2m}$ **C.** $\frac{4m - n}{5n^2 - 2m}$ **D.** $\frac{2}{5n^2}$ **17.** _____

Final Test *(continued)*

(Chapters 1–14)

18. If y varies inversely as x and $y = 6$ when $x = 3$, find y when $x = 36$.

A. 72 **B.** 2 **C.** $\frac{1}{2}$ **D.** 18 **18.** _____

19. Write the equation $4^{-3} = \frac{1}{64}$ in logarithmic form.

A. $\log 64 = 4^3$ **B.** $\log_{-3} 64 = 4$

C. $\log_4 \frac{1}{64} = -3$ **D.** $\log_4 (-3) = 64$ **19.** _____

20. Solve $6^{n+1} \geq 10$. Round to four decimal places.

A. $n \geq 0.2851$ **B.** $n \geq 0.6667$ **C.** $n \geq 1.2851$ **D.** $n \geq -0.7782$ **20.** _____

21. Find S_n for the arithmetic series in which $a_1 = 29$, $n = 17$, and $a_n = 131$.

A. 2720 **B.** 1360 **C.** 177 **D.** 160 **21.** _____

22. Find the sum of the infinite geometric series $1 + \frac{3}{5} + \frac{9}{25} + \ldots$, if it exists.

A. $\frac{5}{3}$ **B.** $\frac{5}{2}$ **C.** $\frac{3}{5}$ **D.** does not exist **22.** _____

23. Use the Binomial Theorem to find the sixth term in the expansion of $(m + 2p)^7$.

A. $21m^2p^5$ **B.** $672m^2p^5$ **C.** $32m^2p^5$ **D.** $448mp^6$ **23.** _____

24. How many four-digit numerical codes can be created if no digit may be repeated?

A. 10,000 **B.** 24 **C.** 3024 **D.** 5040 **24.** _____

25. A bookshelf holds 4 mysteries, 3 biographies, 1 book of poetry, and 2 reference books. If a book is selected at random from the shelf, find the probability that the book selected is a biography or reference book.

A. $\frac{1}{2}$ **B.** $\frac{1}{6}$ **C.** $\frac{5}{6}$ **D.** $\frac{3}{50}$ **25.** _____

26. Find the standard deviation of the data set to the nearest tenth. {21, 13, 18, 16, 13, 35, 12, 8, 15}

A. 16.8 **B.** 7.8 **C.** 7.3 **D.** 5.7 **26.** _____

27. Rewrite 100° in radian measure.

A. $\frac{5}{9}$ **B.** $\frac{5\pi}{9}$ **C.** $\frac{10}{9}$ **D.** $\frac{10\pi}{9}$ **27.** _____

28. Find the exact value of $\sin 165°$.

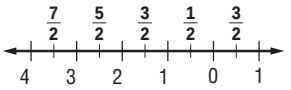
A. $\frac{\sqrt{6} - \sqrt{2}}{4}$ **B.** $\frac{\sqrt{6} + \sqrt{2}}{4}$ **C.** $\frac{\sqrt{2} - \sqrt{6}}{4}$ **D.** $\sqrt{\sqrt{6} - \sqrt{2}}$ **28.** _____

Assessment

Final Test *(continued)*

(Chapters 1–14)

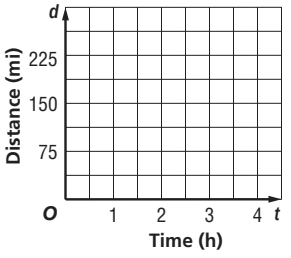
29. Solve $5\,|\,2a + 5\,| - 4 \le 6$ and graph the solution set. **29.** _____



For Questions 30 and 31, use the data in the table below that shows the relationship between the distance traveled and the elapsed time for a trip.

Time *t* (h)	0	1	2	3	4
Distance *d* (mi)	0	55	100	150	260

30. Draw a scatter plot for the data. **30.**



31. Use two ordered pairs to write a prediction equation. Then use your prediction equation to predict the distance traveled in an elapsed time of 6 hours. **31.** _____

32. Classify the system $x + 9y = 10$ and $2x - y = 1$ as *consistent and independent*, *consistent and dependent*, or *inconsistent*. **32.** _____

For Questions 33 and 34, use the following information.

A manufacturer produces badminton and tennis rackets. The profit on each badminton racket is \$10 and on each tennis racket is \$25. The manufacturer can make at most 600 rackets. Of these, at least 100 rackets must be badminton rackets.

33. Let b represent the number of badminton rackets and t represent the number of tennis rackets. Write a system of inequalities to represent the number of rackets that can be produced. **33.** _____

34. How many tennis rackets should the manufacturer produce to maximize profit? **34.** _____

35. Solve the system of equations. **35.** _____

$$\begin{aligned} 2x + y - 3z &= 9 \\ x - 2y + z &= -8 \\ x + 3y - 2z &= 11 \end{aligned}$$

36. Perform the indicated operations. If the matrix does not exist, write *impossible*. **36.** _____

$$\begin{bmatrix} 2 & -1 & 3 \\ 3 & 0 & -4 \end{bmatrix} \cdot \begin{bmatrix} -4 & 2 \\ 0 & -3 \\ -5 & 1 \end{bmatrix} - 4\begin{bmatrix} -5 & 1 \\ 2 & -1 \end{bmatrix}$$

37. Evaluate $\begin{vmatrix} 3 & 4 & 0 \\ 2 & 5 & -1 \\ 0 & 3 & -7 \end{vmatrix}$ using expansion by minors. **37.** _____

Final Test *(continued)*

(Chapters 1–14)

38. Find the inverse of $M = \begin{bmatrix} 1 & 5 \\ -2 & 0 \end{bmatrix}$, if it exists. **38.** _____

39. Simplify $\left[(3x^2y^0)^2 + \frac{1}{x^{-1}}\right](2x^2 + 5)$. Assume that no variable equals 0. **39.** _____

40. Simplify $\frac{5}{3 + \sqrt{6}}$. **40.** _____

41. Write the radical $\sqrt[3]{27t^8u^6}$ using rational exponents. **41.** _____

42. Solve $\sqrt{2x - 7} + 2 \geq 5$. **42.** _____

43. Write a quadratic equation with $\frac{2}{3}$ and -3 as its roots. Write the equation in the form $ax^2 + bx + c = 0$, where a, b, and c are integers. **43.** _____

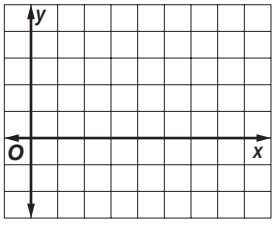
44. Write the equation $y = 4x^2 - 16x + 7$ in vertex form. **44.** _____

45. Use synthetic substitution to find $f(-4)$ for $f(x) = 2x^3 - 5x^2 + 3x - 8$. **45.** _____

46. List all of the possible rational zeros of $f(x) = 3x^4 - 5x^3 + 2x - 12$. **46.** _____

47. Find the inverse of the function $g(x) = 2x - 1$. **47.** _____

48. Graph $y < \sqrt{2x - 6}$. **48.**



49. Write an equation for a circle if the endpoints of a diameter are at $(-1, -5)$ and $(5, 3)$. **49.** _____

50. Write an equation for the hyperbola with vertices $(0, 4)$ and $(0, -4)$ if the length of the conjugate axis is 6 units. **50.** _____

51. Write the equation $y + 12x = 3x^2 + 19$ in standard form. Then state whether the graph of the equation is a *parabola*, *circle*, *ellipse*, or *hyperbola*. **51.** _____

Assessment

Final Test *(continued)*

(Chapters 1–14)

52. Simplify $\dfrac{\frac{f+g}{6}}{\frac{f^2-g^2}{2}}$. **52.** _____

53. Determine the equations of any vertical asymptotes and the values of x for any holes in the graph of $f(x) = \dfrac{x-3}{x^2+x-12}$. **53.** _____

54. Solve $\dfrac{m-4}{m+3} = \dfrac{m+4}{m-3} - \dfrac{2}{m+3}$. **54.** _____

55. Solve $\log_5 n = \frac{1}{4}\log_5 81 + \frac{1}{2}\log_5 64$. **55.** _____

56. In a certain lake, it is estimated that the fish population has been doubling in size every 80 weeks. Write an exponential growth equation of the form $y = ae^{kt}$ that models the growth of the fish population, where t is given in weeks, if the initial population was 5000. **56.** _____

57. Find the eighth term of the arithmetic sequence in which $a_1 = -4$ and $d = 7$. **57.** _____

58. Find the sum of the geometric series for which $a_1 = 2058$, $a_4 = 6$, and $r = \frac{1}{7}$. **58.** _____

59. Find the first three iterates x_1, x_2, x_3 of $f(x) = 7x - 3$ for an initial value $x_0 = 0$. **59.** _____

60. How many different ways can the letters of the word AMERICA be arranged? **60.** _____

61. Three students are selected from a group of four male students and six female students. Find the probability of selecting a male, a female, and another female in that order. **61.** _____

62. The heights of a group of high school students were found to be normally distributed. The mean height was 65 inches and the standard deviation was 2.5 inches. What percent of the students were between 65 inches and 70 inches tall? **62.** _____

63. In $\triangle ABC$, $A = 25°$, $a = 7$, and $b = 4$. Determine whether the triangle has *no* solution, *one* solution, or *two* solutions. Then solve the triangle. Round measure of sides to the nearest tenth and measures of angles to the nearest degree. **63.** _____

64. Find the value of $\cot\left(\text{Cos}^{-1}\dfrac{\sqrt{2}}{2}\right)$. **64.** _____

NAME _____ DATE _____ PERIOD _____

14 Standardized Test Practice

Student Record Sheet *(Use with pages 810–811 of the Student Edition.)*

Part 1 *Multiple Choice*

Select the best answer from the choices given and fill in the corresponding oval.

1 Ⓐ Ⓑ Ⓒ Ⓓ	**4** Ⓐ Ⓑ Ⓒ Ⓓ	**7** Ⓐ Ⓑ Ⓒ Ⓓ	**9** Ⓐ Ⓑ Ⓒ Ⓓ
2 Ⓐ Ⓑ Ⓒ Ⓓ	**5** Ⓐ Ⓑ Ⓒ Ⓓ	**8** Ⓐ Ⓑ Ⓒ Ⓓ	**10** Ⓐ Ⓑ Ⓒ Ⓓ
3 Ⓐ Ⓑ Ⓒ Ⓓ	**6** Ⓐ Ⓑ Ⓒ Ⓓ		

Part 2 *Short Response/Grid In*

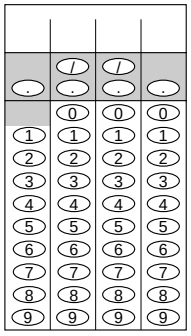
Solve the problem and write your answer in the blank.

For Questions 13–19, also enter your answer by writing each number or symbol in a box. Then fill in the corresponding oval for that number or symbol.

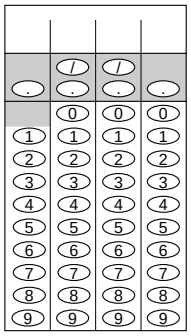
11 _____

12 _____

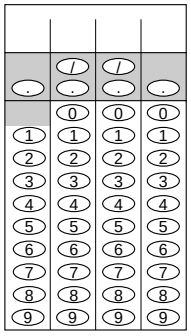
13 _____



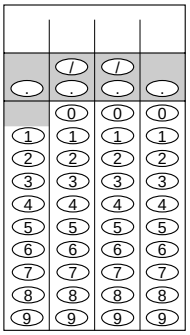
14 _____



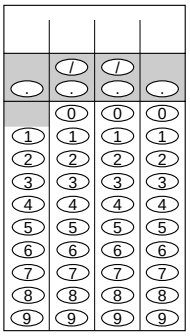
15 _____



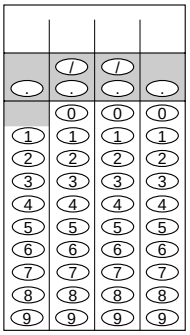
16 _____



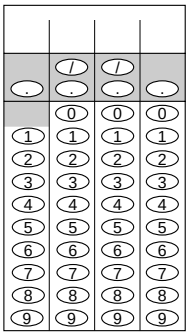
17 _____



18 _____



19 _____



Part 3 *Quantitative Comparison*

Select the best answer from the choices given and fill in the corresponding oval.

20 Ⓐ Ⓑ Ⓒ Ⓓ	**22** Ⓐ Ⓑ Ⓒ Ⓓ	**24** Ⓐ Ⓑ Ⓒ Ⓓ
21 Ⓐ Ⓑ Ⓒ Ⓓ	**23** Ⓐ Ⓑ Ⓒ Ⓓ	

Answers

NAME _____ DATE _____ PERIOD _____

14-1 Study Guide and Intervention

Graphing Trigonometric Functions

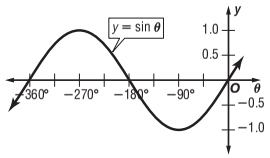
Graph Trigonometric Functions To graph a trigonometric function, make a table of values for known degree measures (0°, 30°, 45°, 60°, 90°, and so on). Round function values to the nearest tenth, and plot the points. Then connect the points with a smooth, continuous curve. The *period* of the sine, cosine, secant, and cosecant functions is 360° or 2π radians.

Amplitude of a Function	The **amplitude** of the graph of a periodic function is the absolute value of half the difference between its maximum and minimum values.

Example **Graph $y = \sin\theta$ for $-360° \le \theta \le 0°$.**

First make a table of values.

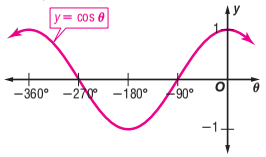
θ	−360°	−330°	−315°	−300°	−270°	−240°	−225°	−210°	−180°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
θ	−150°	−135°	−120°	−90°	−60°	−45°	−30°	0°	
$\sin\theta$	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	−1	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{1}{2}$	0	



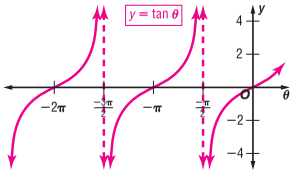
Exercises

Graph the following functions for the given domain.

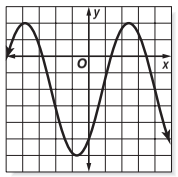
1. $\cos\theta$, $-360° \le \theta \le 0°$



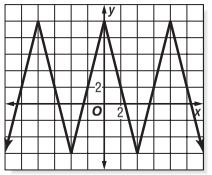
2. $\tan\theta$, $-2\pi \le \theta \le 0$



What is the amplitude of each function?

3. 

4

4. 

8

Lesson 14-1

NAME _____ DATE _____ PERIOD _____

14-1 Study Guide and Intervention *(continued)*

Graphing Trigonometric Functions

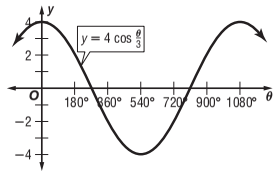
Variations of Trigonometric Functions

Amplitudes and Periods	For functions of the form $y = a \sin b\theta$ and $y = a \cos b\theta$, the amplitude is $\lvert a\rvert$, and the period is $\frac{360°}{\lvert b\rvert}$ or $\frac{2\pi}{\lvert b\rvert}$. For functions of the form $y = a \tan b\theta$, the amplitude is not defined, and the period is $\frac{180°}{\lvert b\rvert}$ or $\frac{\pi}{\lvert b\rvert}$.

Example **Find the amplitude and period of each function. Then graph the function.**

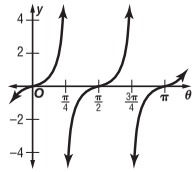
a. $y = 4\cos\frac{\theta}{3}$

First, find the amplitude.
$\lvert a\rvert = \lvert 4\rvert$, so the amplitude is 4.
Next find the period.
$\frac{360°}{\left\lvert\frac{1}{3}\right\rvert} = 1080°$
Use the amplitude and period to help graph the function.



b. $y = -\frac{1}{2}\tan 2\theta$

The amplitude is not defined, and the period is $\frac{\pi}{2}$.

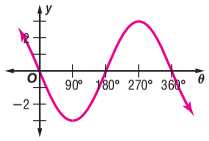


Exercises

Find the amplitude, if it exists, and period of each function. Then graph each function.

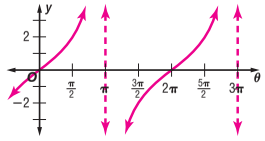
1. $y = -3\sin\theta$

amplitude: 3; period 2π or 360°



2. $y = 2\tan\frac{\theta}{2}$

no amplitude; period 2π or 360°



NAME _____ DATE _____ PERIOD ____

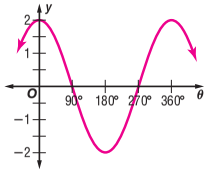
14-1 Skills Practice

Graphing Trigonometric Functions

Find the amplitude, if it exists, and period of each function. Then graph each function.

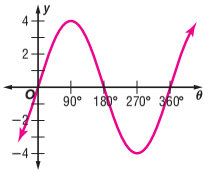
1. $y = 2 \cos \theta$

2; 360°



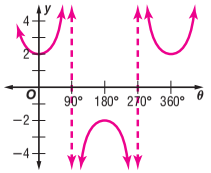
2. $y = 4 \sin \theta$

4; 360°



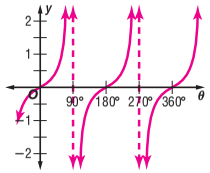
3. $y = 2 \sec \theta$

no amplitude; 360°



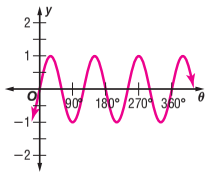
4. $y = \frac{1}{2} \tan \theta$

no amplitude; 180°



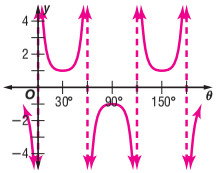
5. $y = \sin 3\theta$

1; 120°



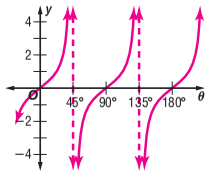
6. $y = \csc 3\theta$

no amplitude; 120°



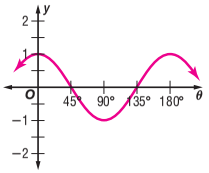
7. $y = \tan 2\theta$

no amplitude; 90°



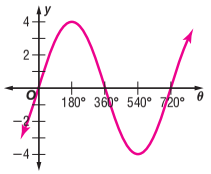
8. $y = \cos 2\theta$

1; 180°



9. $y = 4 \sin \frac{1}{2}\theta$

4; 720°



Lesson 14-1

NAME _____ DATE _____ PERIOD ____

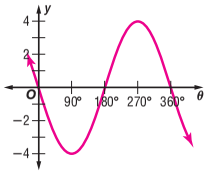
14-1 Practice (Average)

Graphing Trigonometric Functions

Find the amplitude, if it exists, and period of each function. Then graph each function.

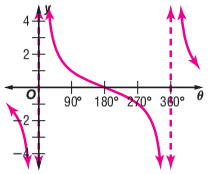
1. $y = -4 \sin \theta$

4; 360°



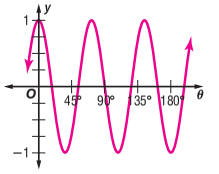
2. $y = \cot \frac{1}{2}\theta$

no amplitude; 360°



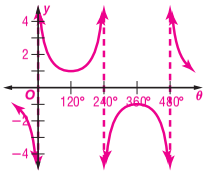
3. $y = \cos 5\theta$

1; 72°



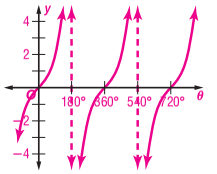
4. $y = \csc \frac{3}{4}\theta$

no amplitude; 480°

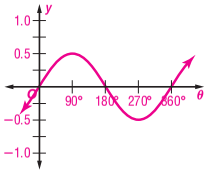


5. $y = 2 \tan \frac{1}{2}\theta$

no amplitude; 360°

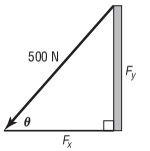


6. $2y = \sin \theta$ **$\frac{1}{2}$; 360°**



FORCE For Exercises 7 and 8, use the following information.

An anchoring cable exerts a force of 500 Newtons on a pole. The force has the horizontal and vertical components F_x and F_y. (A force of one Newton (N), is the force that gives an acceleration of 1 m/sec^2 to a mass of 1 kg.)



7. The function $F_x = 500 \cos \theta$ describes the relationship between the angle θ and the horizontal force. What are the amplitude and period of this function? **500; 360°**

8. The function $F_y = 500 \sin \theta$ describes the relationship between the angle θ and the vertical force. What are the amplitude and period of this function? **500; 360°**

WEATHER For Exercises 9 and 10, use the following information.

The function $y = 60 + 25 \sin \frac{\pi}{6}t$, where t is in months and $t = 0$ corresponds to April 15, models the average high temperature in degrees Fahrenheit in Centerville.

9. Determine the period of this function. What does this period represent?
12; a calendar year

10. What is the maximum high temperature and when does this occur? **85°F; July 15**

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14-1 Reading to Learn Mathematics

Graphing Trigonometric Functions

Pre-Activity **Why can you predict the behavior of tides?**

Read the introduction to Lesson 14-1 at the top of page 762 in your textbook.

Consider the tides of the Atlantic Ocean as a function of time. Approximately what is the period of this function? **12 hours**

Reading the Lesson

1. Determine whether each statement is *true* or *false*.

a. The period of a function is the distance between the maximum and minimum points. **false**

b. The amplitude of a function is the difference between its maximum and minimum values. **false**

c. The amplitude of the function $y = \sin \theta$ is 2π. **false**

d. The function $y = \cot \theta$ has no amplitude. **true**

e. The period of the function $y = \sec \theta$ is π. **false**

f. The amplitude of the function $y = 2 \cos \theta$ is 4. **false**

g. The function $y = \sin 2\theta$ has a period of π. **true**

h. The period of the function $y = \cot 3\theta$ is $\frac{\pi}{3}$. **true**

i. The amplitude of the function $y = -5 \sin \theta$ is -5. **false**

j. The period of the function $y = \csc \frac{1}{4}\theta$ is 4π. **false**

k. The graph of the function $y = \sin \theta$ has no asymptotes. **true**

l. The graph of the function $y = \tan \theta$ has an asymptote at $\theta = 180°$. **false**

m. When $\theta = 360°$, the values of $\cos \theta$ and $\sec \theta$ are equal. **true**

n. When $\theta = 270°$, $\cot \theta$ is undefined. **false**

o. When $\theta = 180°$, $\csc \theta$ is undefined. **true**

Helping You Remember

2. What is an easy way to remember the periods of $y = a \sin b\theta$ and $y = a \cos b\theta$? **Sample answer: The period of the functions $y = \sin \theta$ and $y = \cos \theta$ is 360° or 2π. Divide 360° or 2π by the absolute value of the coefficient of θ, depending on whether you want to find the period in degrees or in radians.**

Lesson 14-1

NAME _____ DATE _____ PERIOD ____

14-1 Enrichment

Blueprints

Interpreting blueprints requires the ability to select and use trigonometric functions and geometric properties. The figure below represents a plan for an improvement to a roof. The metal fitting shown makes a 30° angle with the horizontal. The vertices of the geometric shapes are *not* labeled in these plans. Relevant information must be selected and the appropriate function used to find the unknown measures.

Example **Find the unknown measures in the figure at the right.**

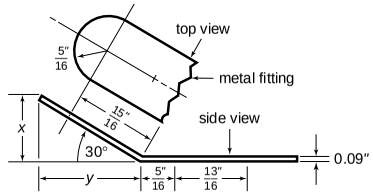
The measures x and y are the legs of a right triangle.

The measure of the hypotenuse is $\frac{15}{16}$ in. + $\frac{5}{16}$ in. or $\frac{20}{16}$ in.

$\frac{y}{\frac{20}{16}} = \cos 30°$ $\quad$ $\frac{x}{\frac{20}{16}} = \sin 30°$

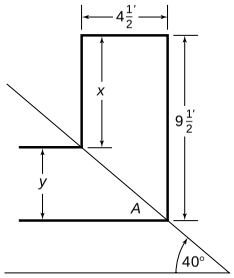
$y = 1.08$ in. $\quad$ $x = 0.63$ in.

Roofing Improvement



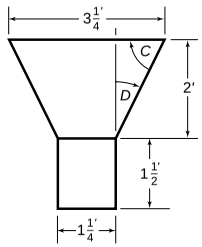
Find the unknown measures of each of the following.

1. Chimney on roof



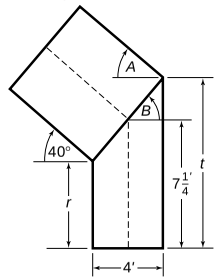
$y = 3.78'$
$x = 5.72'$
$\angle A = 40°$

2. Air vent



$\angle C = 63.43°$
$\angle D = 26.57°$

3. Elbow joint



$\angle A = 40°$
$\angle B = 50°$
$t = 9.63'$
$r = 4.87'$

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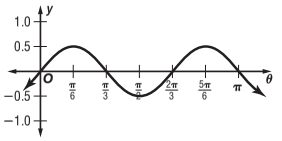
14-2 Study Guide and Intervention

Translations of Trigonometric Graphs

Horizontal Translations When a constant is subtracted from the angle measure in a trigonometric function, a **phase shift** of the graph results.

Phase Shift	The horizontal phase shift of the graphs of the functions $y = a \sin b(\theta - h)$, $y = a \cos b(\theta - h)$, and $y = a \tan b(\theta - h)$ is h, where $b > 0$. If $h > 0$, the shift is to the right. If $h < 0$, the shift is to the left.

Example **State the amplitude, period, and phase shift for $y = \frac{1}{2} \cos 3\left(\theta - \frac{\pi}{2}\right)$. Then graph the function.**



Amplitude: $a = \left|\frac{1}{2}\right|$ or $\frac{1}{2}$

Period: $\frac{2\pi}{|b|} = \frac{2\pi}{|3|}$ or $\frac{2\pi}{3}$

Phase Shift: $h = \frac{\pi}{2}$

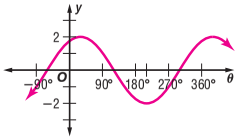
The phase shift is to the right since $\frac{\pi}{2} > 0$.

Exercises

State the amplitude, period, and phase shift for each function. Then graph the function.

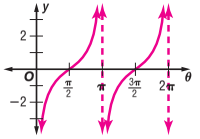
1. $y = 2 \sin (\theta + 60°)$

2; 360°; 60° to the left



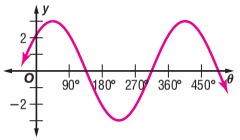
2. $y = \tan\left(\theta - \frac{\pi}{2}\right)$

no amplitude; π; $\frac{\pi}{2}$ to the right



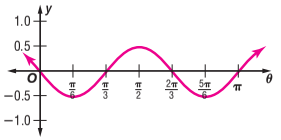
3. $y = 3 \cos (\theta - 45°)$

3; 360°; 45° to the right



4. $y = \frac{1}{2} \sin 3\left(\theta - \frac{\pi}{3}\right)$

$\frac{1}{2}$; $\frac{2\pi}{3}$; $\frac{\pi}{3}$ to the right



Lesson 14-2

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14-2 Study Guide and Intervention *(continued)*

Translations of Trigonometric Graphs

Vertical Translations When a constant is added to a trigonometric function, the graph is shifted vertically.

Vertical Shift	The vertical shift of the graphs of the functions $y = a \sin b(\theta - h) + k$, $y = a \cos b(\theta - h) + k$, and $y = a \tan b(\theta - h) + k$ is k. If $k > 0$, the shift is up. If $k < 0$, the shift is down.

The **midline** of a vertical shift is $y = k$.

Graphing Trigonometric Functions	**Step 1**	Determine the vertical shift, and graph the midline.
	Step 2	Determine the amplitude, if it exists. Use dashed lines to indicate the maximum and minimum values of the function.
	Step 3	Determine the period of the function and graph the appropriate function.
	Step 4	Determine the phase shift and translate the graph accordingly.

Example **State the vertical shift, equation of the midline, amplitude, and period for $y = \cos 2\theta - 3$. Then graph the function.**

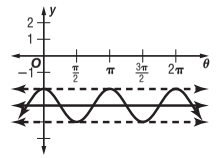
Vertical Shift: $k = -3$, so the vertical shift is 3 units down.

The equation of the midline is $y = -3$.

Amplitude: $|a| = |1|$ or 1

Period: $\frac{2\pi}{|b|} = \frac{2\pi}{|2|}$ or π

Since the amplitude of the function is 1, draw dashed lines parallel to the midline that are 1 unit above and below the midline. Then draw the cosine curve, adjusted to have a period of π.

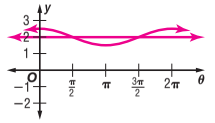


Exercises

State the vertical shift, equation of the midline, amplitude, and period for each function. Then graph the function.

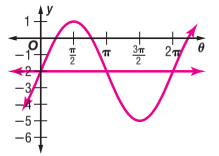
1. $y = \frac{1}{2} \cos \theta + 2$

2 up; $y = 2$; $\frac{1}{2}$; 2π



2. $y = 3 \sin \theta - 2$

2 down; $y = -2$; 3; 2π



Answers

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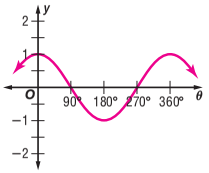
14-2 Skills Practice

Translations of Trigonometric Graphs

State the amplitude, period, and phase shift for each function. Then graph the function.

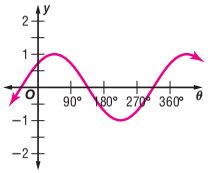
1. $y = \sin(\theta + 90°)$

1; 360°; −90°



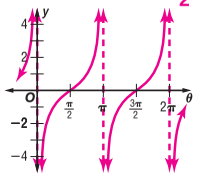
2. $y = \cos(\theta - 45°)$

1; 360°; 45°



3. $y = \tan\left(\theta - \frac{\pi}{2}\right)$

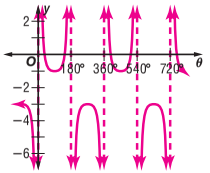
no amplitude; π; $\frac{\pi}{2}$



State the vertical shift, equation of the midline, amplitude, and period for each function. Then graph the function.

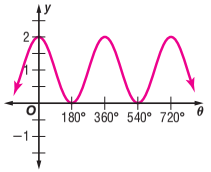
4. $y = \csc\theta - 2$

−2; $y = -2$; 1; 360°

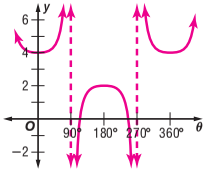


5. $y = \cos\theta + 1$

1; $y = 1$; 1; 360°



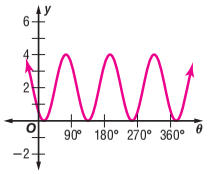
6. $y = \sec\theta + 3$ **3; $y = 3$; no amplitude; 360°**



State the vertical shift, amplitude, period, and phase shift of each function. Then graph the function.

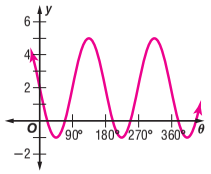
7. $y = 2\cos[3(\theta + 45°)] + 2$

2; 2; 120°; −45°



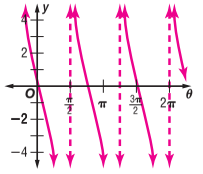
8. $y = 3\sin[2(\theta - 90°)] + 2$

2; 3; 180°; 90°



9. $y = 4\cot\left[\frac{4}{3}\left(\theta + \frac{\pi}{4}\right)\right] - 2$

−2; no amplitude; $\frac{3\pi}{4}$; $-\frac{\pi}{4}$



Lesson 14-2

NAME _____ DATE _____ PERIOD ____

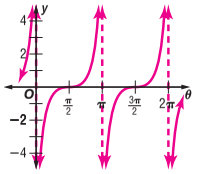
14-2 Practice (Average)

Translations of Trigonometric Graphs

State the vertical shift, amplitude, period, and phase shift for each function. Then graph the function.

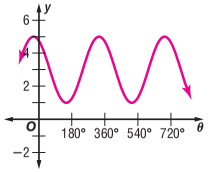
1. $y = \frac{1}{2}\tan\left(\theta - \frac{\pi}{2}\right)$

no vertical shift; no amplitude; π; $\frac{\pi}{2}$



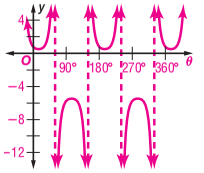
2. $y = 2\cos(\theta + 30°) + 3$

3; 2; 360; −30°



3. $y = 3\csc(2\theta + 60°) - 2.5$

−2.5; no amplitude; 180°; −60°



ECOLOGY For Exercises 4–6, use the following information.

The population of an insect species in a stand of trees follows the growth cycle of a particular tree species. The insect population can be modeled by the function $y = 40 + 30\sin 6t$, where t is the number of years since the stand was first cut in November, 1920.

4. How often does the insect population reach its maximum level? **every 60 yr**

5. When did the population last reach its maximum? **1995**

6. What condition in the stand do you think corresponds with a minimum insect population? **Sample answer: The species on which the insect feeds has been cut.**

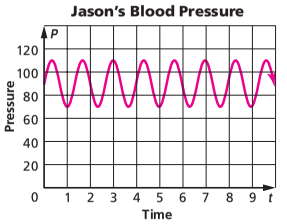
BLOOD PRESSURE For Exercises 7–9, use the following information.

Jason's blood pressure is 110 over 70, meaning that the pressure oscillates between a maximum of 110 and a minimum of 70. Jason's heart rate is 45 beats per minute. The function that represents Jason's blood pressure P can be modeled using a sine function with no phase shift.

7. Find the amplitude, midline, and period in seconds of the function. **20; $P = 90$; $1\frac{1}{3}$ s**

8. Write a function that represents Jason's blood pressure P after t seconds. **$P = 20\sin 270t + 90$**

9. Graph the function.



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14-2 Reading to Learn Mathematics

Translations of Trigonometric Graphs

Pre-Activity **How can translations of trigonometric graphs be used to show animal populations?**

Read the introduction to Lesson 14-2 at the top of page 769 in your textbook.

According to the model given in your textbook, what would be the estimated rabbit population for January 1, 2005? **1200**

Reading the Lesson

1. Determine whether the graph of each function represents a shift of the parent function *to the left*, *to the right*, *upward*, or *downward*. (Do not actually graph the functions.)

a. $y = \sin(\theta + 90°)$ **to the left**

b. $y = \sin\theta + 3$ **upward**

c. $y = \cos\left(\theta - \frac{\pi}{3}\right)$ **to the right**

d. $y = \tan\theta - 4$ **downward**

2. Determine whether the graph of each function has an *amplitude change*, *period change*, *phase shift*, or *vertical shift* compared to the graph of the parent function. (More than one of these may apply to each function. Do not actually graph the functions.)

a. $y = 3\sin\left(\theta + \frac{5\pi}{6}\right)$ **amplitude change and phase shift**

b. $y = \cos(2\theta + 70°)$ **period change and phase shift**

c. $y = -4\cos 3\theta$ **amplitude change and period change**

d. $y = \sec\frac{1}{2}\theta + 3$ **period change and vertical shift**

e. $y = \tan\left(\theta - \frac{\pi}{4}\right) - 1$ **phase shift and vertical shift**

f. $y = 2\sin\left(\frac{1}{3}\theta + \frac{\pi}{6}\right) - 4$ **amplitude change, period change, phase shift, and vertical shift**

Helping You Remember

3. Many students have trouble remembering which of the functions $y = \sin(\theta + \alpha)$ and $y = \sin(\theta - \alpha)$ represents a shift to the left and which represents a shift to the right. Using $\alpha = 45°$, explain a good way to remember which is which.

Sample answer: Although sine curves are infinitely repeating periodic graphs, think of $y = \sin x$ starting a period or cycle at (0, 0). Then $y = \sin(\theta + 45°)$ "starts early" at (−45°), a shift of 45° to the left, while $y = \sin(\theta - 45°)$ "starts late" at 45°, a shift of 45° to the right.

Lesson 14-2

NAME _____ DATE _____ PERIOD _____

14-2 Enrichment

Translating Graphs of Trigonometric Functions

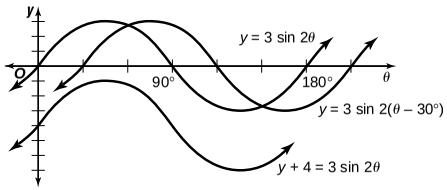
Three graphs are shown at the right:

$y = 3\sin 2\theta$

$y = 3\sin 2(\theta - 30°)$

$y + 4 = 3\sin 2\theta$

Replacing θ with $(\theta - 30°)$ translates the graph to the right. Replacing y with $y + 4$ translates the graph 4 units down.

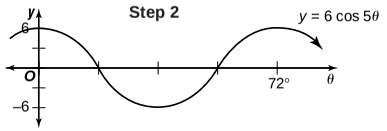


Example **Graph one cycle of $y = 6\cos(5\theta + 80°) + 2$.**

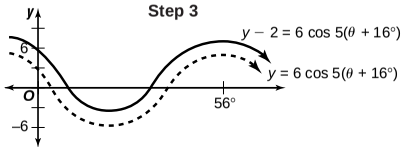
Step 1 Transform the equation into the form $y - k = a\cos b(\theta - h)$.

$y - 2 = 6\cos 5(\theta + 16°)$

Step 2 Sketch $y = 6\cos 5\theta$.



Step 3 Translate $y = 6\cos 5\theta$ to obtain the desired graph.



Sketch these graphs on the same coordinate system. **See students' graphs.**

1. $y = 3\sin 2(\theta + 45°)$ **2.** $y - 1 = 3\sin 2\theta$ **3.** $y + 5 = 3\sin 2(\theta + 90°)$

On another piece of paper, graph one cycle of each curve. **See students' graphs.**

4. $y = 2\sin 4(\theta - 50°)$ **5.** $y = 5\sin(3\theta + 90°)$

6. $y = 6\cos(4\theta + 360°) + 3$ **7.** $y = 6\cos 4\theta + 3$

8. The graphs for problems 6 and 7 should be the same. Use the sum formula for cosine of a sum to show that the equations are equivalent.

$$\begin{aligned}\cos(4\theta + 360°) &= (\cos 4\theta)(\cos 360°) - (\sin 4\theta)(\sin 360°)\\ &= (\cos 4\theta)(1) - (\sin 4\theta)(0)\\ &= \cos 4\theta\end{aligned}$$

So, $y = 6\cos(4\theta + 360°) + 3$ and $y = 6\cos 4\theta + 3$ are equivalent.

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14-3 Study Guide and Intervention

Trigonometric Identities

Find Trigonometric Values A **trigonometric identity** is an equation involving trigonometric functions that is true for all values for which every expression in the equation is defined.

Basic Trigonometric Identities	**Quotient Identities**	$\tan\theta = \frac{\sin\theta}{\cos\theta}$	$\cot\theta = \frac{\cos\theta}{\sin\theta}$	
	Reciprocal Identities	$\csc\theta = \frac{1}{\sin\theta}$	$\sec\theta = \frac{1}{\cos\theta}$	$\cot\theta = \frac{1}{\tan\theta}$
	Pythagorean Identities	$\cos^2\theta + \sin^2\theta = 1$	$\tan^2\theta + 1 = \sec^2\theta$	$\cot^2\theta + 1 = \csc^2\theta$

Example **Find the value of $\cot\theta$ if $\csc\theta = -\frac{11}{5}$; $180° < \theta < 270°$.**

$\cot^2\theta + 1 = \csc^2\theta$ — Trigonometric identity

$\cot^2\theta + 1 = \left(-\frac{11}{5}\right)^2$ — Substitute $-\frac{11}{5}$ for $\csc\theta$.

$\cot^2\theta + 1 = \frac{121}{25}$ — Square $-\frac{11}{5}$.

$\cot^2\theta = \frac{96}{25}$ — Subtract 1 from each side.

$\cot\theta = \pm\frac{4\sqrt{6}}{5}$ — Take the square root of each side.

Since θ is in the third quadrant, $\cot\theta$ is positive, Thus $\cot\theta = \frac{4\sqrt{6}}{5}$.

Exercises

Find the value of each expression.

1. $\tan\theta$, if $\cot\theta = 4$; $180° < \theta < 270°$ $\frac{1}{4}$
2. $\csc\theta$, if $\cos\theta = \frac{\sqrt{3}}{2}$; $0° \le \theta < 90°$ 2
3. $\cos\theta$, if $\sin\theta = \frac{3}{5}$; $0° \le \theta < 90°$ $\frac{4}{5}$
4. $\sec\theta$, if $\sin\theta = \frac{1}{3}$; $0° \le \theta < 90°$ $\frac{3\sqrt{2}}{4}$
5. $\cos\theta$, if $\tan\theta = -\frac{4}{3}$; $90° < \theta < 180°$ $-\frac{3}{5}$
6. $\tan\theta$, if $\sin\theta = \frac{3}{7}$; $0° \le \theta < 90°$ $\frac{3\sqrt{10}}{20}$
7. $\sec\theta$, if $\cos\theta = -\frac{7}{8}$; $90° < \theta < 180°$ $-\frac{8}{7}$
8. $\sin\theta$, if $\cos\theta = \frac{6}{7}$; $270° \le \theta < 360°$ $-\frac{\sqrt{13}}{7}$
9. $\cot\theta$, if $\csc\theta = \frac{12}{5}$; $90° < \theta < 180°$ $-\frac{\sqrt{119}}{5}$
10. $\sin\theta$, if $\csc\theta = -\frac{9}{4}$; $270° < \theta < 360°$ $-\frac{4}{9}$

Lesson 14-3

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14-3 Study Guide and Intervention *(continued)*

Trigonometric Identities

Simplify Expressions The simplified form of a trigonometric expression is written as a numerical value or in terms of a single trigonometric function, if possible. Any of the trigonometric identities on page 849 can be used to simplify expressions containing trigonometric functions.

Example 1 **Simplify $(1 - \cos^2\theta)\sec\theta\cot\theta + \tan\theta\sec\theta\cos^2\theta$.**

$$\begin{aligned}(1 - \cos^2\theta)\sec\theta\cot\theta + \tan\theta\sec\theta\cos^2\theta &= \sin^2\theta \cdot \frac{1}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta} \cdot \frac{1}{\cos\theta} \cdot \cos^2\theta \\ &= \sin\theta + \sin\theta \\ &= 2\sin\theta\end{aligned}$$

Example 2 **Simplify $\frac{\sec\theta \cdot \cot\theta}{1 - \sin\theta} - \frac{\csc\theta}{1 + \sin\theta}$.**

$$\begin{aligned}\frac{\sec\theta \cdot \cot\theta}{1 - \sin\theta} - \frac{\csc\theta}{1 + \sin\theta} &= \frac{\frac{1}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta}}{1 - \sin\theta} - \frac{\frac{1}{\sin\theta}}{1 + \sin\theta} \\ &= \frac{\frac{1}{\sin\theta}(1 + \sin\theta) - \frac{1}{\sin\theta}(1 - \sin\theta)}{(1 - \sin\theta)(1 + \sin\theta)} \\ &= \frac{\frac{1}{\sin\theta} + 1 - \frac{1}{\sin\theta} + 1}{1 - \sin^2\theta} \\ &= \frac{2}{\cos^2\theta}\end{aligned}$$

Exercises

Simplify each expression.

1. $\frac{\tan\theta \cdot \csc\theta}{\sec\theta}$ 1
2. $\frac{\sin\theta \cdot \cot\theta}{\sec^2\theta - \tan^2\theta}$ $\cos\theta$
3. $\frac{\sin^2\theta - \cot\theta \cdot \tan\theta}{\cot\theta \cdot \sin\theta}$ $-\cos\theta$
4. $\frac{\cos\theta}{\sec\theta - \tan\theta}$ $1 + \sin\theta$
5. $\frac{\tan\theta \cdot \cos\theta}{\sin\theta} + \cot\theta \cdot \sin\theta \cdot \tan\theta \cdot \csc\theta$ 2
6. $\frac{\csc^2\theta - \cot^2\theta}{\tan\theta \cdot \cos\theta}$ $\csc\theta$
7. $3\tan\theta \cdot \cot\theta + 4\sin\theta \cdot \csc\theta + 2\cos\theta \cdot \sec\theta$ 9
8. $\frac{1 - \cos^2\theta}{\tan\theta \cdot \sin\theta}$ $\cos\theta$

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14-3 Skills Practice

Trigonometric Identities

Find the value of each expression.

1. $\sin\theta$, if $\cos\theta = -\frac{4}{5}$ and $90° < \theta < 180°$ $\frac{3}{5}$

2. $\cos\theta$, if $\tan\theta = 1$ and $180° < \theta < 270°$ $-\frac{\sqrt{2}}{2}$

3. $\sec\theta$, if $\tan\theta = 1$ and $0° \le \theta < 90°$ $\sqrt{2}$

4. $\cos\theta$, if $\tan\theta = \frac{1}{2}$ and $0° \le \theta < 90°$ $\frac{2\sqrt{5}}{5}$

5. $\tan\theta$, if $\sin\theta = -\frac{\sqrt{2}}{2}$ and $180° < \theta < 270°$ 1

6. $\cos\theta$, if $\sec\theta = 2$ and $270° < \theta < 360°$ $\frac{1}{2}$

7. $\cos\theta$, if $\csc\theta = -2$ and $180° < \theta < 270°$ $-\frac{\sqrt{3}}{2}$

8. $\tan\theta$, if $\cos\theta = -\frac{2\sqrt{5}}{5}$ and $180° < \theta < 270°$ $\frac{1}{2}$

9. $\cos\theta$, if $\cot\theta = -\frac{3}{2}$ and $90° < \theta < 180°$ $-\frac{3\sqrt{13}}{13}$

10. $\csc\theta$, if $\cos\theta = \frac{8}{17}$ and $0° < \theta < 90°$ $\frac{17}{15}$

11. $\cot\theta$, if $\csc\theta = -2$ and $180° < \theta < 270°$ $\sqrt{3}$

12. $\tan\theta$, if $\sin\theta = -\frac{5}{13}$ and $180° < \theta < 270°$ $\frac{5}{12}$

Simplify each expression.

13. $\sin\theta\sec\theta$ $\tan\theta$

14. $\csc\theta\sin\theta$ 1

15. $\cot\theta\sec\theta$ $\csc\theta$

16. $\frac{\cos\theta}{\sec\theta}$ $\cos^2\theta$

17. $\tan\theta + \cot\theta$ $\frac{1}{\cos\theta\sin\theta}$

18. $\csc\theta\tan\theta - \tan\theta\sin\theta$ $\cos\theta$

19. $\frac{1-\sin^2\theta}{\sin\theta+1}$ $1 - \sin\theta$

20. $\csc\theta + \cot\theta$ $\frac{1+\cos\theta}{\sin\theta}$

21. $\frac{\sin^2\theta+\cos^2\theta}{1-\cos^2\theta}$ $\csc^2\theta$

22. $1 + \frac{\tan^2\theta}{1+\sec\theta}$ $\sec\theta$

Lesson 14-3

NAME _____ DATE _____ PERIOD ____

14-3 Practice (Average)

Trigonometric Identities

Find the value of each expression.

1. $\sin\theta$, if $\cos\theta = \frac{5}{13}$ and $0° \le \theta < 90°$ $\frac{12}{13}$

2. $\sec\theta$, if $\sin\theta = -\frac{15}{17}$ and $180° < \theta < 270°$ $-\frac{17}{8}$

3. $\cot\theta$, if $\cos\theta = \frac{3}{10}$ and $270° < \theta < 360°$ $-\frac{3\sqrt{91}}{91}$

4. $\sin\theta$, if $\cot\theta = \frac{1}{2}$ and $0° \le \theta < 90°$ $\frac{2\sqrt{5}}{5}$

5. $\cot\theta$, if $\csc\theta = -\frac{3}{2}$ and $180° < \theta < 270°$ $\frac{\sqrt{5}}{2}$

6. $\sec\theta$, if $\csc\theta = -8$ and $270° < \theta < 360°$ $\frac{8\sqrt{7}}{21}$

7. $\sec\theta$, if $\tan\theta = 4$ and $180° < \theta < 270°$ $-\sqrt{17}$

8. $\sin\theta$, if $\tan\theta = -\frac{1}{2}$ and $270° < \theta < 360°$ $-\frac{\sqrt{5}}{5}$

9. $\cot\theta$, if $\tan\theta = \frac{2}{5}$ and $0° \le \theta < 90°$ $\frac{5}{2}$

10. $\cot\theta$, if $\cos\theta = \frac{1}{3}$ and $270° < \theta < 360°$ $-\frac{\sqrt{2}}{4}$

Simplify each expression.

11. $\csc\theta\tan\theta$ $\sec x$

12. $\frac{\sin^2\theta}{\tan^2\theta}$ $\cos^2\theta$

13. $\sin^2\theta\cot^2\theta$ $\cos^2\theta$

14. $\cot^2\theta + 1$ $\csc^2\theta$

15. $\frac{\csc^2\theta - \cot^2\theta}{1-\cos^2\theta}$ $\csc^2\theta$

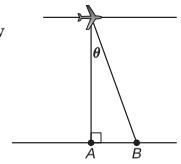
16. $\frac{\csc\theta - \sin\theta}{\cos\theta}$ $\cot\theta$

17. $\sin\theta + \cos\theta\cot\theta$ $\csc\theta$

18. $\frac{\cos\theta}{1-\sin\theta} - \frac{\cos\theta}{1+\sin\theta}$ $2\tan\theta$

19. $\sec^2\theta\cos^2\theta - \tan^2\theta$ $\sec^2\theta$

20. **AERIAL PHOTOGRAPHY** The illustration shows a plane taking an aerial photograph of point A. Because the point is directly below the plane, there is no distortion in the image. For any point B not directly below the plane, however, the increase in distance creates distortion in the photograph. This is because as the distance from the camera to the point being photographed increases, the exposure of the film reduces by $(\sin\theta)(\csc\theta - \sin\theta)$. Express $(\sin\theta)(\csc\theta - \sin\theta)$ in terms of $\cos\theta$ only. $\cos^2\theta$



21. **TSUNAMIS** The equation $y = a\sin\theta t$ represents the height of the waves passing a buoy at a time t in seconds. Express a in terms of $\csc\theta t$. $a = y\csc\theta t$

Answers

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14-3 Reading to Learn Mathematics

Trigonometric Identities

Pre-Activity **How can trigonometry be used to model the path of a baseball?**

Read the introduction to Lesson 14-3 at the top of page 777 in your textbook.

Suppose that a baseball is hit from home plate with an initial velocity of 58 feet per second at an angle of 36° with the horizontal from an initial height of 5 feet. Show the equation that you would use to find the height of the ball 10 seconds after the ball is hit. (Show the formula with the appropriate numbers substituted, but do not do any calculations.)

$h = \left(\frac{-16}{58^2 \cos^2 36°}\right) 10^2 + \left(\frac{\sin 36°}{\cos 36°}\right) 10 + 5$

Reading the Lesson

1. Match each expression from the list on the left with an expression from the list on the right that is equal to it for all values for which each expression is defined. (Some of the expressions from the list on the right may be used more than once or not at all.)

a. $\sec^2\theta - \tan^2\theta$ iii	**i.** $\frac{1}{\sin\theta}$
b. $\cot^2\theta + 1$ v	**ii.** $\tan\theta$
c. $\frac{\sin\theta}{\cos\theta}$ ii	**iii.** 1
d. $\sin^2\theta + \cos^2\theta$ iii	**iv.** $\sec\theta$
e. $\csc\theta$ i	**v.** $\csc^2\theta$
f. $\frac{1}{\cos\theta}$ iv	**vi.** $\cot\theta$
g. $\frac{\cos\theta}{\sin\theta}$ vi	

2. Write an identity that you could use to find each of the indicated trigonometric values and tell whether that value is positive or negative. (Do not actually find the values.)

a. $\tan\theta$, if $\sin\theta = -\frac{4}{5}$ and $180° < \theta < 270°$ $\tan\theta = \frac{\sin\theta}{\cos\theta}$; positive

b. $\sec\theta$, if $\tan\theta = -3$ and $90° < \theta < 180°$ $\tan^2\theta + 1 = \sec^2\theta$; negative

Helping You Remember

3. A good way to remember something new is to relate it to something you already know. How can you use the unit circle definitions of the sine and cosine that you learned in Chapter 13 to help you remember the Pythagorean identity $\cos^2\theta + \sin^2\theta = 1$?
Sample answer: On a unit circle, $x = \cos\theta$ and $y = \sin\theta$. The equation of the unit circle is $x^2 + y^2 = 1$, so this is equivalent to the equation $\cos^2\theta + \sin^2\theta = 1$.

Lesson 14-3

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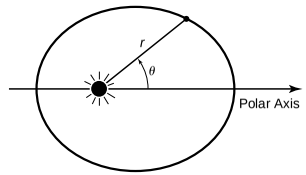
14-3 Enrichment

Planetary Orbits

The orbit of a planet around the sun is an ellipse with the sun at one focus. Let the pole of a polar coordinate system be that focus and the polar axis be toward the other focus. The polar equation of an ellipse is



$r = \frac{2ep}{1 - e\cos\theta}$. Since $2p = \frac{b^2}{c}$ and $b^2 = a^2 - c^2$,

$2p = \frac{a^2 - c^2}{c} = \frac{a^2}{c}\left(1 - \frac{c^2}{a^2}\right)$. Because $e = \frac{c}{a}$,

$2p = a\left(\frac{a}{c}\right)\left(1 - \left(\frac{c}{a}\right)^2\right) = a\left(\frac{1}{e}\right)(1 - e^2)$.

Therefore $2ep = a(1 - e^2)$. Substituting into the polar equation of an ellipse yields an equation that is useful for finding distances from the planet to the sun.

$r = \frac{a(1 - e^2)}{1 - e\cos\theta}$

Note that e is the eccentricity of the orbit and a is the length of the semi-major axis of the ellipse. Also, a is the mean distance of the planet from the sun.

Example **The mean distance of Venus from the sun is 67.24×10^6 miles and the eccentricity of its orbit is .006788. Find the minimum and maximum distances of Venus from the sun.**

The minimum distance occurs when $\theta = \pi$.

$r = \frac{67.24 \times 10^6(1 - 0.006788^2)}{1 - 0.006788\cos\pi} = 66.78 \times 10^6$ miles

The maximum distance occurs when $\theta = 0$.

$r = \frac{67.24 \times 10^6(1 - 0.006788^2)}{1 - 0.006788\cos 0} = 67.70 \times 10^6$ miles

Complete each of the following.

1. The mean distance of Mars from the sun is 141.64×10^6 miles and the eccentricity of its orbit is 0.093382. Find the minimum and maximum distances of Mars from the sun.
max. distance = 15.49×10^7 mi; min. distance = 12.84×10^7 mi

2. The minimum distance of Earth from the sun is 91.445×10^6 miles and the eccentricity of its orbit is 0.016734. Find the mean and maximum distances of Earth from the sun.
max. distance = 93.00×10^6 mi; mean distance = 91.47×10^6 mi

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14-4 Study Guide and Intervention

Verifying Trigonometric Identities

Transform One Side of an Equation Use the basic trigonometric identities along with the definitions of the trigonometric functions to verify trigonometric identities. Often it is easier to begin with the more complicated side of the equation and transform that expression into the form of the simpler side.

Example **Verify that each of the following is an identity.**

a. $\frac{\sin\theta}{\cot\theta} - \sec\theta = -\cos\theta$

Transform the left side.

$$\frac{\sin\theta}{\cot\theta} - \sec\theta \stackrel{?}{=} -\cos\theta$$
$$\frac{\sin\theta}{\frac{\cos\theta}{\sin\theta}} - \frac{1}{\cos\theta} \stackrel{?}{=} -\cos\theta$$
$$\frac{\sin^2\theta}{\cos\theta} - \frac{1}{\cos\theta} \stackrel{?}{=} -\cos\theta$$
$$\frac{\sin^2 - 1}{\cos\theta} \stackrel{?}{=} -\cos\theta$$
$$\frac{-\cos^2\theta}{\cos\theta} \stackrel{?}{=} -\cos\theta$$
$$-\cos\theta = -\cos\theta$$

b. $\frac{\tan\theta}{\csc\theta} + \cos\theta = \sec\theta$

Transform the left side.

$$\frac{\tan\theta}{\csc\theta} + \cos\theta \stackrel{?}{=} \sec\theta$$
$$\frac{\frac{\sin\theta}{\cos\theta}}{\frac{1}{\sin\theta}} + \cos\theta \stackrel{?}{=} \sec\theta$$
$$\frac{\sin^2\theta}{\cos\theta} + \cos\theta \stackrel{?}{=} \sec\theta$$
$$\frac{\sin^2\theta + \cos^2\theta}{\cos\theta} \stackrel{?}{=} \sec\theta$$
$$\frac{1}{\cos\theta} \stackrel{?}{=} \sec\theta$$
$$\sec\theta = \sec\theta$$

Exercises

Verify that each of the following is an identity.

1. $1 + \csc^2\theta \cdot \cos^2\theta = \csc^2\theta$

$$1 + \frac{1}{\sin^2\theta} \cdot \cos^2\theta \stackrel{?}{=} \csc^2\theta$$
$$\frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta} \stackrel{?}{=} \csc^2\theta$$
$$\frac{1}{\sin^2\theta} \stackrel{?}{=} \csc^2\theta$$
$$\csc^2\theta = \csc^2\theta$$

2. $\frac{\sin\theta}{1-\cos\theta} - \frac{\cot\theta}{1+\cos\theta} = \frac{1-\cos^3\theta}{\sin^3\theta}$

$$\frac{\sin\theta(1+\cos\theta) - \frac{\cos\theta}{\sin\theta}(1-\cos\theta)}{(1-\cos\theta)(1+\cos\theta)} \stackrel{?}{=} \frac{1-\cos^3\theta}{\sin^3\theta}$$
$$\frac{\sin\theta + \sin\theta\cdot\cos\theta - \frac{\cos\theta}{\sin\theta} + \frac{\cos^2\theta}{\sin\theta}}{1-\cos^2\theta} \stackrel{?}{=} \frac{1-\cos^3\theta}{\sin^3\theta}$$
$$\frac{\sin^2\theta + \sin^2\theta\cos\theta - \cos\theta + \cos^2\theta}{\sin\theta}{\sin^2\theta} \stackrel{?}{=} \frac{1-\cos^3\theta}{\sin^3\theta}$$
$$\frac{\sin^2\theta + \cos^2\theta + \cos\theta(\sin^2\theta - 1)}{\sin^3\theta} \stackrel{?}{=} \frac{1-\cos^3\theta}{\sin^3\theta}$$
$$\frac{1 + \cos\theta(-\cos^2\theta)}{\sin^3\theta} \stackrel{?}{=} \frac{1-\cos^3\theta}{\sin^3\theta}$$
$$\frac{1-\cos^3\theta}{\sin^3\theta} = \frac{1-\cos^3\theta}{\sin^3\theta}$$

Lesson 14-4

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14-4 Study Guide and Intervention *(continued)*

Verifying Trigonometric Identities

Transform Both Sides of an Equation The following techniques can be helpful in verifying trigonometric identities.

- Substitute one or more basic identities to simplify an expression.
- Factor or multiply to simplify an expression.
- Multiply both numerator and denominator by the same trigonometric expression.
- Write each side of the identity in terms of sine and cosine only. Then simplify each side.

Example **Verify that** $\frac{\tan^2\theta + 1}{\sin\theta\cdot\tan\theta\cdot\sec\theta + 1} = \sec^2\theta - \tan^2\theta$ **is an identity.**

$$\frac{\tan^2\theta + 1}{\sin\theta\cdot\tan\theta\cdot\sec\theta + 1} \stackrel{?}{=} \sec^2\theta - \tan^2\theta$$
$$\frac{\sec^2\theta}{\sin\theta\cdot\frac{\sin\theta}{\cos\theta}\cdot\frac{1}{\cos\theta} + 1} \stackrel{?}{=} \frac{1}{\cos^2\theta} - \frac{\sin^2\theta}{\cos^2\theta}$$
$$\frac{\frac{1}{\cos^2\theta}}{\frac{\sin^2\theta}{\cos^2\theta} + 1} \stackrel{?}{=} \frac{1-\sin^2\theta}{\cos^2\theta}$$
$$\frac{\frac{1}{\cos^2\theta}}{\frac{\sin^2\theta + \cos^2\theta}{\cos^2\theta}} \stackrel{?}{=} \frac{\cos^2\theta}{\cos^2\theta}$$
$$\frac{1}{\sin^2\theta + \cos^2\theta} \stackrel{?}{=} 1$$
$$1 = 1$$

Exercises

Verify that each of the following is an identity.

1. $\csc\theta\cdot\sec\theta = \cot\theta + \tan\theta$

$$\frac{1}{\sin\theta}\cdot\frac{1}{\cos\theta} \stackrel{?}{=} \frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}$$
$$\frac{1}{\sin\theta\cdot\cos\theta} \stackrel{?}{=} \frac{\cos^2\theta + \sin^2\theta}{\sin\theta\cdot\cos\theta}$$
$$\frac{1}{\sin\theta\cdot\cos\theta} = \frac{1}{\sin\theta\cdot\cos\theta}$$

2. $\frac{\tan^2\theta}{1-\cos^2\theta} = \frac{\sec\theta}{\cos\theta}$

$$\frac{\frac{\sin^2\theta}{\cos^2\theta}}{\sin^2\theta} \stackrel{?}{=} \frac{\frac{1}{\cos\theta}}{\cos\theta}$$
$$\frac{1}{\cos^2\theta} = \frac{1}{\cos^2\theta}$$

3. $\frac{\cos\theta\cdot\cot\theta}{\sin\theta} = \frac{\csc\theta}{\sin\theta\cdot\sec^2\theta}$

$$\frac{\cos\theta\cdot\frac{\cos\theta}{\sin\theta}}{\sin\theta} \stackrel{?}{=} \frac{\frac{1}{\sin\theta}}{\sin\theta\cdot\frac{1}{\cos^2\theta}}$$
$$\frac{\cos^2\theta}{\sin^2\theta} = \frac{\cos^2\theta}{\sin^2\theta}$$

4. $\frac{\csc^2\theta - \cot^2\theta}{\sec^2\theta} = \cot^2\theta(1-\cos^2\theta)$

$$\frac{\frac{1}{\sin^2\theta} - \frac{\cos^2\theta}{\sin^2\theta}}{\frac{1}{\cos^2\theta}} \stackrel{?}{=} \frac{\cos^2\theta}{\sin^2\theta}(\sin^2\theta)$$
$$\cos^2\theta\left(\frac{1-\cos^2\theta}{\sin^2\theta}\right) \stackrel{?}{=} \cos^2\theta$$
$$\cos^2\theta\left(\frac{\sin^2\theta}{\sin^2\theta}\right) \stackrel{?}{=} \cos^2\theta$$
$$\cos^2\theta = \cos^2\theta$$

NAME _____ DATE _____ PERIOD ____

14-4 Skills Practice

Verifying Trigonometric Identities

Verify that each of the following is an identity.

1. $\tan\theta\cos\theta = \sin\theta$

$\tan\theta\cos\theta \stackrel{?}{=} \sin\theta$

$\frac{\sin\theta}{\cos\theta}\cdot\cos\theta \stackrel{?}{=} \sin\theta$

$\sin\theta = \sin\theta$

2. $\cot\theta\tan\theta = 1$

$\cot\theta\tan\theta \stackrel{?}{=} 1$

$\frac{\cos\theta}{\sin\theta}\cdot\frac{\sin\theta}{\cos\theta} \stackrel{?}{=} 1$

$1 = 1$

3. $\csc\theta\cos\theta = \cot\theta$

$\csc\theta\cos\theta \stackrel{?}{=} \cot\theta$

$\frac{1}{\sin\theta}\cdot\cos\theta \stackrel{?}{=} \cot\theta$

$\frac{\cos\theta}{\sin\theta} \stackrel{?}{=} \cot\theta$

$\cot\theta = \cot\theta$

4. $\frac{1-\sin^2\theta}{\cos\theta} = \cos\theta$

$\frac{1-\sin^2\theta}{\cos\theta} \stackrel{?}{=} \cos\theta$

$\frac{\cos^2\theta}{\cos\theta} \stackrel{?}{=} \cos\theta$

$\cos\theta = \cos\theta$

5. $(\tan\theta)(1-\sin^2\theta) = \sin\theta\cos\theta$

$(\tan\theta)(1-\sin^2\theta) \stackrel{?}{=} \sin\theta\cos\theta$

$\tan\theta\cos^2\theta \stackrel{?}{=} \sin\theta\cos\theta$

$\frac{\sin\theta}{\cos\theta}\cdot\cos^2\theta \stackrel{?}{=} \sin\theta\cos\theta$

$\sin\theta\cos\theta = \sin\theta\cos\theta$

6. $\frac{\csc\theta}{\sec\theta} = \cot\theta$

$\frac{\csc\theta}{\sec\theta} \stackrel{?}{=} \cot\theta$

$\frac{\frac{1}{\sin\theta}}{\frac{1}{\cos\theta}} \stackrel{?}{=} \cot\theta$

$\frac{\cos\theta}{\sin\theta} \stackrel{?}{=} \cot\theta$

$\cot\theta = \cot\theta$

7. $\frac{\sin^2\theta}{1-\sin^2\theta} = \tan^2\theta$

$\frac{\sin^2\theta}{1-\sin^2\theta} \stackrel{?}{=} \tan^2\theta$

$\frac{\sin^2\theta}{\cos^2\theta} \stackrel{?}{=} \tan^2\theta$

$\left(\frac{\sin\theta}{\cos\theta}\right)^2 \stackrel{?}{=} \tan^2\theta$

$\tan^2\theta = \tan^2\theta$

8. $\frac{\cos^2\theta}{1-\sin\theta} = 1+\sin\theta$

$\frac{\cos^2\theta}{1-\sin\theta} \stackrel{?}{=} 1+\sin\theta$

$\frac{1-\sin^2\theta}{1-\sin\theta} \stackrel{?}{=} 1+\sin\theta$

$\frac{(1+\sin\theta)(1-\sin\theta)}{1-\sin\theta} \stackrel{?}{=} 1+\sin\theta$

$1+\sin\theta = 1+\sin\theta$

Lesson 14-4

NAME _____ DATE _____ PERIOD ____

14-4 Practice (Average)

Verifying Trigonometric Identities

Verify that each of the following is an identity.

1. $\frac{\sin^2\theta+\cos^2\theta}{\cos^2\theta} = \sec^2\theta$

$\frac{\sin^2\theta+\cos^2\theta}{\cos^2\theta} \stackrel{?}{=} \sec^2\theta$

$\frac{1}{\cos^2\theta} \stackrel{?}{=} \sec^2\theta$

$\sec^2\theta = \sec^2\theta$

2. $\frac{\cos^2\theta}{1-\sin^2\theta} = 1$

$\frac{\cos^2\theta}{1-\sin^2\theta} \stackrel{?}{=} 1$

$\frac{\cos^2\theta}{\cos^2\theta} \stackrel{?}{=} 1$

$1 = 1$

3. $(1+\sin\theta)(1-\sin\theta) = \cos^2\theta$

$(1+\sin\theta)(1-\sin\theta) \stackrel{?}{=} \cos^2\theta$

$1-\sin^2\theta \stackrel{?}{=} \cos^2\theta$

$\cos^2\theta = \cos^2\theta$

4. $\tan^4\theta + 2\tan^2\theta + 1 = \sec^4\theta$

$\tan^4\theta + 2\tan^2\theta + 1 \stackrel{?}{=} \sec^4\theta$

$(\tan^2\theta+1)^2 \stackrel{?}{=} \sec^4\theta$

$(\sec^2\theta)^2 \stackrel{?}{=} \sec^4\theta$

$\sec^4\theta = \sec^4\theta$

5. $\cos^2\theta\cot^2\theta = \cot^2\theta - \cos^2\theta$

$\cos^2\theta\cot^2\theta \stackrel{?}{=} \cot^2\theta - \cos^2\theta$

$\cos^2\theta\cot^2\theta \stackrel{?}{=} \frac{\cos^2\theta}{\sin^2\theta} - \cos^2\theta$

$\cos^2\theta\cot^2\theta \stackrel{?}{=} \frac{\cos^2\theta - \cos^2\theta\sin^2\theta}{\sin^2\theta}$

$\cos^2\theta\cot^2\theta \stackrel{?}{=} \frac{(\cos^2\theta)(1-\sin^2\theta)}{\sin^2\theta}$

$\cos^2\theta\cot^2\theta \stackrel{?}{=} \frac{\cos^2\theta}{1}\cdot\frac{\cos^2\theta}{\sin^2\theta}$

$\cos^2\theta\cot^2\theta = \cos^2\theta\cot^2\theta$

6. $(\sin^2\theta)(\csc^2\theta+\sec^2\theta) = \sec^2\theta$

$(\sin^2\theta)(\csc^2\theta+\sec^2\theta) \stackrel{?}{=} \sec^2\theta$

$(\sin^2\theta)\left(\frac{1}{\sin^2\theta}+\frac{1}{\cos^2\theta}\right) \stackrel{?}{=} \sec^2\theta$

$1+\frac{\sin^2\theta}{\cos^2\theta} \stackrel{?}{=} \sec^2\theta$

$1+\tan^2\theta \stackrel{?}{=} \sec^2\theta$

$\sec^2\theta = \sec^2\theta$

7. PROJECTILES The square of the initial velocity of an object launched from the ground is $v^2 = \frac{2gh}{\sin^2\theta}$, where θ is the angle between the ground and the initial path, h is the maximum height reached, and g is the acceleration due to gravity. Verify the identity $\frac{2gh}{\sin^2\theta} = \frac{2gh\sec^2\theta}{\sec^2\theta-1}$.

$$\frac{2gh}{\sin^2\theta} = \frac{2gh}{1-\cos^2\theta} = \frac{2gh}{1-\frac{1}{\sec^2\theta}} = \frac{2gh}{\frac{\sec^2\theta-1}{\sec^2\theta}} = \frac{2gh\sec^2\theta}{\sec^2\theta-1}$$

8. LIGHT The intensity of a light source measured in candles is given by $I = ER^2\sec\theta$, where E is the illuminance in foot candles on a surface, R is the distance in feet from the light source, and θ is the angle between the light beam and a line perpendicular to the surface. Verify the identity $ER^2(1+\tan^2\theta)\cos\theta = ER^2\sec\theta$.

$$ER^2(1+\tan^2\theta)\cos\theta = ER^2\sec^2\theta\cos\theta = ER^2\sec^2\theta\cdot\frac{1}{\sec\theta} = ER^2\sec\theta$$

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14-4 Reading to Learn Mathematics

Verifying Trigonometric Identities

Pre-Activity **How can you verify trigonometric identities?**

Read the introduction to Lesson 14-4 at the top of page 782 in your textbook.

For $\theta = -\pi$, 0, or π, $\sin\theta = \sin 2\theta$. Does this mean that $\sin\theta = \sin 2\theta$ is an identity? Explain your reasoning. **Sample answer: No; an identity is an equation that is true for *all* values of a variable for which the functions involved are defined, not just *some* values. If $\theta = \frac{\pi}{4}$, $\sin\theta = \frac{\sqrt{2}}{2}$, and $\sin 2\theta = 1$.**

Reading the Lesson

1. Determine whether each equation is an *identity* or *not an identity*.

a. $\frac{1}{\sin^2\theta} - \frac{1}{\tan^2\theta} = 1$ **identity**

b. $\frac{\cos\theta}{\sin\theta\tan\theta}$ **not an identity**

c. $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = \cos\theta\sin\theta$ **not an identity**

d. $\cos^2\theta(\tan^2\theta + 1) = 1$ **identity**

e. $\frac{\sin^2\theta}{\cos^2\theta} + \sin\theta\csc\theta = \sec^2\theta$ **identity**

f. $\frac{1}{1-\sin\theta} + \frac{1}{1+\sin\theta} = 2\cos^2\theta$ **not an identity**

g. $\tan^2\theta\cos^2\theta = \frac{1}{\csc^2\theta}$ **identity**

h. $\frac{\sin\theta}{\sec\theta} = \frac{1}{\tan\theta} + \frac{1}{\cot\theta}$ **not an identity**

2. Which of the following is *not* permitted when verifying an identity? **B**

A. simplifying one side of the identity to match the other side

B. cross multiplying if the identity is a proportion

C. simplifying each side of the identity separately to get the same expression on both sides

Helping You Remember

3. Many students have trouble knowing where to start in verifying a trigonometric identity. What is a simple rule that you can remember that you can always use if you don't see a quicker approach? **Sample answer: Write both sides in terms of sines and cosines. Then simplify each side as much as possible.**

Lesson 14-4

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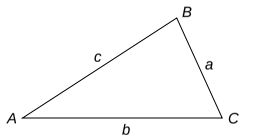
14-4 Enrichment

Heron's Formula

Heron's formula can be used to find the area of a triangle if you know the lengths of the three sides. Consider any triangle ABC. Let K represent the area of $\triangle ABC$. Then



$$K = \frac{1}{2}bc\sin A$$

$$K^2 = \frac{b^2c^2\sin^2 A}{4} \quad \text{Square both sides.}$$

$$= \frac{b^2c^2(1-\cos^2 A)}{4}$$

$$= \frac{b^2c^2(1+\cos A)(1-\cos A)}{4}$$

$$= \frac{b^2c^2}{4}\left(1 + \frac{b^2+c^2-a^2}{2bc}\right)\left(1 - \frac{b^2+c^2-a^2}{2bc}\right) \quad \text{Use the law of cosines.}$$

$$= \frac{b+c+a}{2}\cdot\frac{b+c-a}{2}\cdot\frac{a+b-c}{2}\cdot\frac{a-b+c}{2} \quad \text{Simplify.}$$

Let $s = \frac{a+b+c}{2}$. Then $s - a = \frac{b+c-a}{2}$, $s - b = \frac{a+c-b}{2}$, $s - c = \frac{a+b-c}{2}$.

$$K^2 = s(s-a)(s-b)(s-c) \quad \text{Substitute.}$$

$$K = \sqrt{s(s-a)(s-b)(s-c)}$$

Heron's Formula	The area of $\triangle ABC$ is $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$.

Use Heron's formula to find the area of $\triangle ABC$.

1. $a = 3, b = 4.4, c = 7$
4.1

2. $a = 8.2, b = 10.3, c = 9.5$
36.8

3. $a = 31.3, b = 92.0, c = 67.9$
782.9

4. $a = 0.54, b = 1.32, c = 0.78$
no such triangle

5. $a = 321, b = 178, c = 298$
26,160.9

6. $a = 0.05, b = 0.08, c = 0.04$
0.00082

7. $a = 21.5, b = 33.0, c = 41.7$
351.6

8. $a = 2.08, b = 9.13, c = 8.99$
9.3

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14-5 Study Guide and Intervention

Sum and Difference of Angles Formulas

Sum and Difference Formulas The following formulas are useful for evaluating an expression like sin 15° from the known values of sine and cosine of 60° and 45°.

Sum and Difference of Angles	The following identities hold true for all values of α and β. $\cos(\alpha \pm \beta) = \cos\alpha \cdot \cos\beta \mp \sin\alpha \cdot \sin\beta$ $\sin(\alpha \pm \beta) = \sin\alpha \cdot \cos\beta \pm \cos\alpha \cdot \sin\beta$

Example **Find the exact value of each expression.**

a. cos 345°

$$\begin{aligned}\cos 345° &= \cos(300° + 45°)\\ &= \cos 300° \cdot \cos 45° - \sin 300° \cdot \sin 45°\\ &= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{3}}{2}\right) \cdot \frac{\sqrt{2}}{2}\\ &= \frac{\sqrt{2} + \sqrt{6}}{4}\end{aligned}$$

b. sin (−105°)

$$\begin{aligned}\sin(-105°) &= \sin(45° - 150°)\\ &= \sin 45° \cdot \cos 150° - \cos 45° \cdot \sin 150°\\ &= \frac{\sqrt{2}}{2} \cdot \left(-\frac{\sqrt{3}}{2}\right) - \frac{\sqrt{2}}{2} \cdot \frac{1}{2}\\ &= -\frac{\sqrt{2} + \sqrt{6}}{4}\end{aligned}$$

Exercises

Find the exact value of each expression.

1. sin 105° $\frac{\sqrt{2} + \sqrt{6}}{4}$

2. cos 285° $\frac{\sqrt{6} - \sqrt{2}}{4}$

3. cos (−75°) $\frac{\sqrt{6} - \sqrt{2}}{4}$

4. cos (−165°) $-\frac{\sqrt{2} + \sqrt{6}}{4}$

5. sin 195° $\frac{\sqrt{2} - \sqrt{6}}{4}$

6. cos 420° $\frac{1}{2}$

7. sin (−75°) $-\frac{\sqrt{2} + \sqrt{6}}{4}$

8. cos 135° $-\frac{\sqrt{2}}{2}$

9. cos (−15°) $\frac{\sqrt{2} + \sqrt{6}}{4}$

10. sin 345° $\frac{\sqrt{2} - \sqrt{6}}{4}$

11. cos (−105°) $\frac{\sqrt{2} - \sqrt{6}}{4}$

12. sin 495° $\frac{\sqrt{2}}{2}$

Lesson 14-5

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14-5 Study Guide and Intervention *(continued)*

Sum and Difference of Angles Formulas

Verify Identities You can also use the sum and difference of angles formulas to verify identities.

Example 1 **Verify that $\cos\left(\theta + \frac{3\pi}{2}\right) = \sin\theta$ is an identity.**

$$\cos\left(\theta + \frac{3\pi}{2}\right) \stackrel{?}{=} \sin\theta \quad \text{Original equation}$$

$$\cos\theta \cdot \cos\frac{3\pi}{2} - \sin\theta \cdot \sin\frac{3\pi}{2} \stackrel{?}{=} \sin\theta \quad \text{Sum of Angles Formula}$$

$$\cos\theta \cdot 0 - \sin\theta \cdot (-1) \stackrel{?}{=} \sin\theta \quad \text{Evaluate each expression.}$$

$$\sin\theta = \sin\theta \quad \text{Simplify.}$$

Example 2 **Verify that $\sin\left(\theta - \frac{\pi}{2}\right) + \cos(\theta + \pi) = -2\cos\theta$ is an identity.**

$$\sin\left(\theta - \frac{\pi}{2}\right) + \cos(\theta + \pi) \stackrel{?}{=} -2\cos\theta \quad \text{Original equation}$$

$$\sin\theta \cdot \cos\frac{\pi}{2} - \cos\theta \cdot \sin\frac{\pi}{2} + \cos\theta \cdot \cos\pi - \sin\theta \cdot \sin\pi \stackrel{?}{=} -2\cos\theta \quad \text{Sum and Difference of Angles Formulas}$$

$$\sin\theta \cdot 0 - \cos\theta \cdot 1 + \cos\theta \cdot (-1) - \sin\theta \cdot 0 \stackrel{?}{=} -2\cos\theta \quad \text{Evaluate each expression.}$$

$$-2\cos\theta = -2\cos\theta \quad \text{Simplify.}$$

Exercises

Verify that each of the following is an identity.

1. $\sin(90° + \theta) = \cos\theta$

$$\sin 90° \cdot \cos\theta + \cos 90° \cdot \sin\theta \stackrel{?}{=} \cos\theta$$
$$1 \cdot \cos\theta + 0 \cdot \sin\theta \stackrel{?}{=} \cos\theta$$
$$\cos\theta = \cos\theta$$

2. $\cos(270° + \theta) = \sin\theta$

$$\cos 270° \cdot \cos\theta - \sin 270° \cdot \sin\theta \stackrel{?}{=} \sin\theta$$
$$0 \cdot \cos\theta - (-1) \cdot \sin\theta \stackrel{?}{=} \sin\theta$$
$$\sin\theta = \sin\theta$$

3. $\sin\left(\frac{2\pi}{3} - \theta\right) + \cos\left(\theta - \frac{5\pi}{6}\right) = \sin\theta$

$$\sin\frac{2\pi}{3} \cdot \cos\theta - \cos\frac{2\pi}{3} \cdot \sin\theta + \cos\theta \cdot \cos\frac{5\pi}{6} + \sin\theta \cdot \sin\frac{5\pi}{6} \stackrel{?}{=} \sin\theta$$
$$\frac{\sqrt{3}}{2} \cdot \cos\theta - \left(-\frac{1}{2}\right) \cdot \sin\theta + \cos\theta\left(-\frac{\sqrt{3}}{2}\right) + \sin\theta \cdot \frac{1}{2} \stackrel{?}{=} \sin\theta$$
$$\sin\theta = \sin\theta$$

4. $\cos\left(\frac{3\pi}{4} + \theta\right) - \sin\left(\theta - \frac{\pi}{4}\right) = -\sqrt{2}\sin\theta$

$$\cos\frac{3\pi}{4} \cdot \cos\theta - \sin\frac{3\pi}{4} \cdot \sin\theta - \left(\sin\theta \cdot \cos\frac{\pi}{4} - \cos\theta \cdot \sin\frac{\pi}{4}\right) \stackrel{?}{=} -\sqrt{2}\sin\theta$$
$$\left(-\frac{\sqrt{2}}{2}\right) \cdot \cos\theta - \frac{\sqrt{2}}{2} \cdot \sin\theta - \left(\sin\theta \cdot \frac{\sqrt{2}}{2} - \cos\theta \cdot \frac{\sqrt{2}}{2}\right) \stackrel{?}{=} -\sqrt{2}\sin\theta$$
$$-\sqrt{2}\sin\theta = -\sqrt{2}\sin\theta$$

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14-5 Skills Practice

Sum and Difference of Angles Formulas

Find the exact value of each expression.

1. $\sin 330°$ $-\frac{1}{2}$

2. $\cos(-165°)$ $\frac{-\sqrt{6}-\sqrt{2}}{4}$

3. $\sin(-225°)$ $\frac{\sqrt{2}}{2}$

4. $\cos 135°$ $-\frac{\sqrt{2}}{2}$

5. $\sin(-45°)$ $-\frac{\sqrt{2}}{2}$

6. $\cos 210°$ $-\frac{\sqrt{3}}{2}$

7. $\cos(-135°)$ $-\frac{\sqrt{2}}{2}$

8. $\sin 75°$ $\frac{\sqrt{6}+\sqrt{2}}{4}$

9. $\sin(-195°)$ $\frac{\sqrt{6}-\sqrt{2}}{4}$

Verify that each of the following is an identity.

10. $\sin(90° + \theta) = \cos\theta$

$$\sin(90° + \theta) \stackrel{?}{=} \cos\theta$$
$$\sin 90° \cos\theta + \cos 90° \sin\theta \stackrel{?}{=} \cos\theta$$
$$1\cos\theta + 0\sin\theta \stackrel{?}{=} \cos\theta$$
$$\cos\theta = \cos\theta$$

11. $\sin(180° + \theta) = -\sin\theta$

$$\sin(180° + \theta) \stackrel{?}{=} -\sin\theta$$
$$\sin 180° \cos\theta + \cos 180° \sin\theta \stackrel{?}{=} -\sin\theta$$
$$0\cos\theta + (-1)\sin\theta \stackrel{?}{=} -\sin\theta$$
$$-\sin\theta = -\sin\theta$$

12. $\cos(270° - \theta) = -\sin\theta$

$$\cos(270° - \theta) \stackrel{?}{=} -\sin\theta$$
$$\cos 270° \cos\theta + \sin 270° \sin\theta \stackrel{?}{=} -\sin\theta$$
$$0\cos\theta + (-1)\sin\theta \stackrel{?}{=} -\sin\theta$$
$$-\sin\theta = -\sin\theta$$

13. $\cos(\theta - 90°) = \sin\theta$

$$\cos(\theta - 90°) \stackrel{?}{=} \sin\theta$$
$$\cos\theta \cos 90° + \sin\theta \sin 90° \stackrel{?}{=} \sin\theta$$
$$(\cos\theta)(0) + (\sin\theta)(1) \stackrel{?}{=} \sin\theta$$
$$\sin\theta = \sin\theta$$

14. $\sin\left(\theta - \frac{\pi}{2}\right) = -\cos\theta$

$$\sin\left(\theta - \frac{\pi}{2}\right) \stackrel{?}{=} -\cos\theta$$
$$\sin\theta \cos\frac{\pi}{2} - \cos\theta \sin\frac{\pi}{2} \stackrel{?}{=} -\cos\theta$$
$$(\sin\theta)(0) - (\cos\theta)(1) \stackrel{?}{=} -\cos\theta$$
$$-\cos\theta = -\cos\theta$$

15. $\cos(\pi + \theta) = -\cos\theta$

$$\cos(\pi + \theta) \stackrel{?}{=} -\cos\theta$$
$$\cos\pi \cos\theta - \sin\pi \sin\theta \stackrel{?}{=} -\cos\theta$$
$$-1\cos\theta - 0\sin\theta \stackrel{?}{=} -\cos\theta$$
$$-\cos\theta = -\cos\theta$$

Lesson 14-5

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14-5 Practice (Average)

Sum and Difference of Angles Formulas

Find the exact value of each expression.

1. $\cos 75°$ $\frac{\sqrt{6}-\sqrt{2}}{4}$

2. $\cos 375°$ $\frac{\sqrt{6}+\sqrt{2}}{4}$

3. $\sin(-165°)$ $\frac{\sqrt{2}-\sqrt{6}}{4}$

4. $\sin(-105°)$ $\frac{-\sqrt{2}-\sqrt{6}}{4}$

5. $\sin 150°$ $\frac{1}{2}$

6. $\cos 240°$ $-\frac{1}{2}$

7. $\sin 225°$ $-\frac{\sqrt{2}}{2}$

8. $\sin(-75°)$ $\frac{-\sqrt{2}-\sqrt{6}}{4}$

9. $\sin 195°$ $\frac{\sqrt{2}-\sqrt{6}}{4}$

Verify that each of the following is an identity.

10. $\cos(180° - \theta) = -\cos\theta$

$$\cos(180° - \theta) \stackrel{?}{=} -\cos\theta$$
$$\cos 180° \cos\theta + \sin 180° \sin\theta \stackrel{?}{=} -\cos\theta$$
$$-1\cos\theta + 0\sin\theta \stackrel{?}{=} -\cos\theta$$
$$-\cos\theta = -\cos\theta$$

11. $\sin(360° + \theta) = \sin\theta$

$$\sin(360° + \theta) \stackrel{?}{=} \sin\theta$$
$$\sin 360° \cos\theta + \cos 360° \sin\theta \stackrel{?}{=} \sin\theta$$
$$0\cos\theta + 1\sin\theta \stackrel{?}{=} \sin\theta$$
$$\sin\theta = \sin\theta$$

12. $\sin(45° + \theta) - \sin(45° - \theta) = \sqrt{2}\sin\theta$

$$\sin(45° + \theta) - \sin(45° - \theta)$$
$$\stackrel{?}{=} \sin 45° \cos\theta + \cos 45° \sin\theta - (\sin 45° \cos\theta - \cos 45° \sin\theta)$$
$$\stackrel{?}{=} 2 \cdot \cos 45° \sin\theta$$
$$\stackrel{?}{=} 2 \cdot \frac{\sqrt{2}}{2} \cdot \sin\theta$$
$$= \sqrt{2}\sin\theta$$

13. $\cos\left(x - \frac{\pi}{6}\right) + \sin\left(x - \frac{\pi}{3}\right) = \sin x$

$$\cos\left(x - \frac{\pi}{6}\right) + \sin\left(x - \frac{\pi}{3}\right)$$
$$\stackrel{?}{=} \cos x \cos\frac{\pi}{6} + \sin x \sin\frac{\pi}{6} + \sin x \cos\frac{\pi}{3} - \cos x \sin\frac{\pi}{3}$$
$$\stackrel{?}{=} \frac{\sqrt{3}}{2}\cos x + \frac{1}{2}\sin x + \frac{1}{2}\sin x - \frac{\sqrt{3}}{2}\cos x$$
$$= \sin x$$

14. SOLAR ENERGY On March 21, the maximum amount of solar energy that falls on a square foot of ground at a certain location is given by $E\sin(90° - \phi)$, where ϕ is the latitude of the location and E is a constant. Use the difference of angles formula to find the amount of solar energy, in terms of $\cos\phi$, for a location that has a latitude of ϕ.
$E\cos\phi$

ELECTRICITY In Exercises 15 and 16, use the following information.

In a certain circuit carrying alternating current, the formula $i = 2\sin(120t)$ can be used to find the current i in amperes after t seconds.

15. Rewrite the formula using the sum of two angles. Sample answer: $i = 2\sin(90t + 30t)$

16. Use the sum of angles formula to find the exact current at $t = 1$ second. $\sqrt{3}$ amperes

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14-5 Reading to Learn Mathematics

Sum and Difference of Angles Formulas

Pre-Activity **How are the sum and difference formulas used to describe communication interference?**

Read the introduction to Lesson 14-5 at the top of page 786 in your textbook.

Consider the functions $y = \sin x$ and $y = 2 \sin x$. Do the graphs of these two functions have *constructive* interference or *destructive* interference?
constructive

Reading the Lesson

1. Match each expression from the list on the left with an expression from the list on the right that is equal to it for all values of the variables. (Some of the expressions from the list on the right may be used more than once or not at all.)

a. $\sin(\alpha - \beta)$ **v**	**i.** $\sin \beta$
b. $\cos(\alpha + \beta)$ **vi**	**ii.** $\sin \alpha \cos \beta + \cos \alpha \sin \beta$
c. $\sin(180° + \beta)$ **vii**	**iii.** $-\cos \beta$
d. $\sin(180° - \beta)$ **i**	**iv.** $\cos \alpha \cos \beta + \sin \alpha \sin \beta$
e. $\cos(180° + \beta)$ **iii**	**v.** $\sin \alpha \cos \beta - \cos \alpha \sin \beta$
f. $\sin(\alpha + \beta)$ **ii**	**vi.** $\cos \alpha \cos \beta - \sin \alpha \sin \beta$
g. $\cos(90° - \beta)$ **i**	**vii.** $-\sin \beta$
h. $\cos(\alpha - \beta)$ **iv**	**viii.** $\cos \beta$

2. Which expressions are equal to $\sin 15°$? (There may be more than one correct choice.)

A. $\sin 45° \cos 30° + \cos 45° \sin 30°$ **B.** $\sin 45° \cos 30° - \cos 45° \sin 30°$ **B and C**

C. $\sin 60° \cos 45° - \cos 60° \sin 45°$ **D.** $\cos 60° \cos 45° - \sin 60° \sin 45°$

Helping You Remember

3. Some students have trouble remembering which signs to use on the right-hand sides of the sum and difference of angle formulas. What is an easy way to remember this?

Sample answer: In the *sine* identities, the signs are the *same* on both sides. In the *cosine* identities, the signs are *opposite* on the two sides.

Lesson 14-5

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14-5 Enrichment

Identities for the Products of Sines and Cosines

By adding the identities for the sines of the sum and difference of the measures of two angles, a new identity is obtained.

$$\begin{aligned} \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ \hline (i) \quad \sin(\alpha + \beta) + \sin(\alpha - \beta) &= 2 \sin \alpha \cos \beta \end{aligned}$$

This new identity is useful for expressing certain products as sums.

Example **Write $\sin 3\theta \cos \theta$ as a sum.**

In the identity let $\alpha = 3\theta$ and $\beta = \theta$ so that
$2 \sin 3\theta \cos \theta = \sin(3\theta + \theta) + \sin(3\theta - \theta)$. Thus,
$\sin 3\theta \cos \theta = \frac{1}{2}\sin 4\theta + \frac{1}{2}\sin 2\theta$.

By subtracting the identities for $\sin(\alpha + \beta)$ and $\sin(\alpha - \beta)$, a similar identity for expressing a product as a difference is obtained.

(ii) $\sin(\alpha + \beta) - \sin(\alpha - \beta) = 2 \cos \alpha \sin \beta$

Solve.

1. Use the identities for $\cos(\alpha + \beta)$ and $\cos(\alpha - \beta)$ to find identities for expressing the products $2 \cos \alpha \cos \beta$ and $2 \sin \alpha \sin \beta$ as a sum or difference.
$2 \cos \alpha \cos \beta = \cos(\alpha + \beta) + \cos(\alpha - \beta)$
$2 \sin \alpha \sin \beta = \cos(\alpha - \beta) - \cos(\alpha + \beta)$

2. Find the value of $\sin 105° \cos 75°$ without using tables.
$\frac{1}{2}[\sin(105° + 75°) + \sin(105° - 75°)]$;
$\frac{1}{2}\left(0 + \frac{1}{2}\right)$; $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

3. Express $\cos \theta \sin \frac{\theta}{2}$ as a difference.
$2 \cos \theta \sin \frac{\theta}{2} = \sin\left(\theta + \frac{\theta}{2}\right) - \sin\left(\theta - \frac{\theta}{2}\right)$
$\cos \theta \sin \frac{\theta}{2} = \frac{1}{2} \sin \frac{3\theta}{2} - \frac{1}{2} \sin \frac{\theta}{2}$

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14-6 Study Guide and Intervention

Double-Angle and Half-Angle Formulas

Double-Angle Formulas

Double-Angle Formulas	The following identities hold true for all values of θ. $\sin 2\theta = 2 \sin \theta \cdot \cos \theta$ $\quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta$ $\cos 2\theta = 1 - 2 \sin^2 \theta$ $\cos 2\theta = 2 \cos^2 \theta - 1$

Example **Find the exact values of $\sin 2\theta$ and $\cos 2\theta$ if $\sin \theta = -\frac{9}{10}$ and $180° < \theta < 270°$.**

First, find the value of $\cos \theta$.

$\cos^2 \theta = 1 - \sin^2 \theta$ $\quad \cos^2 \theta + \sin^2 \theta = 1$

$\cos^2 \theta = 1 - \left(-\frac{9}{10}\right)^2$ $\quad \sin \theta = -\frac{9}{10}$

$\cos^2 \theta = \frac{19}{100}$

$\cos \theta = \pm\frac{\sqrt{19}}{10}$

Since θ is in the third quadrant, $\cos \theta$ is negative. Thus $\cos \theta = -\frac{\sqrt{19}}{10}$.

To find $\sin 2\theta$, use the identity $\sin 2\theta = 2 \sin \theta \cdot \cos \theta$.

$\sin 2\theta = 2 \sin \theta \cdot \cos \theta$

$= 2\left(-\frac{9}{10}\right)\left(-\frac{\sqrt{19}}{10}\right)$

$= \frac{9\sqrt{19}}{50}$

The value of $\sin 2\theta$ is $\frac{9\sqrt{19}}{50}$.

To find $\cos 2\theta$, use the identity $\cos 2\theta = 1 - 2 \sin^2 \theta$.

$\cos 2\theta = 1 - 2 \sin^2 \theta$

$= 1 - 2\left(-\frac{9}{10}\right)^2$

$= -\frac{31}{50}$.

The value of $\cos 2\theta$ is $-\frac{31}{50}$.

Exercises

Find the exact values of $\sin 2\theta$ and $\cos 2\theta$ for each of the following.

1. $\sin \theta = \frac{1}{4}$, $0° < \theta < 90°$ $\frac{\sqrt{15}}{8}, \frac{7}{8}$

2. $\sin \theta = -\frac{1}{8}$, $270° < \theta < 360°$ $-\frac{3\sqrt{7}}{32}, \frac{31}{32}$

3. $\cos \theta = -\frac{3}{5}$, $180° < \theta < 270°$ $\frac{24}{25}, -\frac{7}{25}$

4. $\cos \theta = -\frac{4}{5}$, $90° < \theta < 180°$ $-\frac{24}{25}, \frac{7}{25}$

5. $\sin \theta = -\frac{3}{5}$, $270° < \theta < 360°$ $-\frac{24}{25}, \frac{7}{25}$

6. $\cos \theta = -\frac{2}{3}$, $90° < \theta < 180°$ $-\frac{4\sqrt{5}}{9}, -\frac{1}{9}$

Lesson 14-6

NAME _____ DATE _____ PERIOD ____

14-6 Study Guide and Intervention *(continued)*

Double-Angle and Half-Angle Formulas

Half-Angle Formulas

Half-Angle Formulas	The following identities hold true for all values of α. $\sin \frac{\alpha}{2} = \pm\sqrt{\frac{1 - \cos \alpha}{2}}$ $\quad \cos \frac{\alpha}{2} = \pm\sqrt{\frac{1 + \cos \alpha}{2}}$

Example **Find the exact value of $\sin \frac{\alpha}{2}$ if $\sin \alpha = \frac{2}{3}$ and $90° < \alpha < 180°$.**

First find $\cos \alpha$.

$\cos^2 \alpha = 1 - \sin^2 \alpha$ $\quad \cos^2 \alpha + \sin^2 \alpha = 1$

$\cos^2 \alpha = 1 - \left(\frac{2}{3}\right)^2$ $\quad \sin \alpha = \frac{2}{3}$

$\cos^2 \alpha = \frac{5}{9}$ $\quad$ Simplify.

$\cos \alpha = \pm\frac{\sqrt{5}}{3}$ $\quad$ Take the square root of each side.

Since α is in the second quadrant, $\cos \alpha = -\frac{\sqrt{5}}{3}$.

$\sin \frac{\alpha}{2} = \pm\sqrt{\frac{1 - \cos \alpha}{2}}$ $\quad$ Half-Angle formula

$= \pm\sqrt{\frac{1 - \left(-\frac{\sqrt{5}}{3}\right)}{2}}$ $\quad \cos \alpha = -\frac{\sqrt{5}}{3}$

$= \pm\sqrt{\frac{3 + \sqrt{5}}{6}}$ $\quad$ Simplify.

$= \pm\frac{\sqrt{18 + 6\sqrt{5}}}{6}$ $\quad$ Rationalize.

Since α is between 90° and 180°, $\frac{\alpha}{2}$ is between 45° and 90°. Thus $\sin \frac{\alpha}{2}$ is positive and equals $\frac{\sqrt{18 + 6\sqrt{5}}}{6}$.

Exercises

Find the exact value of $\sin \frac{\alpha}{2}$ and $\cos \frac{\alpha}{2}$ for each of the following.

1. $\cos \alpha = -\frac{3}{5}$, $180° < \alpha < 270°$ $\frac{2\sqrt{5}}{5}, -\frac{\sqrt{5}}{5}$

2. $\cos \alpha = -\frac{4}{5}$, $90° < \alpha < 180°$ $\frac{3\sqrt{10}}{10}, \frac{\sqrt{10}}{10}$

3. $\sin \alpha = -\frac{3}{5}$, $270° < \alpha < 360°$ $\frac{\sqrt{10}}{10}, -\frac{3\sqrt{10}}{10}$

4. $\cos \alpha = -\frac{2}{3}$, $90° < \alpha < 180°$ $\frac{\sqrt{30}}{6}, \frac{\sqrt{6}}{6}$

Find the exact value of each expression by using the half-angle formulas.

5. $\cos 22\frac{1}{2}°$ $\frac{\sqrt{2 + \sqrt{2}}}{2}$

6. $\sin 67.5°$ $\frac{\sqrt{2 + \sqrt{2}}}{2}$

7. $\cos \frac{7\pi}{8}$ $-\frac{\sqrt{2 + \sqrt{2}}}{2}$

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14-6 Skills Practice

Double-Angle and Half-Angle Formulas

Find the exact values of $\sin 2\theta$, $\cos 2\theta$, $\sin \frac{\theta}{2}$, and $\cos \frac{\theta}{2}$ for each of the following.

1. $\cos \theta = \frac{7}{25}$, $0° < \theta < 90°$

$\frac{336}{625}, -\frac{527}{625}, \frac{3}{5}, \frac{4}{5}$

2. $\sin \theta = -\frac{4}{5}$, $180° < \theta < 270°$

$-\frac{24}{25}, -\frac{7}{25}, \frac{\sqrt{5}}{5}, -\frac{2\sqrt{5}}{5}$

3. $\sin \theta = \frac{40}{41}$, $90° < \theta < 180°$

$-\frac{720}{1681}, -\frac{1519}{1681}, \frac{5\sqrt{41}}{41}, \frac{4\sqrt{41}}{41}$

4. $\cos \theta = \frac{3}{7}$, $270° < \theta < 360°$

$-\frac{12\sqrt{10}}{49}, -\frac{31}{49}, \frac{\sqrt{14}}{7}, -\frac{\sqrt{35}}{7}$

5. $\cos \theta = -\frac{3}{5}$, $90° < \theta < 180°$

$-\frac{24}{25}, -\frac{7}{25}, \frac{2\sqrt{5}}{5}, \frac{\sqrt{5}}{5}$

6. $\sin \theta = \frac{5}{13}$, $0° < \theta < 90°$

$\frac{120}{169}, \frac{119}{169}, \frac{\sqrt{26}}{26}, \frac{5\sqrt{26}}{26}$

Find the exact value of each expression by using the half-angle formulas.

7. $\cos 22\frac{1}{2}°$ $\frac{\sqrt{2+\sqrt{2}}}{2}$

8. $\sin 165°$ $\frac{\sqrt{2-\sqrt{3}}}{2}$

9. $\cos 105°$ $-\frac{\sqrt{2-\sqrt{3}}}{2}$

10. $\sin \frac{\pi}{8}$ $\frac{\sqrt{2-\sqrt{2}}}{2}$

11. $\sin \frac{15\pi}{8}$ $-\frac{\sqrt{2-\sqrt{2}}}{2}$

12. $\cos 75°$ $\frac{\sqrt{2-\sqrt{3}}}{2}$

Verify that each of the following is an identity.

13. $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$

$\sin 2\theta \stackrel{?}{=} \frac{2 \tan \theta}{1 + \tan^2 \theta}$

$2 \sin \theta \cos \theta \stackrel{?}{=} \frac{2 \tan \theta}{\sec^2 \theta}$

$2 \sin \theta \cos \theta \stackrel{?}{=} 2 \frac{\sin \theta}{\cos \theta} \cdot \cos^2 \theta$

$2 \sin \theta \cos \theta = 2 \sin \theta \cos \theta$

Lesson 14-6

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14-6 Practice (Average)

Double-Angle and Half-Angle Formulas

Find the exact values of $\sin 2\theta$, $\cos 2\theta$, $\sin \frac{\theta}{2}$, and $\cos \frac{\theta}{2}$ for each of the following.

1. $\cos \theta = \frac{5}{13}$, $0° < \theta < 90°$

$\frac{120}{169}, -\frac{119}{169}, \frac{2\sqrt{13}}{13}, \frac{3\sqrt{13}}{13}$

2. $\sin \theta = \frac{8}{17}$, $90° < \theta < 180°$

$-\frac{240}{289}, \frac{161}{289}, \frac{4\sqrt{17}}{17}, \frac{\sqrt{17}}{17}$

3. $\cos \theta = \frac{1}{4}$, $270° < \theta < 360°$

$-\frac{\sqrt{15}}{8}, -\frac{7}{8}, \frac{\sqrt{6}}{4}, \frac{\sqrt{10}}{4}$

4. $\sin \theta = -\frac{2}{3}$, $180° < \theta < 270°$

$\frac{4\sqrt{5}}{9}, \frac{1}{9}, \frac{\sqrt{18+6\sqrt{5}}}{6}, -\frac{\sqrt{18-6\sqrt{5}}}{6}$

Find the exact value of each expression by using the half-angle formulas.

5. $\tan 105°$ $-2 - \sqrt{3}$

6. $\tan 15°$ $2 - \sqrt{3}$

7. $\cos 67.5°$ $\frac{\sqrt{2-\sqrt{2}}}{2}$

8. $\sin \left(-\frac{\pi}{8}\right)$ $-\frac{\sqrt{2-\sqrt{2}}}{2}$

Verify that each of the following is an identity.

9. $\sin^2 \frac{\theta}{2} = \frac{\tan \theta - \sin \theta}{2 \tan \theta}$

$\left(\pm\sqrt{\frac{1-\cos\theta}{2}}\right)^2 \stackrel{?}{=} \frac{\tan\theta - \sin\theta}{2\tan\theta}$;

$\frac{1-\cos\theta}{2} \stackrel{?}{=} \frac{\frac{\tan\theta}{\tan\theta} - \frac{\sin\theta}{\tan\theta}}{2\frac{\tan\theta}{\tan\theta}}$; $\frac{1-\cos\theta}{2} = \frac{1-\cos\theta}{2}$

10. $\sin 4\theta = 4 \cos 2\theta \sin \theta \cos \theta$

$\sin 4\theta \stackrel{?}{=} 4 \cos 2\theta \sin \theta \cos \theta$

$\sin 2(2\theta) \stackrel{?}{=} 4 \cos 2\theta \sin \theta \cos \theta$

$2 \sin 2\theta \cos 2\theta \stackrel{?}{=} 4 \cos 2\theta \sin \theta \cos \theta$

$2(2 \sin \theta \cos \theta)(\cos 2\theta) \stackrel{?}{=} 4 \cos 2\theta \sin \theta \cos \theta$

$4 \cos 2\theta \sin \theta \cos \theta = 4 \cos 2\theta \sin \theta \cos \theta$

11. AERIAL PHOTOGRAPHY In aerial photography, there is a reduction in film exposure for any point X not directly below the camera. The reduction E_θ is given by $E_\theta = E_0 \cos^4 \theta$, where θ is the angle between the perpendicular line from the camera to the ground and the line from the camera to point X, and E_0 is the exposure for the point directly below the camera. Using the identity $2 \sin^2 \theta = 1 - \cos 2\theta$, verify that $E_0 \cos^4 \theta = E_0\left(\frac{1}{2} + \frac{\cos 2\theta}{2}\right)^2$.

$E_0 \cos^4 \theta = E_0(\cos^2 \theta)^2 = E_0(1 - \sin^2 \theta)^2 = E_0\left(1 - \frac{2 \sin^2 \theta}{2}\right)^2 =$
$E_0\left(1 - \frac{1 - \cos 2\theta}{2}\right)^2 = E_0\left(\frac{1}{2} + \frac{\cos 2\theta}{2}\right)^2$

12. IMAGING A scanner takes thermal images from altitudes of 300 to 12,000 meters. The width W of the swath covered by the image is given by $W = 2H' \tan \theta$, where H' is the height and θ is half the scanner's field of view. Verify that $\frac{2H' \sin 2\theta}{1 + \cos 2\theta} = 2H' \tan \theta$.

$\frac{2H' \sin 2\theta}{1 + \cos 2\theta} = \frac{4H' \sin \theta \cos \theta}{1 + (2 \cos^2 \theta - 1)} = \frac{4H' \sin \theta \cos \theta}{2 \cos^2 \theta} = \frac{2H' \sin \theta}{\cos \theta} = 2H' \tan \theta$

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14-6 Reading to Learn Mathematics

Double-Angle and Half-Angle Formulas

Pre-Activity **How can trigonometric functions be used to describe music?**

Read the introduction to Lesson 14-6 at the top of page 791 in your textbook.

Suppose that the equation for the second harmonic is $y = \sin a\theta$. Then what would be the equations for the fundamental tone (first harmonic), third harmonic, fourth harmonic, and fifth harmonic?

$y = \sin 0.5a\theta$; $y = \sin 1.5a\theta$; $y = \sin 2a\theta$; $y = \sin 2.5a\theta$

Reading the Lesson

1. Match each expression from the list on the left with *all* expressions from the list on the right that are equal to it for all values of β.

a. $\sin \frac{\beta}{2}$ **v**

b. $\cos 2\beta$ **ii and iii**

c. $\cos \frac{\beta}{2}$ **iv**

d. $\sin 2\beta$ **i**

i. $2 \sin \beta \cos \beta$

ii. $1 - 2 \sin^2 \beta$

iii. $\cos^2 \beta - \sin^2 \beta$

iv. $\pm\sqrt{\frac{1 + \cos \beta}{2}}$

v. $\pm\sqrt{\frac{1 - \cos \beta}{2}}$

2. Determine whether you would use the *positive* or *negative* square root in the half-angle identities for $\sin \frac{\alpha}{2}$ and $\cos \frac{\alpha}{2}$ in each of the following situations. (Do not actually calculate $\sin \frac{\alpha}{2}$ and $\cos \frac{\alpha}{2}$.)

a. $\sin \frac{\alpha}{2}$, if $\cos \alpha = \frac{2}{5}$ and α is in Quadrant I **positive**

b. $\cos \frac{\alpha}{2}$, if $\cos \alpha = -0.9$ and α is in Quadrant II **positive**

c. $\cos \frac{\alpha}{2}$, if $\sin \alpha = -0.75$ and α is in Quadrant III **negative**

d. $\sin \frac{\alpha}{2}$, if $\sin \alpha = -0.8$ and α is in Quadrant IV **positive**

Helping You Remember

3. Many students find it difficult to remember a large number of identities. How can you obtain all three of the identities for $\cos 2\theta$ by remembering only one of them and using a Pythagorean identity?

Sample answer: Just remember the identity $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$. Using the Pythagorean identity $\cos^2 \theta + \sin^2 \theta = 1$, you can substitute either $1 - \sin^2 \theta$ for $\cos^2 \theta$ or $1 - \cos^2 \theta$ for $\sin^2 \theta$ to get the other two identities for $\cos 2\theta$.

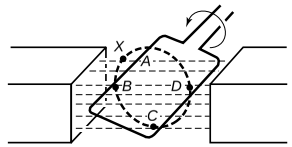
Lesson 14-6

NAME _____ DATE _____ PERIOD _____

14-6 Enrichment

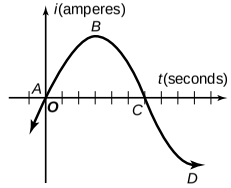
Alternating Current

The figure at the right represents an alternating current generator. A rectangular coil of wire is suspended between the poles of a magnet. As the coil of wire is rotated, it passes through the magnetic field and generates current.



As point X on the coil passes through the points A and C, its motion is along the direction of the magnetic field between the poles. Therefore, no current is generated. However, through points B and D, the motion of X is perpendicular to the magnetic field. This induces maximum current in the coil. Between A and B, B and C, C and D, and D and A, the current in the coil will have an intermediate value. Thus, the graph of the current of an alternating current generator is closely related to the sine curve.

The maximum current may have a positive or negative value.



The actual current, i, in a household current is given by $i = I_M \sin(120\pi t + \alpha)$ where I_M is the maximum value of the current, t is the elapsed time in seconds, and α is the angle determined by the position of the coil at time t_n.

Example **If $\alpha = \frac{\pi}{2}$, find a value of t for which $i = 0$.**

If $i = 0$, then $I_M \sin(120\pi t + \alpha) = 0$. $\quad i = I_M \sin(120\pi t + \alpha)$

Since $I_M \neq 0$, $\sin(120\pi t + \alpha) = 0$. $\quad$ If $ab = 0$ and $a \neq 0$, then $b = 0$.

Let $120\pi t + \alpha = s$. Thus, $\sin s = 0$.

$s = \pi$ because $\sin \pi = 0$.

$120\pi t + \alpha = \pi$ $\quad$ Substitute $120\pi t + \alpha$ for s.

$120\pi t + \frac{\pi}{2} = \pi$ $\quad$ Substitute $\frac{\pi}{2}$ for α.

$= \frac{1}{240}$ $\quad$ Solve for t.

This solution is the first positive value of t that satisfies the problem.

Using the equation for the actual current in a household circuit, $i = I_M \sin(120\pi t + \alpha)$, solve each problem. For each problem, find the first positive value of t.

1. If $\alpha = 0$, find a value of t for which $i = 0$. **$t = \frac{1}{120}$**

2. If $\alpha = 0$, find a value of t for which $i = +I_M$. **$t = \frac{1}{240}$**

3. If $\alpha = \frac{\pi}{2}$, find a value of t for which $i = -I_M$. **$t = \frac{1}{120}$**

4. If $\alpha = \frac{\pi}{4}$, find a value of t for which $i = 0$. **$t = \frac{1}{160}$**

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14-7 Study Guide and Intervention

Solving Trigonometric Equations

Solve Trigonometric Equations You can use trigonometric identities to solve trigonometric equations, which are true for only certain values of the variable.

Example 1 **Find all solutions of $4 \sin^2 \theta - 1 = 0$ for the interval $0° < \theta < 360°$.**

$4 \sin^2 \theta - 1 = 0$

$4 \sin^2 \theta = 1$

$\sin^2 \theta = \frac{1}{4}$

$\sin \theta = \pm\frac{1}{2}$

$\theta = 30°, 150°, 210°, 330°$

Example 2 **Solve $\sin 2\theta + \cos \theta = 0$ for all values of θ. Give your answer in both radians and degrees.**

$\sin 2\theta + \cos \theta = 0$

$2 \sin \theta \cos \theta + \cos \theta = 0$

$\cos \theta (2 \sin \theta + 1) = 0$

$\cos \theta = 0$ or $2 \sin \theta + 1 = 0$

$\sin \theta = -\frac{1}{2}$

$\theta = 90° + k \cdot 180°$; $\theta = \frac{\pi}{2} + k \cdot \pi$

$\theta = 210° + k \cdot 360°$, $330° + k \cdot 360°$; $\theta = \frac{7\pi}{6} + k \cdot 2\pi$, $\frac{11\pi}{6} + k \cdot 2\pi$

Exercises

Find all solutions of each equation for the given interval.

1. $2 \cos^2 \theta + \cos \theta = 1, 0 \le \theta < 2\pi$

$\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

2. $\sin^2 \theta \cos^2 \theta = 0, 0 \le \theta < 2\pi$

$0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

3. $\cos 2\theta = \frac{\sqrt{3}}{2}, 0° \le \theta < 360°$

15°, 165°, 195°, 345°

4. $2 \sin \theta - \sqrt{3} = 0, 0 \le \theta < 2\pi$

$\frac{\pi}{3}, \frac{2\pi}{3}$

Solve each equation for all values of θ if θ is measured in radians.

5. $4 \sin^2 \theta - 3 = 0$

$\frac{\pi}{3} + k \cdot \pi, \frac{2\pi}{3} + k \cdot \pi$

6. $2 \cos \theta \sin \theta + \cos \theta = 0$

$\frac{\pi}{2} + k \cdot 2\pi, \frac{3\pi}{2} + k \cdot 2\pi, \frac{7\pi}{6} + k \cdot 2\pi, \frac{11\pi}{6} + k \cdot 2\pi$

Solve each equation for all values of θ if θ is measured in degrees.

7. $\cos 2\theta + \sin^2 \theta = \frac{1}{2}$

$45° + k \cdot 90°$

8. $\tan 2\theta = -1$

$67.5° + k \cdot 360°, 157.5° + k \cdot 360°$

Lesson 14-7

NAME _____ DATE _____ PERIOD ____

14-7 Study Guide and Intervention *(continued)*

Solving Trigonometric Equations

Use Trigonometric Equations

Example **LIGHT Snell's law says that $\sin \alpha = 1.33 \sin \beta$, where α is the angle at which a beam of light enters water and β is the angle at which the beam travels through the water. If a beam of light enters water at 42°, at what angle does the light travel through the water?**

$\sin \alpha = 1.33 \sin \beta$	Original equation
$\sin 42° = 1.33 \sin \beta$	$\alpha = 42°$
$\sin \beta = \frac{\sin 42°}{1.33}$	Divide each side by 1.33.
$\sin \beta \approx 0.5031$	Use a calculator.
$\beta \approx 30.2°$	Take the arcsin of each side.

The light travels through the water at an angle of approximately 30.2°.

Exercises

1. A 6-foot pipe is propped on a 3-foot tall packing crate that sits on level ground. One foot of the pipe extends above the top of the crate and the other end rests on the ground. What angle does the pipe form with the ground? 36.9°

2. At 1:00 P.M. one afternoon a 180-foot statue casts a shadow that is 85 feet long. Write an equation to find the angle of elevation of the Sun at that time. Find the angle of elevation. $\tan \theta = \frac{180}{85}$; 64.7°

3. A conveyor belt is set up to carry packages from the ground into a window 28 feet above the ground. The angle that the conveyor belt forms with the ground is 35°. How long is the conveyor belt from the ground to the window sill? 48.8 ft

SPORTS **The distance a golf ball travels can be found using the formula $d = \frac{v_0^2}{g} \sin 2\theta$, where v_0 is the initial velocity of the ball, g is the acceleration due to gravity (which is 32 feet per second squared), and θ is the angle that the path of the ball makes with the ground.**

4. How far will a ball travel hit 90 feet per second at an angle of 55°? 237.9 ft

5. If a ball that traveled 300 feet had an initial velocity of 110 feet per second, what angle did the path of the ball make with the ground? 26.3° or 63.7°

6. Some children set up a teepee in the woods. The poles are 7 feet long from their intersection to their bases, and the children want the distance between the poles to be 4 feet at the base. How wide must the angle be between the poles? 33.2°

NAME _____ DATE _____ PERIOD ____

14-7 Skills Practice

Solving Trigonometric Equations

Find all solutions of each equation for the given interval.

1. $\sin\theta = \frac{\sqrt{2}}{2}$, $0° \le \theta < 360°$ **45°, 135°**

2. $2\cos\theta = -\sqrt{3}$, $90° < \theta < 180°$ **150°**

3. $\tan^2\theta = 1$, $180° < \theta < 360°$ **225°, 315°**

4. $2\sin\theta = 1$, $0 \le \theta < \pi$ $\frac{\pi}{6}, \frac{5\pi}{6}$

5. $\sin^2\theta + \sin\theta = 0$, $\pi \le \theta < 2\pi$ $\pi, \frac{3\pi}{2}$

6. $2\cos^2\theta + \cos\theta = 0$, $0 \le \theta < \pi$ $\frac{\pi}{2}, \frac{2\pi}{3}$

Solve each equation for all values of θ if θ is measured in radians.

7. $2\cos^2\theta - \cos\theta = 1$
$0 + 2k\pi, \frac{2\pi}{3} + 2k\pi$, and $\frac{4\pi}{3} + 2k\pi$

8. $\sin^2\theta - 2\sin\theta + 1 = 0$
$\frac{\pi}{2} + 2k\pi$

9. $\sin\theta + \sin\theta\cos\theta = 0$
$k\pi$

10. $\sin^2\theta = 1$
$\frac{\pi}{2} + k\pi$

11. $4\cos\theta = -1 + 2\cos\theta$
$\frac{2\pi}{3} + 2k\pi, \frac{4\pi}{3} + 2k\pi$

12. $\tan\theta\cos\theta = \frac{1}{2}$
$\frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi$

Solve each equation for all values of θ if θ is measured in degrees.

13. $2\sin\theta + 1 = 0$
$210° + k \cdot 360$ and $330° + k \cdot 360°$

14. $2\cos\theta + \sqrt{3} = 0$
$150° + k \cdot 360$ and $210° + k \cdot 360°$

15. $\sqrt{2}\sin\theta + 1 = 0$
$225° + k \cdot 360°$ and $315° + k \cdot 360°$

16. $2\cos^2\theta = 1$
$45° + k \cdot 90°$

17. $4\sin^2\theta = 3$
$60° + k \cdot 180°$ and $120° + k \cdot 180°$

18. $\cos 2\theta = -1$
$90° + k \cdot 180°$

Solve each equation for all values of θ.

19. $3\cos^2\theta - \sin^2\theta = 0$
$\frac{\pi}{3} + k\pi$ and $\frac{2\pi}{3} + k\pi$, or
$60° + k \cdot 180°$ and $120° + k \cdot 180°$

20. $\sin\theta + \sin 2\theta = 0$
$k\pi$ and $\frac{2\pi}{3} + 2k\pi$, or
$k \cdot 180°$ and $120° + k \cdot 360°$

21. $2\sin^2\theta = \sin\theta + 1$
$\frac{\pi}{2} + k\frac{2\pi}{3}$, or $90° + k \cdot 120°$

22. $\cos\theta + \sec\theta = 2$
$2k\pi$, or $k \cdot 360°$

Lesson 14-7

NAME _____ DATE _____ PERIOD ____

14-7 Practice (Average)

Solving Trigonometric Equations

Find all solutions of each equation for the given interval.

1. $\sin 2\theta = \cos\theta$, $90° \le \theta < 180°$
90°, 150°

2. $\sqrt{2}\cos\theta = \sin 2\theta$, $0° \le \theta < 360°$
45°, 90°, 135°, 270°

3. $\cos 4\theta = \cos 2\theta$, $180° \le \theta < 360°$
180°, 240°, 300°

4. $\cos\theta + \cos(90 - \theta) = 0$, $0 \le \theta < 2\pi$
$\frac{3\pi}{4}, \frac{7\pi}{4}$

5. $2 + \cos\theta = 2\sin^2\theta$, $\pi \le \theta \le \frac{3\pi}{2}$
$\frac{4\pi}{3}, \frac{3\pi}{2}$

6. $\tan^2\theta + \sec\theta = 1$, $\frac{\pi}{2} \le \theta < \pi$
$\frac{2\pi}{3}$

Solve each equation for all values of θ if θ is measured in radians.

7. $\cos^2\theta = \sin^2\theta$
$\frac{\pi}{4} + k\frac{\pi}{2}$

8. $\cot\theta = \cot^3\theta$
$\frac{\pi}{2} + k\pi$ and $\frac{\pi}{4} + k\frac{\pi}{2}$

9. $\sqrt{2}\sin^3\theta = \sin^2\theta$
$k\pi, \frac{\pi}{4} + 2k\pi$, and $\frac{3\pi}{4} + 2k\pi$

10. $\cos^2\theta\sin\theta = \sin\theta$
$k\pi$

11. $2\cos 2\theta = 1 - 2\sin^2\theta$
$\frac{\pi}{4} + k\frac{\pi}{2}$

12. $\sec^2\theta = 2$
$\frac{\pi}{4} + k\frac{\pi}{2}$

Solve each equation for all values of θ if θ is measured in degrees.

13. $\sin^2\theta\cos\theta = \cos\theta$
$90° + k \cdot 180°$

14. $\csc^2\theta - 3\csc\theta + 2 = 0$
$30° + k \cdot 360°$, $90° + k \cdot 360°$, and
$150° + k \cdot 360°$

15. $\frac{3}{1 + \cos\theta} = 4(1 - \cos\theta)$
$60° + k \cdot 180°$ and $120° + k \cdot 180°$

16. $\sqrt{2}\cos^2\theta = \cos^2\theta$
$90° + k \cdot 180°$ and $450° + k \cdot 360°$

Solve each equation for all values of θ.

17. $4\sin^2\theta = 3$ $\frac{\pi}{3} + k\pi$ and $\frac{2\pi}{3} + k\pi$,
or $60° + k \cdot 180°$ and $120° + k \cdot 180°$

18. $4\sin^2\theta - 1 = 0$ $\frac{\pi}{6} + k\pi$ and $\frac{5\pi}{6} + k\pi$,
or $30° + k \cdot 180°$ and $150° + k \cdot 180°$

19. $2\sin^2\theta - 3\sin\theta = -1$ $\frac{\pi}{6} + \frac{\pi k}{3}$,
or $30° + k \cdot 60°$

20. $\cos 2\theta + \sin\theta - 1 = 0$ $k\pi$ and $\frac{\pi}{6} + 2k\pi$,
or $k \cdot 180°$ and $30° + k \cdot 360°$

21. WAVES Waves are causing a buoy to float in a regular pattern in the water. The vertical position of the buoy can be described by the equation $h = 2\sin x$. Write an expression that describes the position of the buoy when its height is at its midline. $k\pi$ or $k \cdot 180°$

22. ELECTRICITY The electric current in a certain circuit with an alternating current can be described by the formula $i = 3\sin 240t$, where i is the current in amperes and t is the time in seconds. Write an expression that describes the times at which there is no current. $0.75kt$

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NAME _____ DATE _____ PERIOD ____

14-7 Reading to Learn Mathematics

Solving Trigonometric Equations

Pre-Activity **How can trigonometric equations be used to predict temperature?**

Read the introduction to Lesson 14-7 at the top of page 799 in your textbook.

Describe how you could use a graphing calculator to determine the months in which the average daily high temperature is above 80°F. (Assume that $x = 1$ represents January.) Specify the graphing window that you would use.
Sample answer: Graph the functions $y = 11.56 \sin(0.4516x - 1.641) + 80.89$ (using radian mode) and $y = 80$ on the same screen. Use the window [1, 12] by [60, 100] with Xscl = 1 and Yscl = 4. Note the x values for which the curve is above the horizontal line.

Reading the Lesson

1. Identify which equations have no solution. **C, E, and G**

A. $\sin\theta = 1$ **B.** $\tan\theta = 0.001$ **C.** $\sec\theta = \frac{1}{2}$

D. $\csc\theta = -3$ **E.** $\cos\theta = 1.01$ **F.** $\cot\theta = -1000$

G. $\cos\theta + 2 = -1$ **H.** $\sec\theta - 1.5 = 0$ **I.** $\sin\theta - 0.009 = 0.99$

2. Use a trigonometric identity to write the first step in the solution of each trigonometric equation. (Do not complete the solution.)

a. $\tan\theta = \cos^2\theta + \sin^2\theta, 0 \le \theta < 2\pi$ **$\tan\theta = 1$**

b. $\sin^2\theta - 2\sin\theta + 1 = 0, 0° \le \theta < 360°$ **$(\sin\theta - 1)^2 = 0$**

c. $\cos 2\theta = \sin\theta, 0° \le \theta < 360°$ **$1 - 2\sin^2\theta = \sin\theta$**

d. $\sin 2\theta = \cos\theta, 0 \le \theta < 2\pi$ **$2\sin\theta\cos\theta = \cos\theta$**

e. $2\cos 2\theta + 3\cos\theta = -1, 0° \le \theta < 360°$ **$2(2\cos^2\theta - 1) + 3\cos\theta = -1$**

f. $3\tan^2\theta + 5\tan\theta - 2 = 0$ **$(3\tan\theta - 1)(\tan\theta + 2) = 0$**

Helping You Remember

3. A good way to remember something is to explain it to someone else. How would you explain to a friend the difference between verifying a trigonometric identity and solving a trigonometric equation. **Sample answer: Verifying a trigonometric identity means showing that the two sides are equal for *all* values of the variable for which the functions involved are defined. This is done by transforming one or both sides until the same expression is obtained on both sides. Solving a trigonometric equation means finding the values of the variable for which both sides are equal. This process may require simplifying trigonometric expressions, but it also requires finding the angles for which a trigonometric function has a particular value.**

Lesson 14-7

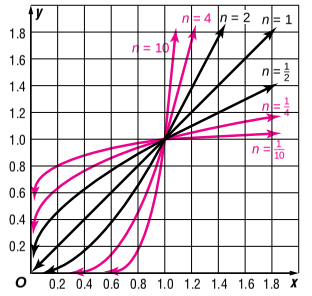
NAME _____ DATE _____ PERIOD ____

14-7 Enrichment

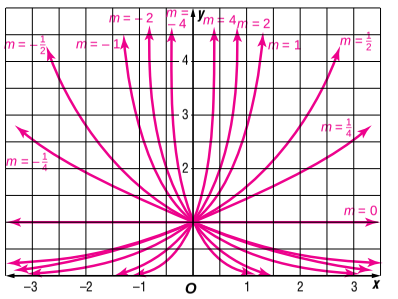
Families of Curves

Use these graphs for the problems below.

The Family $y = x^n$



The Family $y = e^{mx}$



1. Use the graph on the left to describe the relationship among the curves $y = x^{\frac{1}{2}}$, $y = x^1$, and $y = x^2$. **For $n = \frac{1}{2}$ and $n = 2$, the graphs are reflections of one another in the line with equation $y = x^1$.**

2. Graph $y = x^n$ for $n = \frac{1}{10}, \frac{1}{4}, 4$, and 10 on the grid with $y = x^{\frac{1}{2}}$, $y = x^1$, and $y = x^2$.
See students' graphs.

3. Which two regions in the first quadrant contain no points of the graphs of the family for $y = x^n$?
$\{(x, y) | x \ge 1$ and $0 < y \le 1\}$ and $\{(x, y) | 0 < x \le 1$ and $y \ge 1\}$

4. On the right grid, graph the members of the family $y = e^{mx}$ for which $m = 1$ and $m = -1$.
See students' graphs.

5. Describe the relationship among these two curves and the y-axis.
the graphs for $m = 1$ and $m = -1$ are reflections in the y-axis.

6. Graph $y = e^{mx}$ for $m = 0, \pm\frac{1}{4}, \pm\frac{1}{2}, \pm 2$, and ± 4.
See students' graphs.

Chapter 14 Assessment Answer Key

Form 1

Page 879

1. B
2. A
3. D
4. C
5. A
6. A
7. B
8. D
9. C

Page 880

10. C
11. A
12. B
13. D
14. C
15. D
16. C
17. A
18. B
19. A
20. C

B: See students' answers.

Form 2A

Page 881

1. A
2. C
3. B
4. A
5. D
6. C
7. B
8. D
9. B

(continued on the next page)

Answers

Chapter 14 Assessment Answer Key

Form 2A *(continued)*
Page 882

10. C
11. A
12. C
13. A
14. B
15. D
16. D
17. C
18. B
19. C
20. A

B: See students' answers.

Form 2B
Page 883

1. C
2. B
3. D
4. B
5. A
6. A
7. C
8. D
9. B
10. A

Page 884

11. D
12. B
13. C
14. C
15. D
16. A
17. C
18. B
19. A
20. D

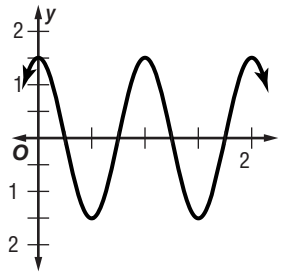
B: See students' answers.

Chapter 14 Assessment Answer Key

Form 2C

Page 885

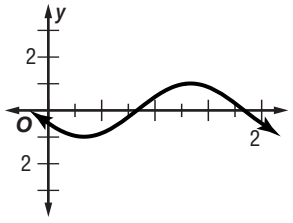
1.



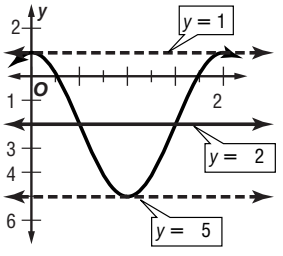
2. 3; 90° or $\frac{\pi}{2}$

3. none; 900° or 5π

4. $-\frac{2\pi}{3}$



5.



6. $\frac{5}{4}$

7. $-\frac{\sqrt{21}}{2}$

8. 1

9. $\tan^2 \theta$

Page 886

10. See students' answers.

11. See students' answers.

12. $\frac{\sqrt{6} - \sqrt{2}}{4}$

13. $\frac{\sqrt{2} - \sqrt{6}}{4}$

14. See students' answers.

15.

16. $\frac{\sqrt{18 + 12\sqrt{2}}}{6}$

17. $\frac{-\sqrt{2 - \sqrt{3}}}{2}$

18. See students' answers.

19.

20. about 15 weeks

B: $\frac{1}{2}$

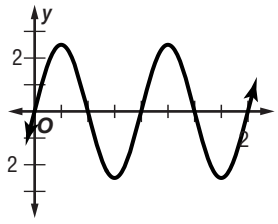
Answers

Chapter 14 Assessment Answer Key

Form 2D

Page 887

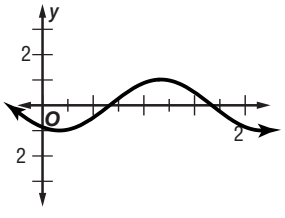
1.



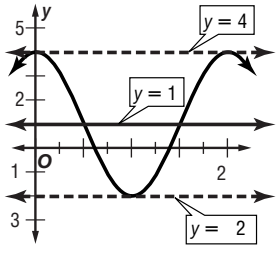
2. 2; 120° or $\frac{2\pi}{3}$

3. none; 720° or 4π

4. $\frac{2\pi}{3}$



5.



6. $\frac{3\sqrt{2}}{4}$

7. $-\frac{\sqrt{21}}{2}$

8. 1

9. $-\sec^2 \theta$

Page 888

10. See students' answers.

11. See students' answers.

12. $\frac{\sqrt{6} - \sqrt{2}}{4}$

13. $\frac{\sqrt{2} + \sqrt{6}}{4}$

14. See students' answers.

15. $-\frac{7}{8}$

16. $\frac{\sqrt{18 - 12\sqrt{2}}}{6}$

17. $-\frac{\sqrt{2 + \sqrt{3}}}{2}$

18. See students' answers.

19. $0° + k \cdot 180°$, $60° + k \cdot 360°$, $300° + k \cdot 360°$

20. about 10 weeks

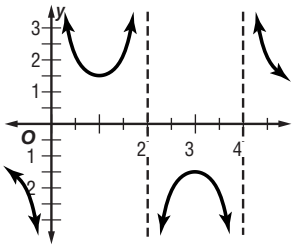
B: 0

Chapter 14 Assessment Answer Key

Form 3

Page 889

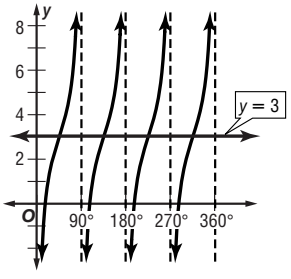
1.



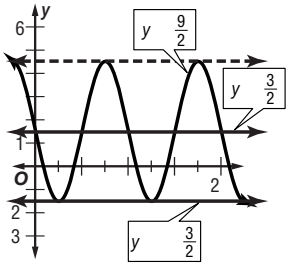
2. $\frac{2}{15}$; 90° or $\frac{\pi}{2}$

3. none; 900° or 5π

4. 3; none; 90°; 45°



5. $\frac{3}{2}$; 3; π; $-\frac{\pi}{4}$



6.

7.

8. 1

9. See students' answers.

Page 890

10. See students' answers.

11.

12. $\frac{\sqrt{6} - \sqrt{2}}{4}$

13. See students' answers.

14. See students' answers.

15. $-\frac{3\sqrt{55}}{32}$

16. $-\frac{\sqrt{32 - 2\sqrt{87}}}{}$

17. $-\frac{\sqrt{2 - \sqrt{3}}}{}$

18. See students' answers.

19. $\frac{\pi}{3} + \frac{2k\pi}{3}$

20. 0.42 sec

B: $\frac{17\sqrt{10}}{75}$

Answers

Chapter 14 Assessment Answer Key

Page 891, Open-Ended Assessment
Scoring Rubric

Score	General Description	Specific Criteria
4	**Superior** A correct solution that is supported by well-developed, accurate explanations	• Shows thorough understanding of the concepts of *trigonometric functions and their translations; using and verifying trigonometric identities; finding values of sine and cosine involving sum and difference, double-angle, and half-angle formulas;* and *solving trigonometric equations.* • Uses appropriate strategies to solve problems. • Computations are correct. • Written explanations are exemplary. • Graphs are accurate and appropriate. • Goes beyond requirements of some or all problems.
3	**Satisfactory** A generally correct solution, but may contain minor flaws in reasoning or computation	• Shows an understanding of the concepts of *trigonometric functions and their translations; using and verifying trigonometric identities; finding values of sine and cosine involving sum and difference, double-angle, and half-angle formulas;* and *solving trigonometric equations.* • Uses appropriate strategies to solve problems. • Computations are mostly correct. • Written explanations are effective. • Graphs are mostly accurate and appropriate. • Satisfies all requirements of problems.
2	**Nearly Satisfactory** A partially correct interpretation and/or solution to the problem	• Shows an understanding of most of the concepts of *trigonometric functions and their translations; using and verifying trigonometric identities; finding values of sine and cosine involving sum and difference, double-angle, and half-angle formulas;* and *solving trigonometric equations.* • May not use appropriate strategies to solve problems. • Computations are mostly correct. • Written explanations are satisfactory. • Graphs are mostly accurate. • Satisfies the requirements of most of the problems.
1	**Nearly Unsatisfactory** A correct solution with no supporting evidence or explanation	• Final computation is correct. • No written explanations or work is shown to substantiate the final computation. • Satisfies minimal requirements of some of the problems.
0	**Unsatisfactory** An incorrect solution indicating no mathematical understanding of the concept or task, or no solution is given	• Shows little or no understanding of most of the concepts of *trigonometric functions and their translations; using and verifying trigonometric identities; finding values of sine and cosine involving sum and difference, double-angle, and half-angle formulas;* and *solving trigonometric equations.* • Does not use appropriate strategies to solve problems. • Computations are incorrect. • Written explanations are unsatisfactory. • Does not satisfy requirements of problems. • Graphs are inaccurate or inappropriate. • No answer may be given.

Chapter 14 Assessment Answer Key

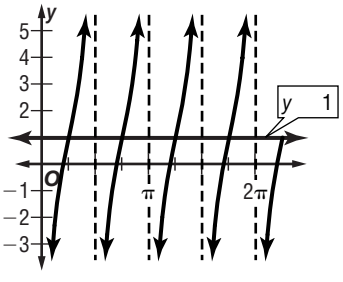
Page 891, Open-Ended Assessment
Sample Answers

In addition to the scoring rubric found on page A28, the following sample answers may be used as guidance in evaluating open-ended assessment items.

1. Students should explain that the answers given by Groups A and B are incorrect. For $\theta = 0°$, cot θ is undefined, so this solution is extraneous. While the answer given by Group D is correct, giving all angles coterminal with 90° and 270°, the Group C answer includes all of these same values for θ in a single expression, so is the most efficient way in which to express the solution.

2. Student responses must have one of the four forms: $y = a \csc 4(\theta + h) + k$, $y = a \sec 4(\theta + h) + k$, $y = a \tan 2(\theta + h) + k$, or $y = a \cot 2(\theta + h) + k$, where a is any real number, $h > 0$, and $k > 0$.

Sample answer: $y = 3 \tan 2\left(\theta - \frac{\pi}{4}\right) + 1$



3. Ideally, students should verify the identity by transforming one side of the equation into the form of the other side (as in 14-4A), and by transforming both sides of the equation separately into a common form (as covered in 14-4B).

Sample answer by method in 14-4A:

$$\frac{1}{1 - \sin^2 \theta} \stackrel{?}{=} \tan^2 \theta + 1$$
$$\frac{1}{1 - (1 - \cos^2 \theta)} \stackrel{?}{=} \tan^2 \theta + 1$$
$$\frac{1}{\cos^2 \theta} \stackrel{?}{=} \tan^2 \theta + 1$$
$$\sec^2 \theta \stackrel{?}{=} \tan^2 \theta + 1$$
$$\tan^2 \theta + 1 = \tan^2 \theta + 1$$

Sample answer by method in 14-4B:

$$\frac{1}{1 - \sin^2 \theta} \stackrel{?}{=} \tan^2 \theta + 1$$
$$\frac{1}{1 - (1 - \cos^2 \theta)} \stackrel{?}{=} \frac{\sin^2 \theta}{\cos^2 \theta} + 1$$
$$\frac{1}{\cos^2 \theta} \stackrel{?}{=} \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta}$$
$$\frac{1}{\cos^2 \theta} \stackrel{?}{=} \frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta}$$
$$\frac{1}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

4. For sin θ to exist, students must select p and q so that $|p| < |q|$. Signs of p and q must be consistent with the quadrant selected and the sign of the sine function in that quadrant. Then, using appropriate values and signs for p and q, students should apply the necessary identities and formulas to evaluate each function.

Sample answer: For $p = -3$ and $q = 5$, and the terminal side of θ in Quadrant III, $\sin \theta = -\frac{3}{5}$. Therefore, $\cos \theta = -\frac{4}{5}$, $\tan \theta = \frac{3}{4}$, $\csc \theta = -\frac{5}{3}$, $\sec \theta = -\frac{5}{4}$, $\cot \theta = \frac{4}{3}$, $\sin 2\theta = \frac{24}{25}$, $\cos 2\theta = \frac{7}{25}$, $\sin \frac{\theta}{2} = \frac{3\sqrt{10}}{10}$, and $\cos \frac{\theta}{2} = -\frac{\sqrt{10}}{10}$.

5. Sample answers:

5a. $\sin 240° = \sin (180° + 60°)$
$= \sin 180° \cos 60° + \cos 180° \sin 60°$
$= -\frac{\sqrt{3}}{2}$

5b. $\sin 240° = \sin (270° - 30°)$
$= \sin 270° \cos 30° - \cos 270° \sin 30°$
$= -\frac{\sqrt{3}}{2}$

5c. $\sin 240° = \sin (2 \cdot 120°)$
$= 2 \sin 120° \cos 120° = -\frac{\sqrt{3}}{2}$

5d. $\sin 240° = \sin \frac{480°}{2}$
$= -\sqrt{\frac{1 - \cos 480°}{2}} = -\frac{\sqrt{3}}{2}$

Answers

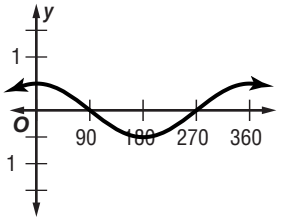
Chapter 14 Assessment Answer Key

Vocabulary Test/Review
Page 892

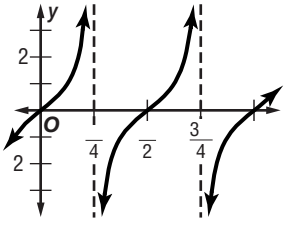
1. false; amplitude
2. false; vertical shift
3. false; midline
4. true
5. false; half-angle formula
6. true
7. true
8. Sample answer: A phase shift is a horizontal translation of the graph of a trigonometric function.

Quiz (Lessons 14–1 and 14–2)
Page 893

1. $\frac{1}{2}$; 360°



2. none; $\frac{\pi}{2}$



3. $-\frac{\pi}{4}$
4. 2; $y = 2$

Quiz (Lessons 14–5 and 14–6)
Page 894

1. $\frac{\sqrt{6}+\sqrt{2}}{4}$
2.
3. $\frac{\sqrt{3}}{3}$
4. See students' answers.
5. See students' answers.
6.
7.
8. $-\frac{\sqrt{50-10\sqrt{21}}}{}$
9. $-\frac{\sqrt{2-\sqrt{2}}}{}$
10. See students' answers.

Quiz (Lessons 14–3 and 14–4)
Page 893

1.
2. $\frac{1}{2}$
3. -4
4. $\tan^2 \theta$
5. A

Quiz (Lesson 14–7)
Page 894

1. 30°, 150°, 270°
2. $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$
3. $0° + k \cdot 120°$
4. $\frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi$
5. $s = \frac{40}{\tan \theta}$; about 53°

Chapter 14 Assessment Answer Key

Mid-Chapter Test
Page 895

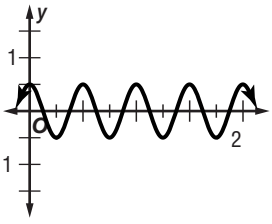
1. C

2. A

3. B

4. A

5. C

6. 

7. none; 45° or $\frac{\pi}{4}$

8. $\frac{\sqrt{7}}{4}$

9. $\cos \theta$

10. 1

11. See students' answers.

Cumulative Review
Page 896

1. $x \mid -2 \le x \le 3$

2. $x^2 - 3x + 9 - \frac{16}{x - 1}$

3. $13(n^2)^2 - 52(n^2) = 0$; $-2, 0, 2$

4. $\frac{\log 20}{\log 8}$; 1.4406

5. 21

6. $\frac{11}{56}$

7. $\frac{11}{32}$

8. Sample answers: $\frac{15\pi}{11}, -\frac{29\pi}{11}$

9. $\frac{\sqrt{3}}{2}$

10. $\sin \theta = -\frac{1}{2}$, $\cos \theta = -\frac{\sqrt{3}}{2}$

11. 2; 6π

12. $-\frac{5}{12}$

13. $\frac{\sqrt{98 + 28\sqrt{10}}}{14}$

14.

Answers

Chapter 14 Assessment Answer Key

Standardized Test Practice

Page 897

1. A B **C** D

2. E F **G** H

3. A B C **D**

4. E **F** G H

5. **A** B C D

6. E F **G** H

7. A B C **D**

8. **E** F G H

Page 898

9.

3	0		
.	/ .	/ .	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

10.

1	5		
.	/ .	/ .	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	**5**	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

11.

4	5		
.	/ .	/ .	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	**5**	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

12.

1	.	7	0
.	/ **.**	/ .	.
	0	0	**0**
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	**7**	7
8	8	8	8
9	9	9	9

13. A B **C** D

14. **A** B C D

15. **A** B C D

Chapter 14 Assessment Answer Key

Unit 5 Test

Page 899

1. $A = 70°$, $a \approx 27.5$, $c \approx 29.2$

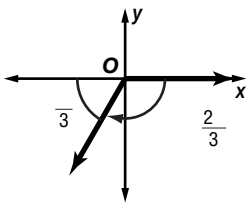
2. $\frac{-5\pi}{36}$

3. 324°

4. Sample answers: 50°, −670°

5. $\sin\theta = -\frac{4\sqrt{41}}{41}$; $\cos\theta = -\frac{5\sqrt{41}}{41}$; $\tan\theta = \frac{4}{5}$; $\csc\theta = -\frac{\sqrt{41}}{4}$; $\sec\theta = -\frac{41}{5}$; $\cot\theta = \frac{5}{4}$

6.



7. $-\sqrt{3}$

8. $\frac{\sqrt{2}}{2}$

9. 0

10. $\frac{\sqrt{3} - 1}{2}$

11. 99.5 ft

12. one; $B \approx 129.5°$, $C \approx 15.5°$, $b \approx 57.8$

13. Law of Cosines; $A \approx 87.1°$, $B \approx 54.2°$, $C \approx 38.6°$

14. Law of Sines; $C \approx 86°$, $b \approx 9.7$, $c \approx 15.6$

15. $\sin\theta = -\frac{8}{17}$; $\cos\theta = -\frac{15}{17}$

16. $-\frac{\pi}{3}$ or −60°

17. See students' work.

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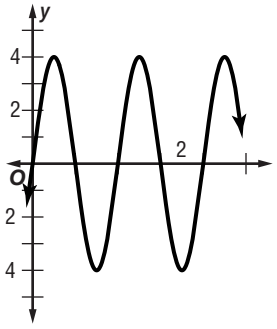
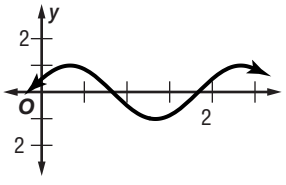
18.



19. 1; 120° or $\frac{2\pi}{3}$

20. none; 720° or 4π

21. $\frac{\pi}{3}$



22. −1; $y = -1$

23. $\frac{5}{3}$

24. 1

25. $\frac{\sqrt{2}}{2}$

26. $\frac{\sqrt{2} - \sqrt{6}}{4}$

27. See students' answers.

28. $\frac{\sqrt{35}}{18}$

29. $\frac{\sqrt{21}}{6}$

30. about 12 weeks

Answers

Chapter 14 Assessment Answer Key

Second Semester Test

Page 901

1. B
2. A
3. C
4. C
5. D
6. A
7. A
8. D
9. B
10. C

Page 902

11. D
12. B
13. D
14. A
15. B
16. C
17. C
18. D
19. B
20. A

(continued on the next page)

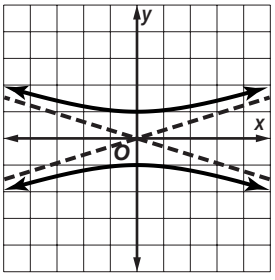
Second Semester Test *(continued)*

Page 903

21. $y = \frac{1}{8}(x - 2)^2 + 3$

22. $(x - 10)^2 + (y + 3)^2 = \frac{1}{25}$

23. $(0, \pm 1)$; $(0, \pm\sqrt{10})$; $y = \pm\frac{1}{3}x$



24. $(x + 1)^2 + (y + 1)^2 = 25$; circle

25.

26. inverse; 3.1

27. $\frac{9}{2}$

28. $-\frac{4}{3}$

29. 3.1945

30. 27

31. $\frac{\log 32}{\log 7} \approx 1.7810$

32. about 0.00012; $y = ae^{-0.00012t}$; about 32,600 years ago

33. 1, 4, 7

Page 904

34. 192

35. $\frac{8}{11}$

36. $243x^5 - 405x^4y + 270x^3y^2 - 90x^2y^3 + 15xy^4 - y^5$

37. Sample answer: $n = 2$

38. 35

39. $\frac{1}{18}$

40. $\frac{21}{32}$

41. positively skewed

42. $\frac{25}{65{,}536}$

43. $\sin\theta = -\frac{15}{17}$; $\cos\theta = -\frac{8}{17}$; $\tan\theta = \frac{15}{8}$; $\csc\theta = -\frac{17}{15}$; $\sec\theta = -\frac{17}{8}$; $\cot\theta = \frac{8}{15}$

44. one; $B \approx 22°$, $C \approx 123°$, $c \approx 29.2$

45. See students' answers.

46. $-\frac{7}{18}$

Answers

Chapter 14 Assessment Answer Key

Final Test

Page 905

1. A
2. C
3. A
4. B
5. D
6. D
7. B
8. D

Page 906

9. A
10. D
11. C
12. B
13. A
14. B
15. D
16. D
17. A

(continued on the next page)

Final Test *(continued)*

Page 907

18. C

19. C

20. A

21. B

22. B

23. B

24. D

25. A

26. C

27. B

28. A

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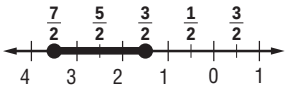
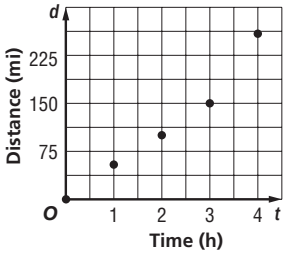
29. $\left\{a \mid -\frac{7}{2} \le a \le -\frac{3}{2}\right\}$



30.



31. Sample answer using (2, 100) and (3, 150): $y = 50x$; 300 mi

32. consistent and independent

33. $t \ge 0$; $b \ge 100$;

34. 500

35. (−1, 2, −3)

36. $\begin{bmatrix} -3 & 6 \\ 0 & 6 \end{bmatrix}$

37. −40

(continued on the next page)

Answers

Final Test *(continued)*

Page 909

38. $\frac{1}{10}\begin{bmatrix} 0 & -5 \\ 2 & 1 \end{bmatrix}$

39. $18x^6 + 45x^4 + 2x^3 + 5x$

40. $\frac{15 - 5\sqrt{6}}{3}$

41. $3t^{\frac{8}{3}}u^2$

42.

43. $3x^2 + 7x - 6 = 0$

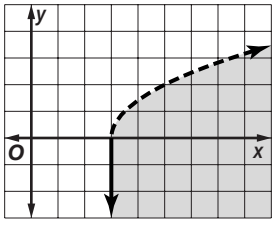
44. $y = 4(x - 2)^2 - 9$

45. -228

46. $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}$

47. $g^{-1}(x) = \frac{x + 1}{2}$

48.



49. $(x - 2)^2 + (y + 1)^2 = 25$

50.

51. $y = 3(x - 2)^2 + 7$; parabola

Page 910

52.

53. asymptote: $x = -4$; hole: $x = 3$

54. $-\frac{1}{2}$

55. 24

56. $y = 5000e^{0.0087t}$

57. 45

58. 2400

59. $-3, -24, -171$

60. 2520

61. $\frac{1}{6}$

62. 47.5%

63. one; $B \approx 14°$, $C \approx 141°$, $c \approx 10.4$

64. 1